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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 226

Planning and Design of Airport Terminal Restrooms and Ancillary Spaces

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

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FOREWORD

By Joseph D. Navarrete
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ACRP Research Report 226 provides a thoughtful, step-by-step process to help airport industry practitioners plan, design, and implement terminal restroom and other ancillary amenity projects. It is an updated and expanded version of *ACRP Report 130* and reflects the latest thinking in this quickly evolving topic. The guidebook effectively uses graphics, icons, and callouts to help readers quickly find the information they need to develop facilities that balance cost, aesthetics, space limitations, and day-to-day maintenance requirements. It will be of interest to anyone whose goal is to meet customer expectations for airport restrooms and related facilities.

Customer satisfaction has become one of the primary drivers for the success of an airport, and restrooms and ancillary facilities often provide the first and last impression of a destination. Since the publication of *ACRP Report 130* in 2015, design requirements and customer expectations have continued to evolve. Examples of activities and services that are now being accommodated at airports include service animal relief, lactation/nursing, companion care, children's play areas, sensory rooms, meditation, worship, ablution, sleeping, fitness, showers, and health/urgent care. Research was needed to provide the industry with guidelines to create facilities to meet these activities and services.

The research, led by Alliiance, began with a review of current restroom and amenities planning and design resources. The team then gathered extensive input from key stakeholder groups, airport staff, product developers, and representatives from other entities with facilities of similar scale and scope. Detailed case studies were conducted for restrooms and amenities at U.S. and international airports to review recent layouts and designs. The research team then used their findings and expertise to develop the guidebook.

The guidebook's recommendations are presented in three sections: planning, design, and implementation. One innovative and broad-ranging recommendation in the planning section (Chapter 2) is for airports to form a "restroom team" consisting of representatives from key stakeholder groups. This team would not only lead restroom projects but also serve as a forum for addressing the occasional stakeholder conflicts that can arise. Chapter 3 presents an updated "restroom of the future" prototype that addresses timely concerns regarding accessibility, gender, communal hygiene, and maintenance. The design section (Chapter 4) features steps for evaluating options based on considerations for initial cost, life-cycle cost, warranty, maintenance, and sustainability. Next, in the implementation section (Chapter 5), the guidebook gives practitioners suggestions for project phasing, delivery methods, and maintaining standards for various products (e.g., fixtures and finishes).

Appendices A through I of this report and interactive versions of worksheets presented in Tables 4-1 through 4-3 are available on the TRB website by searching on "ACRP Research Report 226".

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SUMMARY

Planning and Design of Airport Terminal Restrooms and Ancillary Spaces

There is a change underway in the aviation industry. It is a grassroots movement—passengers have spoken through blogs, media articles, and surveys. Restrooms in airport terminals that have lines or are cramped, dirty, smelly, or ugly are no longer tolerable. If travelers can have a massage and grab a coffee en route to their gate, they feel it is reasonable to expect restrooms with a few touches of hospitality such as soft lighting, warm water, and calming music. Such small conveniences are an easy accommodation. The real challenge for airports is to provide restrooms with enough space for people to move around in and offer secure, clean, and dry places for their belongings. The most frequent complaints airports receive concern these issues, and the industry is in catch-up mode.

The driving force behind the development of this guidebook is customer service. When people travel, few things dampen their experience more than confronting a barrier to fulfilling their needs. Airport terminal restrooms are used by all people no matter their age, gender, culture, or mobility. The further an airport goes to accommodate the spectrum of traveler needs, the more likely it is that a traveler will choose its location over another. Because of the diversity in ages, health, and origins of the traveling public, the cleanliness of restrooms is a primary concern. Promoting and accommodating personal hygiene and providing clean surfaces help prevent the spread of both common ailments and pandemics.

ACRP Report 130: Guidebook for Airport Terminal Restroom Planning and Design, published in 2015, presents a step-by-step process to help airport practitioners plan, design, and implement airport restroom projects with customer service in mind. The research for that project revealed that additional restroom-related topics needed to be considered, such as lactation rooms, pet relief areas, and shower facilities, and suggested these as topics for future research. This guidebook, *ACRP Research Report 226: Planning and Design of Airport Terminal Restrooms and Ancillary Spaces*, addresses planning, design, and implementation of airport spaces such as lactation rooms, pet relief areas, and shower facilities, in addition to restrooms. This research suggests that while restrooms are the mainstay of an airport's supportive infrastructure, a collection of amenity spaces are rapidly becoming commonplace in airports. These amenity spaces are organized into three groups in this guidebook:

- **Regulated/essential amenities.** Regulated amenity spaces are those required by building codes and governing bodies such as aviation authorities or those amenity spaces deemed essential to the needs of travelers based on this research. These include the following:
  - Service Animal Relief Area (SARA)
  - Lactation

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  - Nursing Mothers
  - Companion Care Restroom
  - Changing Table Restroom
- **Waiting-related amenities.** These spaces provide benefits and relief primarily to travelers who arrive early for their flight or who have a short layover between flights. These are
  - Companion Waiting Area
  - Children's Play Area
  - Sensory Room
  - Meditation/Quiet Area
  - Yoga Area
  - Worship
  - Ablution
  - Smoking
- **Layover-related amenities**. This group of spaces provides accommodations for travelers who have a long layover (e.g., overnight) and include
  - Sleeping
  - Fitness
  - Public Showers
  - Clothes Changing
  - Health/Urgent Care
  - Business Centers

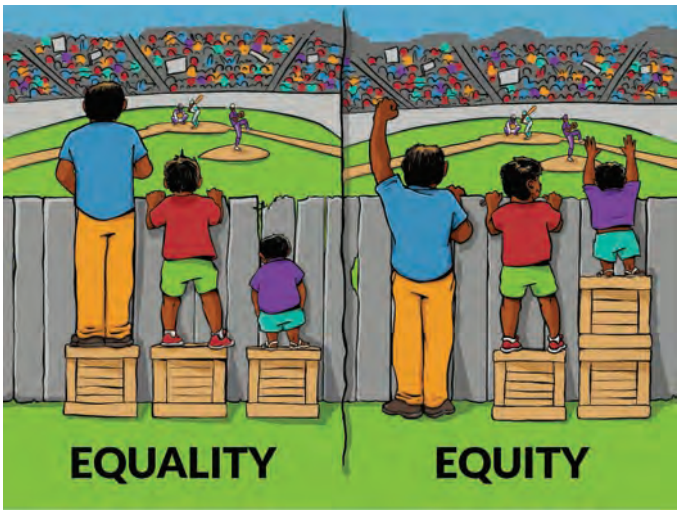
ACRP Research Report 226 continues the mission of *ACRP Report 130* by providing airport managers and planning/design professionals the tools to meet the increasing demands of today's traveling public by utilizing the following:

- Findings regarding the state of the industry based on a literature review, surveys, focus groups, and case studies.
- Planning methods for evaluating existing restrooms to determine what is needed and prototypes to guide the development of a restroom and amenities master plan.
- Design guidelines for airports of different sizes and budgets to compare and prioritize restroom and amenity components and features, including considerations for new construction/renovation and the benefits of standardization.
- Implementation strategies for handling construction impacts and post-construction review.

As this guidebook will demonstrate, restrooms and amenity spaces that are successful find a balance among accommodation of the needs and expectations of the traveler, the efficiency of airport operations, and the cost of building and managing these spaces. Although aesthetics are important and certainly affect a traveler's perceptions of cleanliness and safety, function still rules. Functional trends in restroom design, for example, still include touch-free environments, open entryways without doors, large-format materials with a minimum of joints, concealed trash, space for belongings, hand-drying options, and considerations for sustainability.

One of the root drivers for this guidebook was to advance accessibility—not just in terms of American with Disabilities Act (ADA) requirements, but complete access for everyone. On the left side of the graphic in Figure S-1, "Equality" is shown as providing individuals with the same accommodation. In the illustration of equality, the tallest person doesn't need a boost to see, the middle person now has a view, but the third person still needs the additional accommodation of a viewing hole to see. On the right side of the graphic, labeled "Equity," each person is provided with the accommodation they need to experience an unobstructed view of the game without any modification to the fence.

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Graphic provided courtesy of Interaction Institute for Social Change | Artist: Angus Maguire.

Figure S-1. Equality vs. equity.

But what if you use a wheelchair or crutches? Stacked boxes don't help. The issue is not the height of the people; it's the fence. That is the barrier.

As airport restroom and amenities teams have found (more information on the team is provided in Chapter 2), eliminating the "fence" is a daunting and sometimes unattainable task. Nearly every good design decision creates a conflict with something else. For example, a fixed, fold-down step at a sink provides a child or person of short stature access to the faucet but renders the sink inaccessible because the step prevents a wheelchair from pulling in.

This research has shown that the starting goal should always be adjacency for any given function or task. This means, for example, that when a person is at a sink, everything should be within reach for that person (e.g., water, soap, drying) without that person having to move to a different location. This provides equality for an able-bodied adult, a child, a person in a wheelchair, and so forth. To make people's experience at the sink equitable, though, an adjustable-height sink that will accommodate every height should be provided. However, adjustable-height sinks often have a flexible drain with a lot of ridges, which is a maintenance nightmare. This example illustrates the unlikelihood of achieving perfect solutions. But it's also the challenge.

Every airport has a unique context and experiences issues that will require compromise among a wide range of stakeholders. The hope is that the findings regarding the state of the industry, planning methods, design guidelines, and implementation strategies presented in this guidebook will be helpful to airport restroom and amenities teams as they plan and design restrooms and amenities that are the best fit for their airport and its travelers.

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CHAPTER 1

Introduction

1.1 How to Use This Guidebook

The task of planning and designing traveler-responsive restrooms and amenities is complicated. Nearly every expectation, requirement, or initiative conflicts with another. Adding enough space to maneuver in the toilet stall with a carry-on bag, for example, takes away potential income-generating square footage for concessions or holdrooms. However, compressing the plumbing chase footprint to increase usable restroom area creates a hardship for the plumbers. Piecing together this puzzle requires more than just one or two facilities staff and an architect. It requires a team: the restroom and amenities team (see Chapter 2). Representatives of all airport stakeholders and outside experts should work together to prioritize customer accommodations so that the planning and design are fiscally responsible and do not adversely compromise airport operations.

The scope of this guidebook focuses on public airport restrooms in both airside and landside areas. However, many of the considerations apply to airport staff restrooms as well. In addition to restrooms, the research team has identified the following amenity spaces as being more frequently implemented and providing important and necessary accommodations for the traveling public. The amenities addressed in this guidebook include

- **Regulated/essential amenities.** These are spaces currently required by the FAA and building codes or are essential for the needs of every traveler:
  - Service Animal Relief Area (SARA)
  - Lactation
  - Nursing Mothers
  - Companion Care Restroom
  - Changing Table Restroom (regulation anticipated)
- **Waiting-related amenities.** These spaces provide either distraction or accommodation while waiting for a flight, particularly if it is delayed:
  - Companion Waiting Area
  - Children's Play Area
  - Sensory Room
  - Meditation/Quiet Area
  - Yoga Area
  - Worship
  - Ablution
  - Smoking
- **Layover-related amenities.** These spaces accommodate primarily travelers with an extensive or overnight stay at the airport between connecting flights:
  - Sleeping
  - Fitness

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- Public Showers
- Clothes Changing
- Health/Urgent Care
- Business Centers

This guidebook is organized chronologically to follow the process that the restroom and amenities team navigates, from determining drivers and goals through post-construction evaluation.

Planning is the first of the three primary stages in a restroom project. It is the walk-before-running stage. It is a big picture view of what the airport has and what it needs in terms of where restrooms and amenity spaces should be located, how many fixtures are required, and how much space each location will need. Planning also considers timing—whether all restrooms will be completed at once or in phases. This is different from the design effort, which focuses on the quality of the components rather than the quantity.

Design balances the character of the spaces with the durability and maintenance of the materials and facility operations. This requires prioritizing the most impactful components that meet cost and maintenance requirements. When the scope of the work to be done is finalized and documented, the implementation effort begins.

Although the implementation or construction period is relatively straightforward, it is an opportunity to monitor which aspects of the design are difficult to build or obtain as well as to refine details both for the project at hand and for future work. With the grand opening of the new spaces comes the compilation of the restroom and amenities standards, which will streamline future restroom initiatives. Figure 1-1 illustrates the restroom development process.

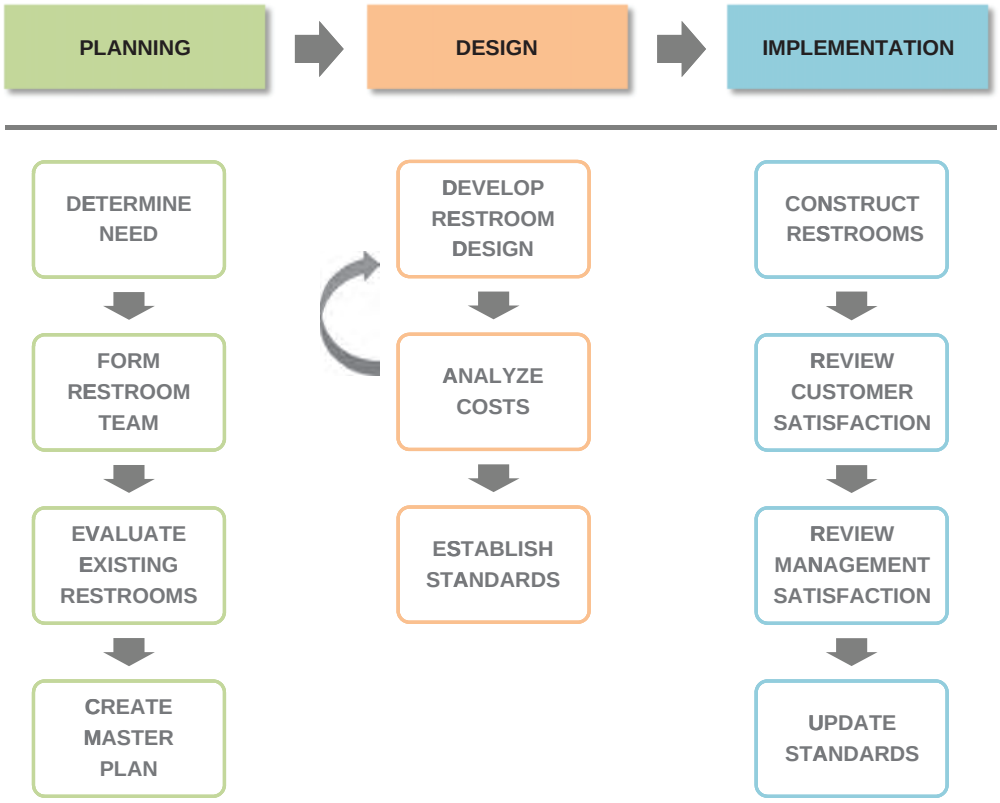


Figure 1-1. Restroom and amenities development process.

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Supporting forms and reference materials for this guidebook are provided in Appendices A through I, which are available on the TRB website by searching "ACRP Research Report 226".

Throughout this guidebook, the following icons have been used to highlight significant points of interest:


Significant Customer Service Impact


Accessibility


Sustainability


Product Development Opportunity


Potential Conflict with Other Needs

1.2 Research Approach

The development of this guidebook began with reviewing current resources available for the planning and design of restrooms and amenities, soliciting input from significant stakeholders—including the traveler and the airport management perspectives—and conducting case studies on restrooms of various sizes and amenities at airports of various sizes throughout the United States, Canada, Europe, and the Middle East. In the process of studying the evolution of public restrooms and current trends in airport restroom design, the research team envisioned the airport restroom of the future (see Chapter 3) to provoke contemplation and discussion.

With an understanding of the state of the industry, the team focused on developing comprehensive methods for evaluating an airport's current restroom and amenity facilities to determine the airport's actual needs. Building on these data, the team developed formulas for restroom sizing along with prototypes and distribution strategies for both restrooms and amenities to aid the restroom and amenities team in its development of a master plan. Considering airports of different sizes, the team developed design considerations to guide the process of selecting components by prioritizing such aspects as costs, maintenance, and sustainability. Guidelines on the impacts of new construction versus renovations were also created. Finally, a process for evaluating the success of a restroom project was formulated to refine an airport's restroom standards for future projects.

1.3 State of the Industry

In 2008, Airport Interviewing & Research, Inc., conducted an extensive survey of airport travelers' restroom concerns and desires titled "In Search of the Perfect Restroom." The top findings were

1. Concern about cleanliness and germs
2. Concern about privacy
3. Concern for security of personal belongings
4. Expectations for convenience in terms of amenities and proximity

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Travelers with disabilities also included desires for

- Restroom(s) reserved for travelers with disabilities, so they do not have to wait for the accessible stall to be free
- Clear signage
- Large enough toilet stalls for both maneuvering and belongings

Other concerns and desires included preventing the intrusion of natural and fixture noises, providing adequate ventilation and lighting, considering broader fixture features, and providing light colors, as well as using fewer or no grout joints that tend to show dirt.

The focus groups and surveys conducted for this guidebook confirmed that these issues remain. It was also confirmed that the airport industry is aware of the customer service issues, although perhaps not to the depth revealed in the outreach efforts conducted for this research. Aviation industry surveys tend to lack enough detail related to restrooms for airport managers to fully appreciate the extent of their customers' concerns. The aim of this guidebook is to support airport managers by providing the necessary level of detail.

Some relatively new amenity spaces, like SARAs and lactation rooms, have to meet similar requirements as restrooms in terms of comfort, durability, and being welcoming. In this guidebook, amenity spaces are considered with the same focus on customer service and airport maintenance as restrooms.

1.4 Attributes of Successful Airport Restrooms and Amenities

Ideally, a visit to a restroom is one of many unremarkable daily tasks. Travelers are thinking about that sunny beach that awaits, the opening words of their big presentation, or seeing their kids after a month at the grandparents. The restroom should not jar travelers from their reverie. At most, they should pause as they enter the restroom and think, "Oh, isn't this pleasant!" Public restrooms in airports often give travelers their first and last impression of their destination. No wonder the passenger's experience is so high on customer service surveys. In addition, restrooms are the one space in our built world that everyone has an opinion about, and the opinion is often an emotional one.

As this guidebook will demonstrate, restrooms and amenities that are successful find a balance among the accommodation of the needs and expectations of the traveler, the efficiency of airport operations, and the costs of building and managing these spaces. While aesthetics are important and certainly impact a traveler's perceptions of cleanliness and safety, function still rules. Toilets must work and be clean; dispensers need to have ample paper supply; there should be a clean, dry location to place belongings at the sink; the soap dispensers must work; and amenity spaces need to be located where they are needed with the necessary accommodations.

Trends in restroom design include touch-free environments, open entryways without doors, large-format materials with minimal joints, concealed trash, alternative hand-drying options, and sustainability. This includes water- and energy-saving flush valves and faucets, energy-efficient lighting, and occupancy sensors. As can be seen in the amenity space types listed in Section 1.1, providing the traveler with the most engaging and accommodating traveling experience is also a rapidly growing trend.

The biggest trend in the last decade, however, has been a focus on customer service, which is the primary driver for most of these initiatives and the impetus for this guidebook. Nearly every airport has some means of collecting travelers' comments—from old school suggestion boxes to social media. The key is to track the pulse of the airport's heart—its traveling public.

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CHAPTER 2

Planning

2.1 Drivers and Goals

Over the last couple of decades, complaints to airports about restrooms have been mounting. The most frequent comments from travelers are that restrooms are filthy and look outdated, and travelers always seem to experience a long line to use the facilities. Other comments mention that bathroom tiles are either chipped or cracked; tile grout looks mop-water gray; and the toilet stalls are so small that passengers with carry-on baggage have difficulty getting the door closed, and then end up using their bags to keep the door closed because the latch doesn't work.

More recently, travelers' expectations have been broadening, and they now sometimes expect certain amenities. Newer needs include spaces for lactation and nursing mothers and restrooms with changing tables for older children and adults. Passengers are asking for restroom space for service animals, and not just space outdoors by the parking lot, but also indoors on the secure airside so they have time to stop for Fido during a brief layover instead of dealing with a potentially long walk and time-consuming re-screening procedures afterward.

To meet these customer expectations, airports are having to make significant changes to their facilities. However, with the prospect of big changes come opinions, and then studies followed by regulatory considerations, and options. Updating restrooms has become extremely complicated. Agencies such as the FAA and National Council on Disabilities, creator of the Americans with Disabilities Act (ADA), are stakeholders in the process, and there are many regulations to be addressed in the process. To help bring about this change, airport managers, planners, and designers need to navigate the first step of any restroom or amenities project: planning.

As with every capital improvement at an airport, there are needs that drive a project's initiation. These drivers might include the following:

- A change in airport vision/mission
- Customer service complaints
- A related project, either new construction or renovation
- A change in operation (e.g., increased passengers)
- Security/safety issues
- Building code/accessibility compliance
- Sustainability master planning
- Aesthetics (typically dated finishes)
- Failing materials/products
- Standardization

Restroom projects are often incorporated into a larger renovation, expansion, or new building. Occasionally, however, an airport will decide to focus only on restrooms, ensuring

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they all reach a consistent standard. Depending on the type of space, either amenities are added to a restroom-related scope or they become a standalone project. Regardless, goals need to be developed to generate an architectural program. This guidebook details the various elements that can become part of an architectural program, including the types and sizes of spaces, the quantity of each space and approximate locations, level of material quality, schedule, and budget. To develop a program, however, the existing facility requires assessments to understand its current state.

Two independent efforts are required to assess an airport's existing restrooms. One is a physical survey of each restroom space to determine its age, condition, number of fixtures, critical dimensions, and so forth. This effort will document what the airport currently has.

A parallel effort calculates what the airport needs. This effort looks at the airport's passenger flow, which determines how many fixtures are needed (and their distribution throughout the airport) to accommodate the calculated passenger load. Comparing these calculated quantities with the physically surveyed quantities and considering any increases in scoping under current federal, state, and local ADA regulations will determine the number of fixtures and restroom locations that will need to be added and/or relocated.

Before either of these efforts is launched, however, a group of stakeholders needs to be assembled: the restroom and amenities team.

2.2 The Restroom and Amenities Team

There is one attribute every member of this group will need for the process to succeed—a sense of humor. Besides the inevitable bathroom jokes, the process of designing airport restrooms brings a barrage of regulations that often conflict, various must-have lists, and goals. Team members need to be resilient and open to compromise, and then support the final program while staying up to date on potential improvements and refinements for the future.

From day one, the team should include every manager with a vested interest in the planning, implementation, and maintenance of a restroom or amenity space. From within an airport, this should include representatives from the following groups:

- Facilities/planning
- Customer service
- ADA coordinator
- Carpentry
- Heating, ventilating, and air conditioning (HVAC)
- Plumbing
- Electrical
- Information systems
- Cleaners
- Airport police

Consulting experts should include the following:

- Aviation planners
- Architects
- Interior designers
- Mechanical engineers
- Electrical engineers
- Technology systems designers
- Accessibility design specialists

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Everyone on the team should be committed to the success of the project from planning through construction and beyond. Ideally, a core group of members should meet periodically after the project's completion to assess customer service and maintenance as well as operational issues and opportunities for improvement. Continued group participation is recommended because challenges continually arise, circumstances change as the travel market fluctuates, restrooms and amenity spaces experience wear and tear, and new products and services are introduced.

A key benefit for team members committing to this type of project is the broadening of their expertise. They develop an awareness of initiatives in restrooms at other airports as well as other public facilities, and they become familiar with the range of products and systems available while monitoring new developments through relationships with manufacturers and while serving in relevant professional and industry organizations and conferences.

Eventually, cost and schedule estimates of various design options will be needed, so one or more of the following (depending on the airport's procurement requirements) also should be invited to the team:

- Cost estimator
- Construction manager
- General contractor
- Primary subcontractors

2.3 Existing Restroom Evaluation

With the restroom and amenities team in place, the easiest task to initiate, but also the most time intensive, is determining the current state of an airport's restrooms and amenity spaces. This process involves visiting each of these spaces in an airport as well as associated spaces (e.g., janitor's closets, pipe chases, and related storage rooms) and documenting existing conditions.

Restrooms documented for this guidebook can serve as models for an airport's space evaluation. Evaluation forms provided in Appendix B: Existing Restroom Evaluation Forms include the components specific to each amenity space (see Chapter 4). For restrooms, the following types of components are listed in the Appendix B evaluation forms:

- Proximity to other airport functions
- Signage
- Surfaces
- Hardware
- Accessories
- Amenities
- Plumbing
- HVAC
- Power
- Lighting
- Technology
- Accessibility

The Appendix B evaluation forms can be edited to create a list specific to an airport's unique restroom features. Ideally, a representative from each of the departments responsible for the management or maintenance of restrooms will fill out these forms. Forms should be updated as modifications are made to any space to keep inventory current. In addition to surveying

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the existing restroom facilities, it is suggested that maintenance data from the last 5 years be compared with customer service data, as described in the next section.

2.3.1 Customer Service Process



An important means of understanding how well restrooms and amenities are operating and being received by the public is to offer travelers a way to communicate their comments. This can be as simple as setting up a rack with comment cards and a deposit box near or in airport restrooms. Technologies, however, provide more streamlined and sustainable methods for obtaining this type of information. For instance, numerous airports provide a number for texting or tweeting comments, and the new restrooms at Singapore's Changi Airport provide a touch screen with a five-button scale that goes from happy (excellent) to sad (very poor) to rate a passenger's restroom experience. Some airports even ask for feedback when connecting to an airport's Wi-Fi. The brief survey questions are often tailored to the person's location, so if a traveler is standing near a restroom when logging on, the questions relate to the restroom experience. One aspect to consider regarding feedback devices that require a person to touch a button to indicate their satisfaction is hygiene.

Data from these sources are typically collected by an airport's customer service staff for monitoring. It is also beneficial to have digital comments sent directly to the facilities department so cleaning or maintenance crews can be immediately dispatched to take care of things like empty paper dispensers or clogged sinks. It is important to be timely in the review of comments to minimize recurring issues.

There are also airport industry surveys, typically conducted annually, that address customer service issues with restrooms among other aspects of the terminal. Airports Council International's (ACI's) Airport Service Quality (ASQ) is currently the most broadly used survey. Presumably, questions about amenities spaces will soon be included in these surveys as well. If an airport does not implement some form of feedback collection, one or more of these survey options should be considered.

When changes to an airport's restrooms are being contemplated, external feedback is invaluable in confirming customer service issues and evaluating priorities. The research team's surveys of travelers and airport managers revealed that there is often a disconnect between the perceptions of these two groups.

2.3.2 Maintenance Process

The survey of existing restrooms and amenities provides an invaluable opportunity to interview maintenance crews to determine which products have not met expectations in terms of durability and ease of maintenance and which procedures have proved to be inefficient or ineffective, requiring change. The survey can provide an opportunity to ask maintenance-related questions concerning the frequency and procedure for cleaning restrooms, how the trades monitor the need for repairs, and whether there is ample attic stock for quick fixes.

Numerous airports, both large and small, report that they periodically walk around with clipboards in hand to note fixtures or surfaces that need attention. Examples of the forms used in this type of process are provided in the case study on Los Angeles International Airport (LAX) (see Appendix C.1). At Hartsfield–Jackson Atlanta International Airport (ATL), a full-time cleaner is assigned to each women's restroom to wipe up spills and keep paper stocked. ATL has found that this full-time attendant, who manages a small number of restrooms, reduces the frequency of completely shutting down a restroom for basic upkeep. Another airport keeps a strike team poised to clean any restroom immediately after a surge.

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Regardless of the strategy, the key to an effective maintenance program is to create a standard operating procedure (SOP) for each restroom and amenity space for all maintenance staff to reference. SOPs help ensure that the expected level of quality control is maintained in areas of cleaning and repairs. SOPs should be reviewed annually to confirm that new products are included and that all customer service issues are being addressed. A sample cleaning SOP from LAX is included in Appendix C.1.

2.4 Quantity of Spaces Needed

In tandem with evaluating the condition of the restrooms and amenities, the quantity and distribution of these spaces should be analyzed. For restrooms, there are formulas and data available to evaluate the complex attributes of the ebb and flow of traffic through these spaces. Central to this analysis is determining the fixture count in both airside and landside amenities. After this exercise is complete, an airport should determine the distribution of restrooms and amenity spaces throughout the airport. This is determined first by the development of restroom prototypes (as highlighted in Section 2.5) and amenity prototypes (as indicated in Section 2.6), which is followed by test fitting the prototypes to determine the restroom and amenity locations (as noted in Section 2.7). The last step in the process is creating a master plan for current and future restroom and amenity projects.

2.4.1 Restroom Fixtures

Something important to consider before delving into calculating the size, locations, and fixture quantities of restrooms is a restroom's level of service (LoS). The original measure of LoS, as developed by International Air Transport Association (IATA), quantifies passenger flow through an airport, both by space and time. LoS accounts for the capacities of areas such as check-in, security checkpoints, holdrooms, and so forth. *ACRP Report 55: Passenger Level of Service and Spatial Planning for Airport Terminals* provides an overview of how LoS is applied to airport planning. Restroom sizes and locations, however, are not specifically addressed in *ACRP Report 55*.

It is a subjective exercise to correlate LoS with airport restrooms and is outside the scope of this guidebook. However, some guidelines can be suggested based on general assumptions. *ACRP Report 25: Airport Passenger Terminal Planning and Design* states that a passenger's perception of LoS can be influenced by the availability and cleanliness of restrooms. From this, it can be inferred that an airport striving for LoS A (excellent) versus LoS C (good) should provide well-appointed restrooms that provide most of the amenities recommended here including adequate space for belongings and circulation, accommodations for individuals with additional needs, and highly maintained facilities.

The planning principles described in this chapter equate mostly with LoS B (high). LoS B indicates a high level of customer service with room for improvement if an airport strives to provide the highest levels of customer service. In terms of capacity, this means increasing the number of restroom fixtures and the frequency of restroom locations.

Airport restroom sizes are typically calculated based on the portion of the terminal they serve and its projected passenger demand. Airside, concourse, or secure (post-security) restroom sizes are calculated based on aircraft seat capacity. Landside, terminal, or non-secure (pre-security) restroom sizes are based on passenger peak hours and the number of expected visitors.

Landside and airside restrooms accommodate a different mix of users. While restrooms in both areas are visited by travelers and airport employees, landside locations are also frequented by meeters and greeters, well-wishers, and, in some airports, transportation chauffeurs. Outbound or departing travelers, concerned about making their flights and unsure how long

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it will take to pass through security, often wait to use the restroom until they are on the secure side. Likewise, inbound or arriving passengers tend to wait until they've landed to avoid the cramped aircraft restrooms.

Comments from the research team's case studies and focus groups suggest that the impact of these scenarios on terminal restrooms is that landside restrooms tend to get more abuse since they are used by a broader spectrum of the non-traveling public. Some comments suggested that these locations are more akin to subway station restrooms. While the durability of the materials used for building restrooms remains paramount, higher-end materials may not be as appreciated on the landside, indicating that more of the overall budget could be diverted to providing extra amenities in airside restrooms, as described later in this chapter.

The following sections describe the fixture count calculation process for airside and landside loads. It is important to note that areas within an airport that are not greatly influenced by passenger flow should use local building code requirements to determine the fixture counts.

2.4.2 Airside Calculations

Calculations for secure restroom locations are typically based on the types of aircraft serving the adjacent concourse. In order to provide reasonable walking distances from the gates, planning standards use the FAA's equivalent aircraft (EQA) factor to convert an existing gate size to an EQA: 1 EQA ≈ 145 Seats (typical narrowbody aircraft).

Volume 1 of *ACRP Report 25* recommends providing one restroom module (one restroom for males and one for females) for every eight EQA. Depending on the fleet mix, this translates to one module located centrally to a reasonable number of gates (approximately four narrowbody gates on each side of a double-loaded concourse) where the farthest a passenger must walk is approximately 450 to 550 ft. This distance should be used as a recommendation only because local building codes may have more stringent guidelines depending on occupancy classifications.

Historical observations indicate deplaning (arriving) passengers produce the greatest demand on the concourse restrooms. Because most passengers on short-haul domestic flights wait until arrival to use restroom facilities, it is important to provide adequate capacity to serve these arriving passengers. This is especially important where passengers from flights that simultaneously arrive from adjacent gates produce a surge effect on the restrooms located nearby. Taking surges into account, the peak 20-minute arrival period of the peak arrival hour is a good indicator for calculating peak passenger capacity.

Observations also indicate that arriving passengers typically use the first restroom location they pass between their arrival gate and baggage claim or a connecting gate. Passenger behaviors suggest this is true even if queues are present and another location may be only a short distance away. This occurs primarily because passengers are uncertain about the distance to the next restroom. Due to these passenger tendencies, a good planning rule, as stated in *ACRP Report 25*, is to provide fewer restroom locations with larger capacities. It is also recommended that restroom modules be placed adjacent to major concession nodes within airside concourses.

Today's restrooms should be designed for at least an equal split between male (toilets/urinals) and female fixtures because the trend of increasing female travelers is expected to continue (refer to Appendix E: Surveys and Appendix H: Bibliography). However, it is recommended that 25 to 50 percent more fixtures should be provided for females than for males. This is due in part to longer utilization time and, as voiced in the restroom focus groups, because female travelers typically bring children into restrooms with them, which can slow them down. Women also do not experience the quick turn-around advantage that urinals provide for men (see Chapter 3). Conducting usage surveys, especially at restrooms known to have lines, can help in understanding passenger loads unique to concourse configurations, and

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Potty Parity

A report by The Royal Society for Public Health in the UK argued that there should be twice as many female toilets for every one male toilet/urinal because of "time consuming factors related to clothing, menstruation, and anatomical differences." The argument is that women do not have equal access because of longer wait times, in part due to the inherited male bias in the creation of building code requirements. Females also still tend to have a stroller and/or small children in tow.

Source: "Women need twice as many public washrooms as men . . ." *Global News*, May 28, 2019.

airline practices at gates such as high turn frequency, and so forth. Results may even suggest a higher female-to-male ratio at some locations within the airport.

It is important to consult local building codes as well as airport authorities, which may mandate specific requirements. Keep in mind that building codes provide minimum requirements. To provide exceptional customer service, especially in larger hubs, code minimums have proven to be inadequate.

A family restroom should also be included in each module to provide a space for individuals, companions, or families with special needs (see Chapter 4).

The following four-step process (determine EQA factor, determine number of modules, determine peak passenger capacity, and determine total number of male and female fixtures) can be used to determine the number of airside restroom modules and fixtures needed.

Step 1: Determine Equivalent Aircraft Factor

Utilize Table 2-1 to determine the concourse EQA factor by multiplying the number of aircraft in each airplane design group (ADG) by the EQA index. Sum all resulting EQA values to calculate the total EQA.

As an example, consider a hypothetical concourse that contains eight A320, two B757, and two B767 aircraft. Therefore, based on Table 2-1:

8 (A320) × 1.0	8.0
2 (B757) × 1.3	2.6
2 (B767) × 1.9	3.8
Total EQA	**14.4**

Table 2-1. EQA index.

FAA AIRPLANE DESIGN GROUP (ADG)		TYPICAL SEATS	TYPICAL AIRCRAFT	EQA INDEX
I	Small Regional	25	Metro	0.2
II	Medium Regional	50	SF340/CRJ	0.4
III	Large Regional	75	DHC8/E175	0.5
III	Narrowbody	145	A320/B377/MD80	1.0
IIIa	B757 (winglets)	185	B757	1.3
IV	Widebody	280	B767/MD11	1.9
V	Jumbo	400	B747, 777, 787/A330, 340	2.8
VI	Super Jumbo	525	A380/B747-8	3.6

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Note: A general planning rule is to utilize the design aircraft for each gate. However, it is important to understand wingtip adjacency conflicts, if any. Therefore, utilizing the largest aircraft that each gate is capable of accommodating, while producing no impacts to adjacent gates, will provide a good capacity estimate for the concourse.

Step 2: Determine Number of Modules

To determine the number of modules, divide the EQA factor (value from Step 1) by eight. Generally, rounding up will provide the appropriate number of locations. However, the concourse layout should be carefully considered when determining the number of modules (e.g., single- vs. double-loaded concourses—see Figure 2-1).

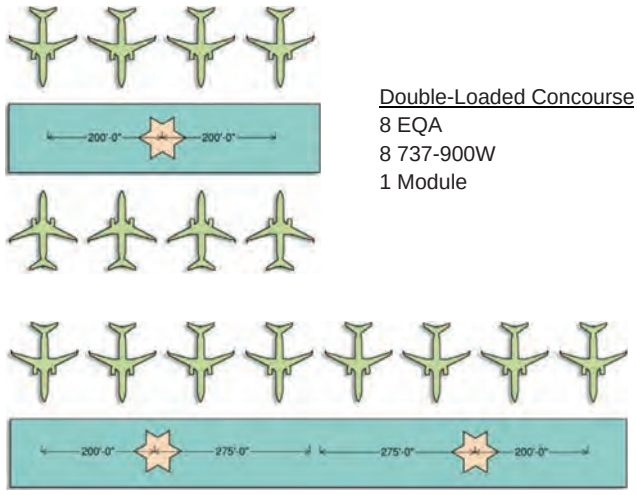
Step 3: Determine Peak Passenger Capacity

Peak passenger capacity (design passengers) is calculated using the following formula:

Design Passengers = **EQA** (value from Step 1) × **145 Seats** (1.0 EQA) × **90%** (load factor)

Note: Utilizing EQA seats will typically produce the greatest capacity scenario (all gates in use). However, this value should be compared and, in some cases, balanced against the design peak hour (typically calculated by the airport's planning consultant) to gauge effects on total calculated fixtures. It may be appropriate to adjust EQA seat values based on the expected fleet mix in use for each given airport.

For example, if the ADG III fleet mix expected to use the terminal is only 737-800 aircraft, seat capacity, depending on class configuration, can range from 160 to 185 seats. Load factor (LF) is based on latest industry planning standards and should be compared with factors unique to each airport.



Double-Loaded Concourse
8 EQA
8 737-900W
1 Module

Single-Loaded Concourse
8 EQA
8 737-900W
2 Modules

Figure 2-1. Locations based on concourse type.

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Next, use the design passenger capacity to determine passenger demand:

Peak 20-Minute Passenger Demand = **Design Passengers** × **Peak 20-Minute %**

> Note: The following peak 20-minute % values are recommended:
>
> - Concourses with hub activity = 60%
> - Concourses with origin and destination (O&D) activity = 50%
>
> In some cases, an O&D airport's passenger activity may exhibit peaks consistent with hub-type operations. Therefore, the above values should be used as guidelines. It is recommended that peaking characteristics unique to each airport be utilized (typically obtained from the airport's master plan forecast data) and compared to these values.

Last, calculate the design factor, which will be used to determine the required fixture count:

Design Factor = **Peak 20-Minute Passenger Demand** × **% Using Restrooms** (utilization)

> Note: The general utilization rate planning standard = 50% to 60%.

Step 4: Determine Total Number of Male and Female Fixtures

Since the number of female fixtures can fluctuate (with a suggested minimum being equal to the number of male fixtures, but potentially increasing the number of female fixtures based on the airport's customer service philosophy), the count of male fixtures is used for base calculations. The recommended count of male fixtures is determined as follows:

Male Fixtures = **Design Factor** (from Step 3)
× **Male %** ÷ **13** (13 equals peak 20-minute male passengers per fixture ratio)

> Note: When the passenger gender mix is unknown, assume a 50%/50% ratio (see Potty Parity text box). The peak 20-minute male passenger ratio is based on previous factors (LF, gender split, peak 20-minute percentage, utilization percentage) and includes an average 1.5-minute male and 2-minute female dwell time at a fixture. It should be noted that if known data unique to the airport are used, the ratio will fluctuate as the time is adjusted. For instance, as dwell time at the fixture decreases, the ratio increases and vice versa.

The female recommended fixture count is determined as follows:

Female Fixtures = **Male Fixtures** × **Female Increase Factor**

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Note: Table 2-2 illustrates the resulting male/female ratios for typical female increase factors. For facilities where the known gender mix is higher for females, the actual calculated female fixture ratio should be utilized, which is based on dwell time at the fixture. Example: 55% female @ 2 min/person and 45% male @1.5 min/person = (2.0 × 0.55) / (1.5 × 0.45) = 1.6

Table 2-2. Female increase factors for fixtures.

FEMALE INCREASE FACTOR	FEMALE/MALE FIXTURE % RATIO
1.25 (25%)	56%/44%
1.50 (50%)	60%/40%
2.00 (100%)	67%/33%

Finally, the number of fixtures per module (determined in Step 2) is calculated. Each module is equivalent to a typical restroom set.

Fixtures/Module = Total Fixtures ÷ Total # of Locations

Table 2-3 lists typical airside *male* fixture ranges for EQA factors categorized by O&D or hub-type facilities, passenger utilization rates, and peak 20-minute percentage factors. Female increase factors can be utilized with this table to calculate the number of *female* fixtures.

Important: An airside fixture count of six per gender is the minimum recommended for any facility type.

For non-hub facilities or concourses with limited activity, the fixture count should be compared to local building codes.

2.4.3 Landside Calculations

Landside restrooms are typically located within the major terminal areas such as check-in, baggage claim, and concession areas. Calculations are based on the total O&D peak-hour passenger (PHP) demand and the visitors of those passengers. One approach, as described in *ACRP Report 25*, is to provide one fixture per gender for every 100 individuals; however, many variables, such as the terminal layout, can influence this factor. Local building codes should be consulted as should airport authorities, which may mandate specific requirements.

Table 2-3. Total number of male fixtures per EQA based on 50%/50% gender mix, 1.5-minute dwell (male), 2.0-minute dwell (female), and 90% LF. Minimum six male fixtures for all airport classifications. Number of fixtures rounded.

		TOTAL MALE FIXTURES BY EQA			
		O&D		Hub	
Pax Utilization		50%	60%	50%	60%
Peak 20 min %		33% to 50%	33% to 50%	60%+	60%+
EQA	3.0	6	6	6	6-7
	4.0	6	6	6-8	7-9
	5.0	6	6-7	7-10	9-12
	6.0	6-7	6-9	9-12	11-14
	7.0	6-9	7-10	10-14	12-16
	8.0*	**7-10**	**8-12**	**12-16**	**14-19**
	9.0	7-11	9-13	13-18	16-21
	10.0	8-12	10-15	15-20	18-24
	11.0	9-14	11-16	16-22	19-26
	12.0	10-15	12-18	18-24	21-28
	13.0	11-16	13-19	19-25	23-31
	14.0	11-17	14-21	21-27	25-33
	15.0	12-18	15-22	22-29	26-35
	16.0	13-20	16-24	24-31	28-38

*The EQA of 8.0 relates to the 8.0 used in the Figure 2-1 example.

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Visitor ratios are typically provided with airport survey data or are compiled by an airport or its planning consultant and should be used to calculate the total visitor demand, when available. However, if this information is unknown, an increase factor of 20% for well-wishers (WWs—those sending off travelers) and 30% for meeters and greeters (M&Gs—those welcoming travelers) is a typical industry-wide planning standard.

For terminal facilities with enplaning/deplaning passenger processing functions (such as check-in and baggage claim, respectively) on separate levels, utilizing the associated visitor ratios with their respective passenger processing functions is appropriate:

WW + enplaning passengers

and

M&G + deplaning passengers

For terminal facilities with both check-in and baggage claim functions on a single level, a visitor ratio that is the average of the WW and M&G ratios may be more appropriate.

The following two-step process (determine design passenger demand and determine number of fixtures) can be used to determine the number of landside restroom modules and fixtures needed.

Step 1: Determine Design Passenger Demand

For a multi-level facility:

Check-in:

Design Demand = **Total Enplaning** (departing) **Peak-Hour O&D Passengers**
× **WW Ratio** (e.g., 1.20)

Baggage Claim:

Design Demand = **Total Deplaning** (arriving) **Peak-Hour O&D Passengers**
× **M&G Ratio** (e.g., 1.30)

For a single-level facility:

Design Demand = **Total O&D** (enplaning/deplaning) **PHPs** × **Visitor Ratio** (e.g., 1.25)

Step 2: Determine Number of Fixtures

Once the number of fixtures for males is determined, use the female increase factor used in the airside calculations.

Total Male Fixtures = Design Demand (based on gender percentage) ÷ Ratio

> Note: Ratio = 1 fixture per 70 PHP for first 400 passengers + 1 fixture per 200 PHP in excess of 400 passengers.

Total Female Fixtures = Total Male Fixtures × Female Increase Factor

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Additional Considerations

Following completion of Steps 1 and 2, fixtures per location per level should be distributed. One method is to distribute the required fixtures in direct proportion to each existing location's percentage of the total terminal fixture count (for a single-level facility) or each level's total fixture count (for a multi-level facility). However, this should be studied carefully as some locations may exhibit lines during peak passenger demand. As a result, a higher percentage distribution for those areas may be required.

For terminals with integrated multimodal ground transportation centers (GTCs) including taxi, limousine, rental car, and rail, a ground-access or curb-front survey can provide valuable information on passenger modal splits. This information can then be used to identify the appropriate passenger demand for those enplaning passengers that are dropped off at the curb, thereby reducing the actual demand placed on the restroom facilities located in the check-in area. Enplaning passengers utilizing GTCs, depending on location, should have access to restroom facilities prior to arriving in the check-in area.

Restrooms should be in proximity to the major passenger processing functions such as check-in, baggage claim, and security screening and to locations such as the meeter/greeter area, GTC, and major landside concessions nodes. Locations should also be visible from major passenger flow areas.

2.5 Restroom Prototypes

Whether for new construction or renovation, prototypes provide building blocks that allow the airport's restroom and amenities team to develop a master plan for locating public restrooms and amenity spaces throughout an airport terminal over a long-term timeframe. In the previously described process of evaluating an airport's restroom needs, the planning formulas provide the number of fixtures needed to accommodate anticipated passenger demand. However, it is important to understand how those numbers translate into actual physical space requirements.

In traditional airport planning, restroom blocks are assigned a square footage based on broad planning formulas and rules of thumb. Also, traditionally, restrooms have been placed near the bottom of the priority list. Thus, a 1,500-square-ft rectangular-shaped restroom block might be stretched, compressed, or hacked to fit the odd-sized spaces left over from the primary terminal planning design. As airport designers and managers routinely discover during their renovation projects, these spaces rarely function well with today's toilet stall or circulation space requirements. The result is often cramped restroom spaces that have inconvenient columns, impractical utility chases, and so forth.

Sections 2.5.3 and 2.5.4 provide not just a base square footage for two common prototype restroom plans, but also appropriate dimensions to efficiently streamline the planning process. Through case studies and professional experience, the research team has observed that nearly all public airport restrooms are based on one of these two layout plans—the room plan (see Section 2.5.3), which has the sink and toilet areas distinctly separated, or the galley plan (see Section 2.5.4), which lines up toilets and urinals on one side of the circulation aisle and sinks on the other. These two plans are relatively equal in size, although each has advantages and disadvantages that will be highlighted in the discussion that follows. These two prototypes work for a hub or O&D airport of any size and can be modified to accommodate any airport's unique terminal layouts, although space proportions may dictate one prototype plan over the other. It is not uncommon for a restroom team to develop two or three basic prototype variations for their airport. See Chapter 3 for an alternate prototype of the restroom layout.

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It should also be noted that concourse restrooms are typically in the same block as holdrooms. These usually range from 30 to 45 ft deep. Holdrooms serving international flights tend toward the larger depths.

Sections 2.5.3 and 2.5.4 show basic plan layouts of the room and galley prototypes, respectively, with three-dimensional images to illustrate how proposed components might appear in practice.

To provide a complete understanding of the restroom component space requirements, in Section 2.5.1 recommended sizing and configurations for each of the following spatial notes are provided:

- Entry
- Sink
- Baby diaper changing
- Toilet stall
- Accessible stall
- Urinal
- Grooming
- Chase
- Janitor/storage

In Section 2.5.6, additional diagrams that indicate locations of the following restroom elements are presented:

- Spatial nodes
- External restroom components
- Internal restroom components
- Signage
- Accessories
- HVAC
- Lighting
- Power
- Technology

A final consideration when developing prototype layouts pertains to the landside restrooms found in the ticketing area, baggage claim, GTC, remote parking structures, and so forth. For continuity and ease of maintenance, these restrooms will have the same fixtures, finish pallet, and so forth as the airside locations. However, given the behavior noted in Section 2.4.1—travelers typically using airside restrooms—the landside restrooms might be scaled back to the less burdened needs (no baggage) of M&Gs, WWs, and employees. This can save money and space that can be used to provide a greater value in the airside restrooms.

2.5.1 Spatial Nodes

Spatial nodes are stations of activity found in restrooms and some amenities. For example, the sink node is where there's a sink to wash up, paper towels for drying, trash, a shelf, mirror, coat hooks, a convenience outlet, and task lighting. The following spatial nodes are discussed in this section: entry, sink, baby diaper changing, toilet stall, accessible stall, urinal, grooming, chase, and janitor/storage.

Entry

The entry, as with other areas of circulation within a restroom, needs to accommodate two-way traffic for passengers with luggage, in wheelchairs, pushing strollers, and so forth.

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A clear width of 6 ft and 6 inches is a reasonable minimum; 7 ft is preferred. Avoid locating anything in the entry corridor that will cause a bottleneck such as information signs and drinking fountains. Designers should ensure that there are no sightlines into the restroom from the external public spaces, paying special attention to reflective surfaces and materials. A tensile barrier strap should be provided to close the restroom for maintenance. Artwork or some other element is useful to identify the entry, draw people into the restroom, and enhance the calming aspect of the restroom experience. Entry nodes should be consistent throughout the airport to serve as a macro-wayfinding element.

Sink

The most desirable arrangement of sinks is a pair with a drying station at each end that consists of a paper towel dispenser, a hand dryer (if used), and a trash receptacle. This locates each sink position next to a drying station (see Figure 2-2), thus eliminating water dripping on the floor as users access a remote station. While a drying station may be located between two sink nodes, drying time is shorter than washing, so conflicts should be minimal.



It is desirable to have a shelf behind the sink on which to place belongings while washing and drying hands. The shelf should be slightly higher than the sink top to remain dry, but not higher than the bottom of the mirror (ANSI A117.1 requires 40 inches from the floor to the reflective surface). The shelf should be 8 to 12 inches deep for purses, bags, hats, folders, and so forth.

Traditionally, sinks and sink tops have been approximately 24 inches deep. This seems to be a standard based on residential kitchen counters and commercial break room millwork, but it uses up valuable floor space without benefit. If a shelf is provided, as recommended previously, reaching beyond the 24-inch sink depth is not possible for persons with mobility impairments. Reaching a faucet with this sink depth can be difficult for children and even harder for travelers with impaired reach.



ANSI A117.1 requires that in restrooms with six or more sinks at least one sink is to have an "enhanced reach range." This means that the location where controls or sensors for water and soap are activated cannot be more than 11 inches back from the front edge of the counter and/or sink. The recommended sink depth to accommodate these requirements is 20 inches from the face of the raised shelf. From a universal design perspective, every sink should have these provisions, so everyone always has equal access.



The drying station should have a paper towel dispenser and trash receptacle. For the time being, paper towels are needed to clean up spills and messes. Hand dryers are an optional add-on, although their desirability is currently in question (see the text box on paper towels vs. hand dryers). If hand dryers are used, they should also be located next to a sink. This presents a design challenge in that paper towels, trash, and hand dryers all vie for the same space. The recent innovation of providing a hand dryer on a sink, either as part of the faucet or as a standalone device alongside a faucet and soap dispenser, is promising in that it simplifies the dispenser–hand, dryer–trash configuration. It also puts washing and drying in one location, speeding up the throughput.

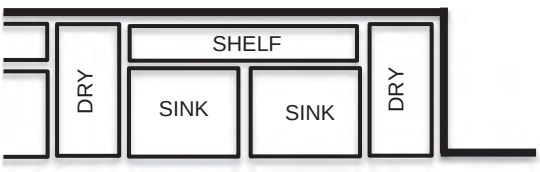


Figure 2-2. Sink node.

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Paper Towels vs. Hand Dryers

Results from the focus groups and surveys undertaken in this research (see Appendices D.1, D.2, and E) suggest that people are not enamored with hand dryers because they are loud (especially in a restroom where there are multiple dryers) and have the potential to cause issues with bacteria and dripping water. In addition, some users are reluctant to insert their hands into the narrow slot. For persons using wheelchairs or scooters, it can be awkward to reach hand dryers, and the noise of hand dryers has been known to seize hearing aids at certain frequencies, leaving the user "deaf" for several minutes afterward. Hand dryers are also an issue for persons with autism or other heightened sensory sensitivity. Because of this, hand dryers are especially not recommended in companion care and changing table restrooms. It was observed in our research that the "green" advantage of hand dryers vs. paper towels tended to focus on the positive attributes of the sponsoring industry.

A current trend is the development of a variety of hand dryer styles that are attempting to address customer concerns with lower sound levels, more open hand access, easier maintenance, and other features. Clearly, this is a technology in transition. Whether hand dryers are at a stage of development that meets customer and maintenance needs is something the restroom and amenities team will have to evaluate.

Locating sink stations throughout the airport would hopefully encourage more frequent handwashing and thus, from a hygiene viewpoint, would seem beneficial. However, the reality of overflowing sinks, wadded up paper towels, and so forth would increase maintenance needs as well as look unsightly. The placement of attractive, high-capacity, commercial-grade, hand-sanitizing stations can mitigate those issues as well as add to the perception of a clean terminal.

If trash is collected in containers hidden from view, allow space for access to empty the trash. While it may seem desirable to empty trash from the pipe chase, cleaners do not favor this since it requires entry into an additional space, which reduces their efficiency.

Baby Diaper Changing

A baby diaper changing table should be in a private corner of the restroom, out of the primary circulation path if possible. Locating a changing table in the entry area is to be avoided so the person changing the infant is not on display and to reduce unpleasant odors for those entering the restroom. Also, a changing table should not be in an accessible stall as the use of the changing table precludes use of the toilet and vice versa. In fact, building codes typically no longer allow this arrangement. It is recommended that a changing station have its own sink or be within easy reach of one (see Figure 2-3).

Toilet Stall

With an in-swinging door at the toilet stall, travelers must maneuver around their carry-on, the open door, and the toilet to enter the stall and then try to close the door. While there is a risk of out-swinging stall doors opening onto passersby, the benefits of these doors outweigh this potential hazard by providing a clear space for individuals to maneuver themselves and their

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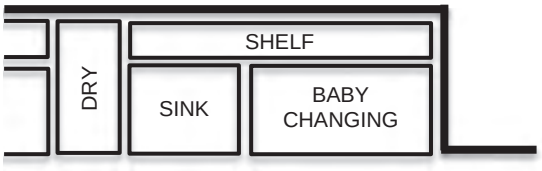


Figure 2-3. Baby diaper changing node.

belongings with the door in any position. Also, the open position of out-swinging stall doors, which can be a few inches out from a stall, can allow travelers to readily see if a stall is occupied.

Numerous new products are coming on the market that use a red or green light to indicate whether a stall is occupied. These are problematic because they use the colors red and green, which prohibit those who are color blind from deciphering the indicator. However, these lights typically have a communication interface so that airports can potentially connect them to their facilities management system and monitor restroom usage. Such practices eliminate the need for "occupied/vacant" indicators (which also have the red/green issue) and reduce violations of the ANSI A117.1 rule that doors must be operable with one hand.

With the growth of our aging population, it is recommended that every typical stall is set up as an ambulatory stall (see Figure 2-4). The current International Building Code requires that in any restroom with six or more toilets/urinals (both prototypes discussed herein fall under this requirement), one shall be an ambulatory stall in addition to the required wheelchair-accessible compartment. An ambulatory stall has grab bars on both sides to aid those with limited mobility such as older adults, someone with a broken leg, stroke survivors, etc. It is also recommended that a vertical grab bar, like that required in an accessible stall, be provided on both sides to help people pull themselves up or ease themselves down.

ANSI A117.1 requires an ambulatory stall to have a clear width of 3 ft and a clear length of 5 ft. With an extra foot in length for a carry-on, this is a perfect size for airport stalls. In addition, this is the same depth as the accessible stall (see Figure 2-5), which simplifies restroom layouts.

Accessible Stall

The current International Building Code requires every public restroom to have a wheelchair-accessible compartment. This compartment requires a space large enough for a person in a wheelchair to turn completely around, which is typically a 60-inch-diameter circle. Note that the next version of ANSI A117.1 will increase the diameter to 67 inches. This will provide more space for powered wheelchairs. The code has allowances for overlap with fixtures (but usually not doors), and some jurisdictions offer a modification if there is an open gap below the stall partition versus a partition to the floor.

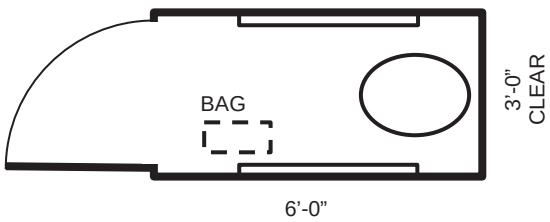


Figure 2-4. Typical (ambulatory) stall layout.

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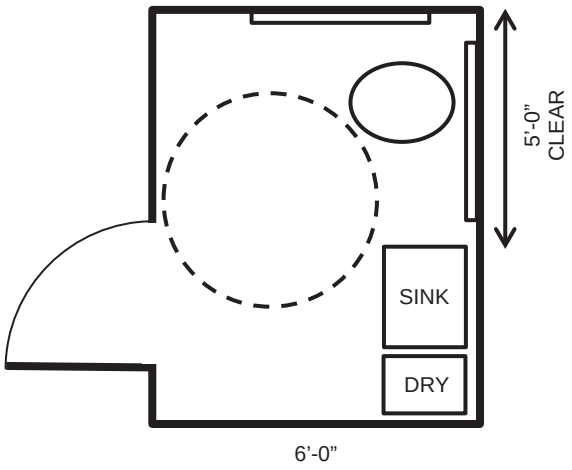


Figure 2-5. Wheelchair-accessible stall layout.

The prototypes show a sink node within the wheelchair-accessible stall. In the prevalent arrangement, where people must leave an accessible stall to wash and dry their hands, wheelchair users must touch their wheels and the door hardware on the way. This creates a potentially humiliating experience that is unhygienic for the user and future occupants of the stall, especially for those with a leg ostomy bag.

Urinal

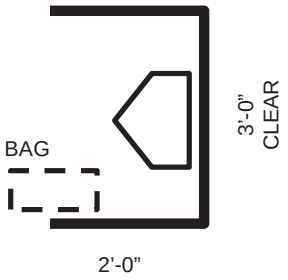


Figure 2-6. Typical urinal layout.

Security and privacy are primary considerations for urinals. It is recommended to provide a space 3-ft wide at the urinal to match the width of the recommended toilet stall. This provides a common fixture module for laying out restrooms. More importantly, it provides floor space for travelers' carry-ons, so they can be put to the side rather than behind a person (see Figure 2-6).

For wall-hung urinal screens, the maximum depth recommended by partition manufacturers is 18 inches. However, a minimum 24-inch depth is recommended to keep carry-ons within a user's peripheral view. Deeper screens are also more private. End support pilasters that match the stall pilasters can be provided to support this depth. The height of a urinal screen should match the height of stall side panels, both for aesthetic continuity and for additional privacy.

Grooming

There are two fixtures for grooming that should be considered: (1) a small shelf with a mirror above it that provides a remote place to put on make-up, fix up hair, and so forth and (2) a full-height mirror for checking that a traveler's clothes are in place. The latter should be placed near the exit, but should not be in heavily trafficked areas. Also, placement of the mirror should avoid creating unintended sightlines into the restroom from outside the restroom. Some airports are placing a sign above the exit from the restroom that reads, "EXIT," because some people with dementia can be confused by the full-height mirror reflection, thinking it's a hallway. The grooming node can be placed anywhere, although ideally in an alcove. The bottom of the full-height mirror should not be placed below the wall base to prevent water from floor cleaning leaching up behind the mirror and damaging the silvering.

Chase

For a plumbing chase to be usable, it needs to be at least wide enough for a person to comfortably move around and work. It is recommended that the chase (and access door)

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be wide enough to wheel in a plumber's tool cart as the TSA does not permit tools to be left unsupervised in public areas. With narrow plumbing chases, additional maintenance staff is required to watch the tools while another works within the chase. Larger chases should be considered when piping is located on both sides of the pipe chase. It is important to place piping and conduits in the chase as close to walls as possible. A 1-ft pipe zone is recommended. Access to pipe chases should be provided from public corridors or concourses so that maintenance staff members of the opposite gender of the restroom do not have to close the restroom for most toilet, urinal, and sink maintenance.

Janitor/Storage

A janitor's closet should provide space for a mop sink, hose bibb, and related tools. A hose bibb can be used both for the mop sink and for a hose to run into the restrooms for periodic deep cleaning efforts. This is also a good location for soap reservoirs if this type of soap-dispensing system is utilized.

Some airports provide a central storage space for paper stock and cleaning supplies. In larger airports, the travel distance can make this kind of storage inefficient, so verify with cleaning managers preferred locations as well as the recommended size and quantity of shelving required for storage if it is located within a restroom block.

If janitorial and storage functions are co-located, verify also with the plumbing and cleaning managers whether there is a need to provide a segregated and locked space for each trade. This is typically done with chain-link fencing. The door to the space may also need to be wider to accommodate the wheeled "whales" used for collecting trash.

2.5.2 External Restroom Components

There are several components that are recommended for the area outside restrooms:



- **Drinking fountain.** Building codes typically require a drinking fountain, and it's logical to locate these with other plumbing (and this colocation is often required). Each fountain location should have a high and low fountain to accommodate those in wheelchairs and those with different heights. It is common to provide a water-bottle filling station at the drinking fountains since liquids cannot be brought through security. A single water-bottle filling station can be used on a high-low fountain if the station is mounted on the low, wheelchair-accessible fountain. If the high-low fountains are separate units, each will require a water-bottle filler.
- **Automated external defibrillator (AED)/fire extinguisher/courtesy phone.** These components are often scattered throughout the airport, and it can be difficult for travelers to remember their location when they are in an emergency. Grouping them at each restroom block makes it particularly easy for airport employees, who are most likely to respond first, to remember where to go for help.
- **Waiting area.** Many travelers fly with a companion or family. Providing seating by the restroom for those finished early or waiting is a welcome accommodation, especially for aging adults or those traveling with children. It can be as simple as a bench, a small sitting group, and/or a leaning rail. A variety of seating types is ideal for people with reduced mobility, e.g., those who may need a higher seat or armrests to sit or stand. Regardless of the seating provided, space for a wheelchair should also be provided.
- **Power/charging station.** It is debatable whether to provide power and USB. This is becoming commonplace in most new airport seating areas. The drawback is that it invites people to camp out with their laptops, thus negating the waiting area.
- **Flight information display system (FIDS).** A restroom block is a great location for an adjacent FIDS. It can be checked by travelers while waiting for companions. Overhead

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directional signage is also important outside restrooms since travelers may lose orientation and need a reminder as they exit of which direction leads to baggage claim or their gate. This is especially the case where the restroom has exits to two concourses.

- **Art display.** Space permitting, a spot where either permanent or rotating artwork from the airport's art program is displayed is another great attraction for those waiting for companions at restrooms.
- **Vending.** Those waiting for their companions are a captive audience, so there is an opportunity to locate vending machines at the restroom block if space permits.

2.5.3 Restroom "Room" Prototype

The room prototype layout separates the toilet and washing functions into "rooms" (see Figure 2-7). It requires a little more space than the galley prototype (Section 2.5.4) but tends to feel less congested because occupants are not traversing the main circulation path when moving between the toilets and sinks. For this reason, this prototype is preferred over the galley prototype whenever feasible. This layout expands easily by adding opposing pairs of stalls and sink pairs.

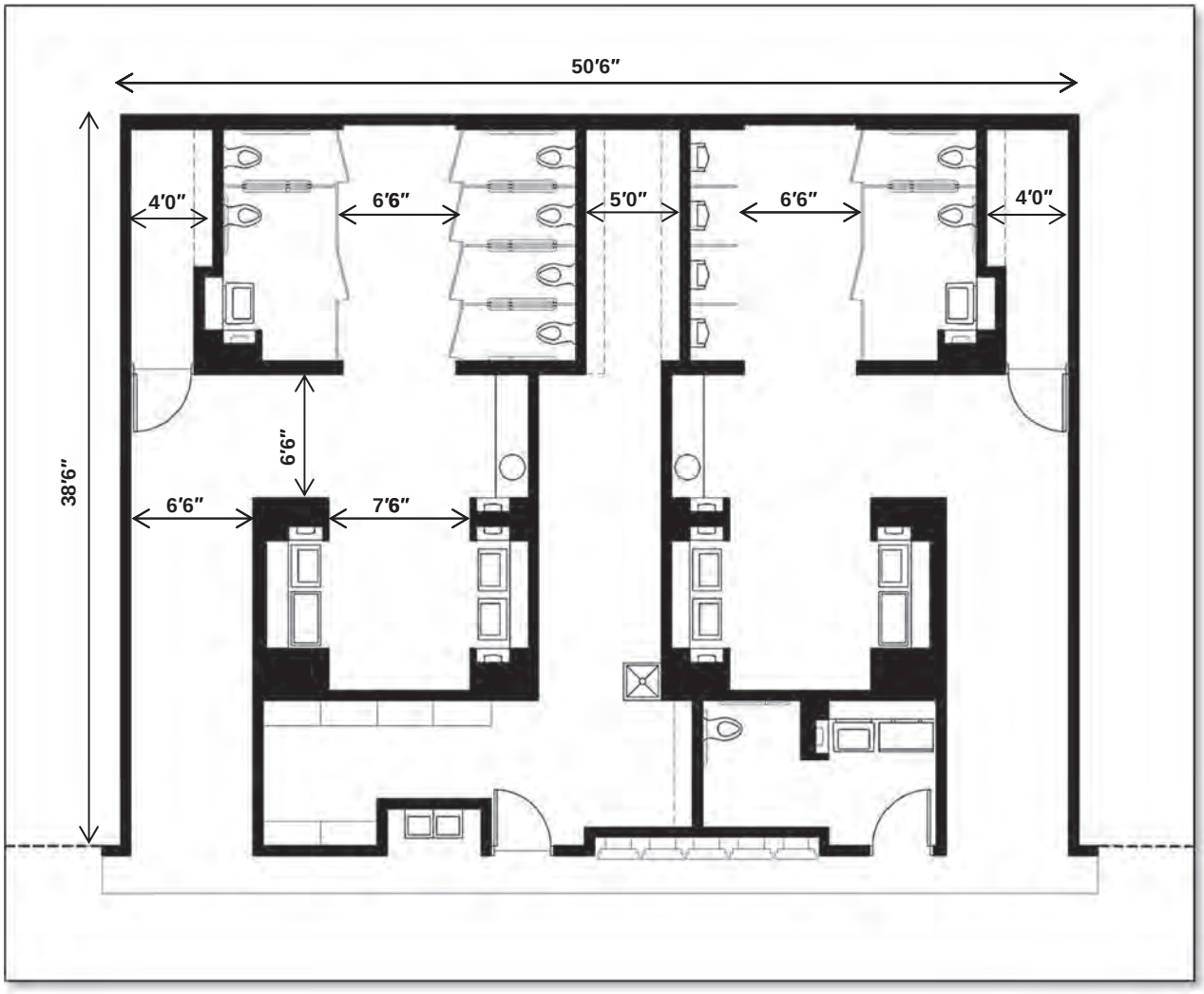


Figure 2-7. Room prototype layout—1,950 square ft (approximately 160 square ft per fixture).

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Figure 2-8. Room prototype—view toward entry.

Another feature of this layout is the potential for a large "grooming" node that greets travelers as they turn into the restroom proper from the entry (see Figures 2-8 through 2-10). This buffer adds a welcoming, hospitable feel to the space and works well for people who just need to quickly wash their hands. Other options for this space are an art wall or another sink grouping. A baby diaper changing table is discouraged here, as noted previously.

Variations of the room prototype are used at Hartsfield–Jackson Atlanta International Airport (ATL), Dallas/Fort Worth International Airport (DFW), Long Beach Airport (LGB), and Jackson Hole Airport (JAC) as discussed in case studies for this research (see Appendix C.1).

2.5.4 Restroom "Galley" Prototype

The galley layout is a more efficient plan, spatially (see Figures 2-11 through 2-13). The toilets and urinals stretch along one side of the circulation space, and sinks line the other. A set of male and female restrooms can share a single plumbing chase. If a restroom needs to expand to more than two sink groups and a baby diaper changing table group, the width of circulation space should be increased to 7 ft and 6 inches to accommodate the extra traffic. The accessible stall and changing table remain on the shorter, quieter end of the restroom in expanded versions.



Figure 2-9. Room prototype—view toward grooming node.

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Figure 2-10. Room prototype—view from sink node to grooming and toilet nodes with entry node to the right.

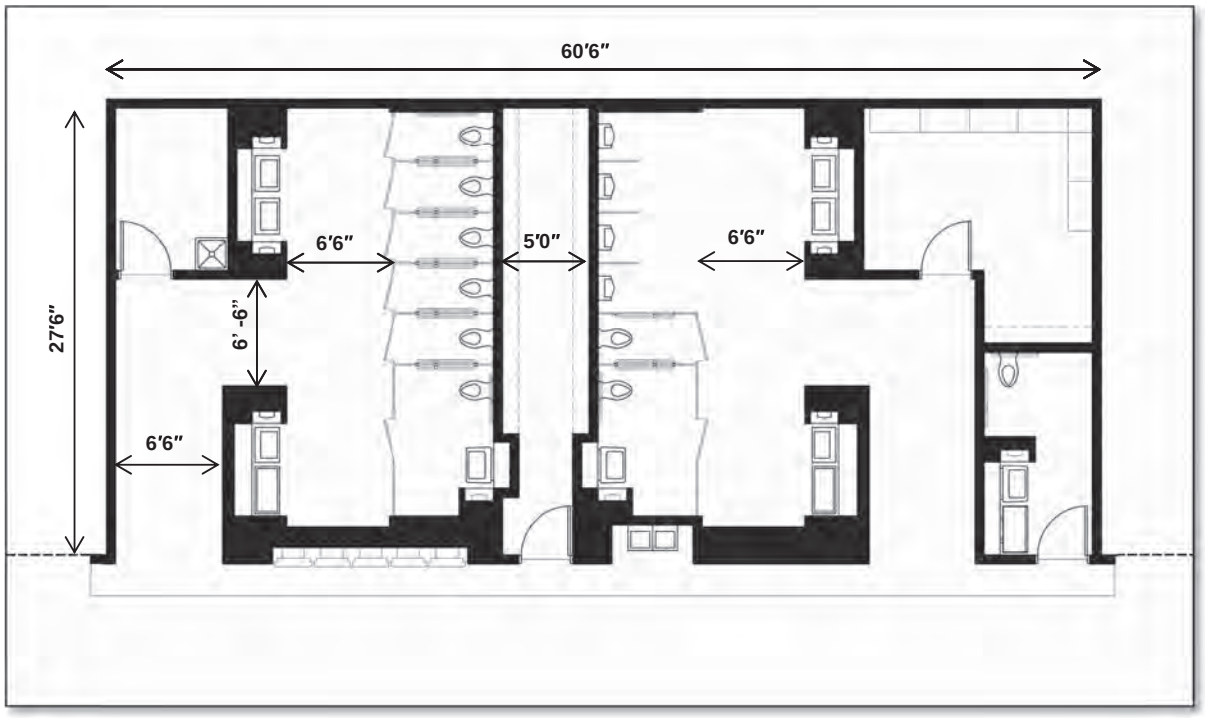


Figure 2-11. Galley prototype layout—1,650 square ft (approximately 138 square ft per fixture).



Figure 2-12. Galley prototype—view toward entry node.

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Figure 2-13. Galley prototype—view toward toilet nodes with the entry node to the left.

One drawback of this type of layout is that the entry point directly approaches the stalls and urinals. This condition can be remedied by eliminating two or three stalls and creating a blank wall for art or creating a grooming alcove like the room prototype. This enhancement requires more floor area, however (using only slightly more floor area than the preferred room prototype).

Variations of the galley prototype are used at LAX, Minneapolis–Saint Paul International Airport (MSP), Sacramento International Airport (SMF), John Wayne Airport (SNA), Blue Grass Airport (LEX), and Duluth International Airport (DLH) as discussed in case studies for this research (see Appendix C.1).

2.5.5 Prototype Layout Expansion

The galley and room prototypes in Figures 2-14 and 2-15, respectively, are shown with the preferred minimum restroom size and gender mix of six fixtures per gender stated in Section 2.4. Figures 2-16 and 2-17 show how both prototypes can expand for larger fixture counts. The dashed-line stalls and sinks show how the restroom can be expanded while

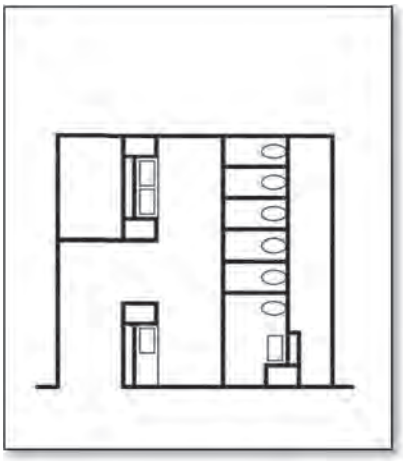


Figure 2-14. Basic galley prototype.

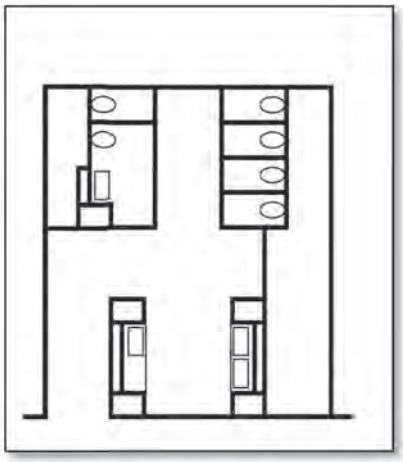


Figure 2-15. Basic room prototype.

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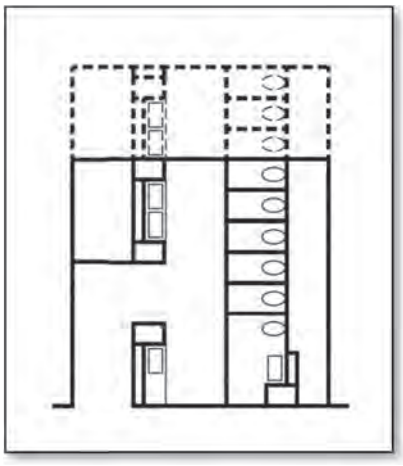


Figure 2-16. Expanded galley prototype.

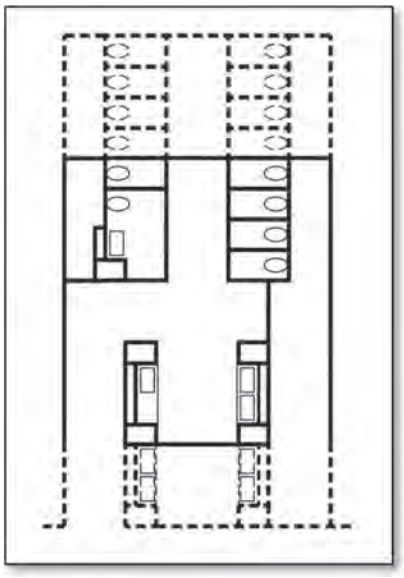


Figure 2-17. Expanded room prototype.

maintaining the fixture/sink balance. The recommended mix is two stalls or urinals for each sink station.

Many airports have a column bay spacing of approximately 30-by-30 ft, so a typical concourse might be three bays wide. This limits expansion outward. However, a variation of the prototype might expand the restroom along the concourse length. Alternatively, a larger restroom block might have a pair of female and male restrooms grouped together. This arrangement facilitates cleaning by closing only half a gender's restroom in each location. The drawback is that if both gender sets are open, it may be unclear whether one is full. This can lead to customer frustration. A technology that monitors and indicates stall occupancy with an electronic sign at the restroom entry would alleviate this issue.

2.5.6 Restroom Components

The placement of the restroom components (described in more detail in Chapter 4) needs careful consideration to provide travelers with what they need in locations that are intuitive and efficient. These components include

- Spatial nodes from Section 2.5.1
- External restroom components from Section 2.5.2
- Internal restroom components
- Signage
- Accessories
- HVAC
- Lighting
- Power
- Technology

Figures 2-18 through 2-26 show the recommended placement for each of these nodes and components as well as providing brief commentary. The diagrams are based on the room prototype. However, the placement applies similarly to the galley prototype.

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Figure 2-18. Spatial nodes arrangement.

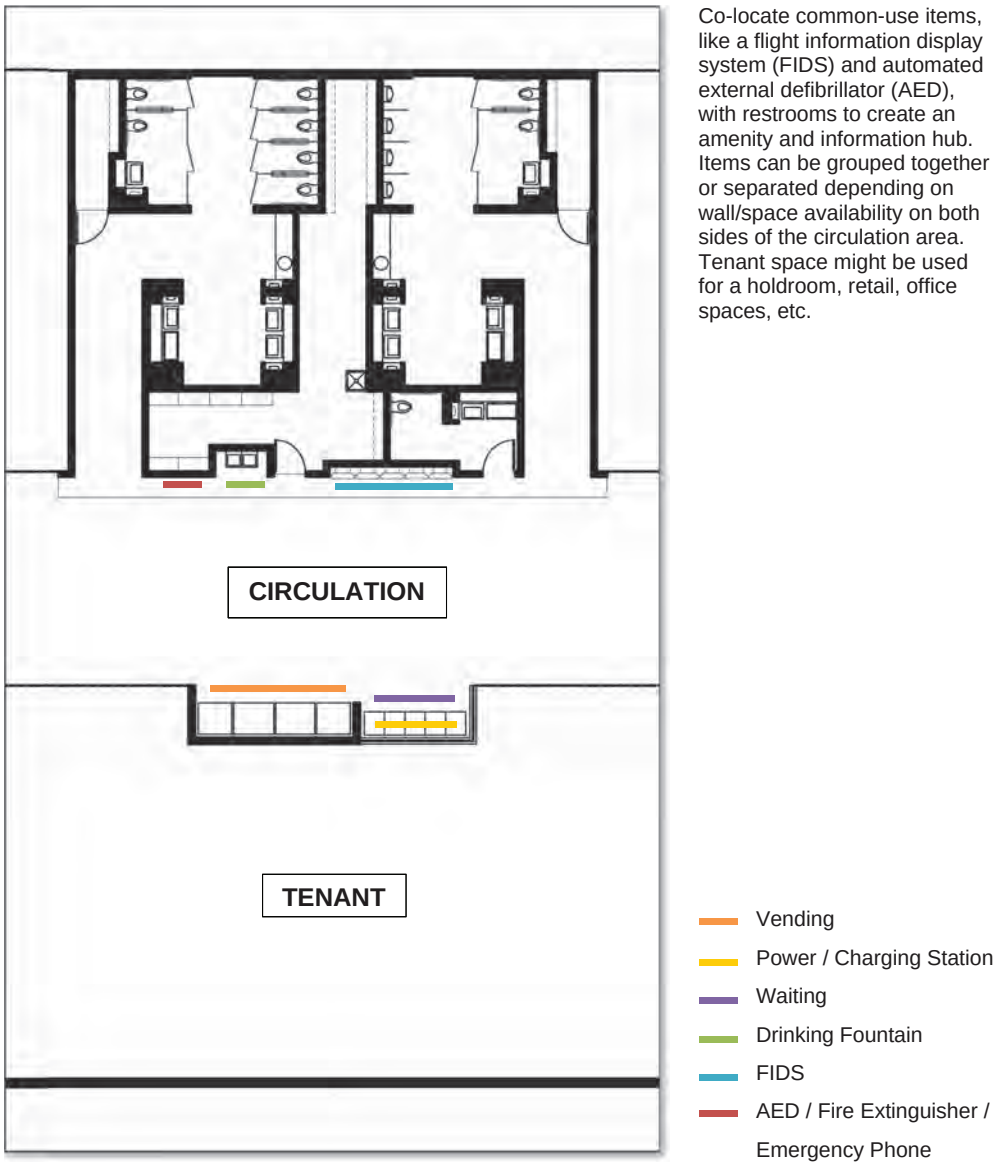


Figure 2-19. Locations of external restroom components.

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Figure 2-20. Locations of internal components.



Figure 2-21. Signage locations.

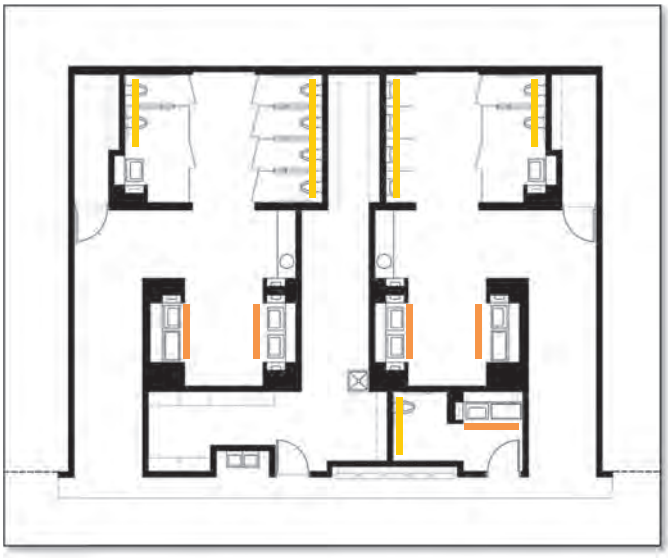
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Provide recessed accessories wherever possible to reduce surfaces requiring cleaning, to minimize potential corners to bump into, and to streamline restroom appearance.

- Mirror
- Toilet Paper / Disposal
- Seat Paper Dispenser
- Paper Towels / Trash
- Biohazard Disposal
- Sanitary Products Vendor
- Utility Shelf / Rack

Figure 2-22. Locations of accessories.



Supplying warm air from vents under sinks helps to dry wet floors quickly. Locating exhaust vents above toilets will draw out air supplied from low vents and quickly pull out odors.

Note: Providing supply air under sinks requires a thicker wall to accommodate insulated ductwork.

- Air Supply (Low)
- Exhaust (High)

Figure 2-23. HVAC locations.

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Wall lights provide even ambient light and minimize fixture clutter on ceilings. Side lighting at mirrors illuminates faces without shadows. A downlight in each stall eliminates shadows from light panels behind occupants. Lighting at restroom exterior should complement lighting concept of the surrounding public space.

- Lighted Panels
- Recessed Downlight
- Mirror Light
- Utility Light
- Motion Detector

Figure 2-24. Lighting locations.



A receptacle at each sink allows use of a shaver or hair dryer. Provide central receptacle for cleaning equipment. Coordinate power requirements in chase and storage spaces with equipment and airport needs.

- Duplex Receptacle

Figure 2-25. Power locations.

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A FIDS helps accommodate the FAA requirement for visual paging and allows a waiting companion to check flight status and potentially other airport information. A thermal people-counter tracks the number of people entering and exiting the restroom. Data can be sent to cleaners to alert when use-threshold has been reached. See Section 4.3.8 for information on the RMS.

Figure 2-26. Technology locations.

2.6 Amenity Prototypes and Considerations

The essential/regulated amenity spaces are either regulated by the FAA (SARAs and areas for lactation) or deemed essential (spaces for nursing mothers and companion care and changing table restrooms) based on this research. Prototypes for these spaces are provided in the sections that follow. See Chapter 4 for detailed recommendations on related components.

The waiting-related and layover-related amenity space prototypes demonstrate considerations for each space. In addition, the case studies in Appendix C.2 provide a good starting point for considering all of these spaces via photos and other airport-specific information.

2.6.1 Regulated/Essential Amenity Spaces

Service Animal Relief Area Prototype

This FAA-required space (on the secure side) is going through an evolution as most airports are providing their best interpretation of current requirements (based on FAA Advisory Circular 150/5360-14A—Access to Airports by Individuals with Disabilities). Solutions range from 2-ft by 4-ft portable boxes (not recommended) to large spaces for numerous animals. SARAs are ideally located adjacent to restroom blocks to access plumbing. Also, they are located there because if their pet has to use the facilities, the travelers likely do too.

As the name suggests, this space is designed for service animals, primarily dogs, as they are most the most common service animals and are specifically referenced in the FAA regulations. As shown in Figure 2-27, the space should have an accessible, artificial turf area, up to approximately 12 ft by 12 ft, with an automatic flushing system and floor drain below. There are two predominant training philosophies for guide dogs. One has the dog owner located in the center of the turf area while their dog walks in a circle around them on a 6-ft leash (12-ft by 12-ft turf area). The second, newer variant has the owner located at the edge of the hard floor/turf transition (to avoid stepping in a leave-behind they may not be able to see) while their dog walks in a line back and forth in front of them (12-ft by 6-ft turf area). The restroom and amenity team should confirm with their local service animal organizations (as required by the FAA Advisory Circular) which configuration is preferred.

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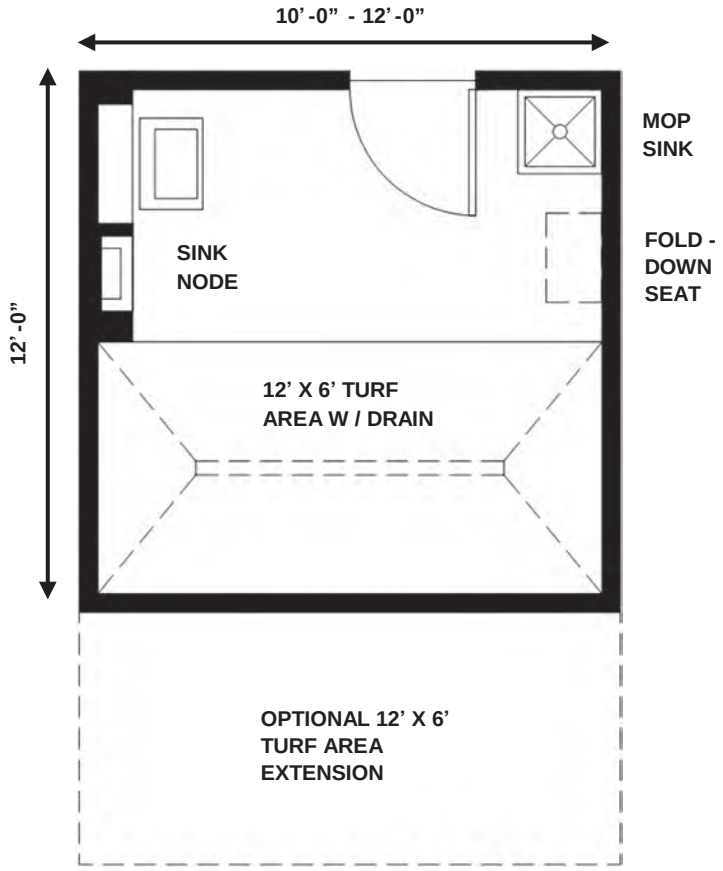


Figure 2-27. SARA prototype.

A sink node is provided and a mop sink with a fold-down seat to rinse soiled paws is recommended. The mop sink should have a sprayer head to allow for cleaning appropriate areas. A nearby storage room should be provided for attic stock of additional turf rolls. The space is typically enclosed to manage odors.

Lactation/Nursing Mothers Prototype

Spaces for lactation and nursing mothers have two distinct functions and are treated separately in this guidebook. However, they are ideally located together as a suite, as shown in Figure 2-28. A space for lactation is dedicated to a mother so she can express breast milk, preferably in private. A space for nursing mothers has a lounge area with comfortable seating for nursing babies. These tend to be more communal and have multiple chairs.

Spaces for lactation and nursing mothers are in high demand. Many airports use portable pods. These come in single-user and accessible models, which also accommodate multiple traveling companions. Note, however, that an inaccessible pod that is not placed in a cluster with an accessible model violates ADA requirements. As pods are considered a temporary solution, and their features are readily available on websites, the focus of this write-up is on built spaces.

Lactation rooms typically feature a counter, a chair on wheels, and a baby diaper changing node. Nursing mothers’ spaces also have a baby diaper changing node. They typically have one or more chairs, usually with side tables. Sometimes there are small play areas for older children traveling too.

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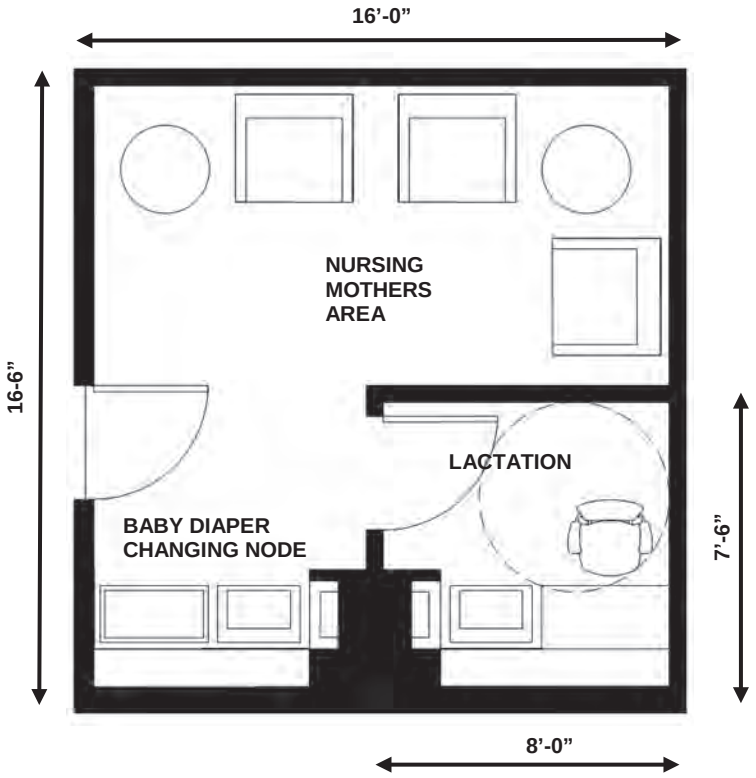


Figure 2-28. Prototype of space for lactation/nursing mothers.

Companion Care Restroom Prototype

The companion care restroom is similar in layout to the wheelchair-accessible stall except that a divider is placed between the toilet and sink. This provides privacy for the person using the toilet if traveling with a companion, children, etc. (See "room" and "galley" prototypes in Sections 2.5.3 and 2.5.4). As shown in Figure 2-29, this space has an accessible toilet node and baby diaper changing node. Some airports provide a fold-down bench or toddler seat. The biggest challenge with these spaces is the naming: unisex, family room, assisted care, washroom toilette, and most commonly, companion care. Some airports provide a separate baby diaper changing space next to the companion care restroom (see Chapter 3).

Changing Table Restroom Prototype

This is a new space that several airports are now providing. England has a strong organization, called "Changing Spaces," that provides resources for developing these areas. As shown in Figure 2-30, these spaces are essentially companion care restrooms with the addition of an adult-sized, powered change table; a hand-held shower; and a ceiling-mounted multidirectional hoist for assisting individuals from a wheelchair onto the table or a toilet. Some airports also provide a fold-down bench.

2.6.2 Waiting-Related Amenity Spaces

Companion Waiting Area

For the traveler waiting for a companion, doing so in a small alcove off of the busy concourse where it might even be possible to sit down is a welcome relief, especially if the traveler is

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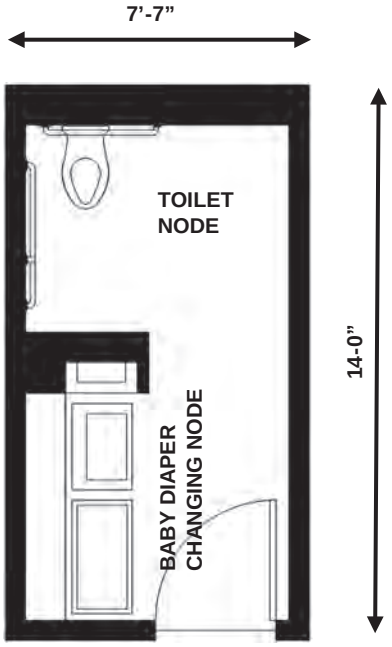


Figure 2-29. Companion care restroom prototype.

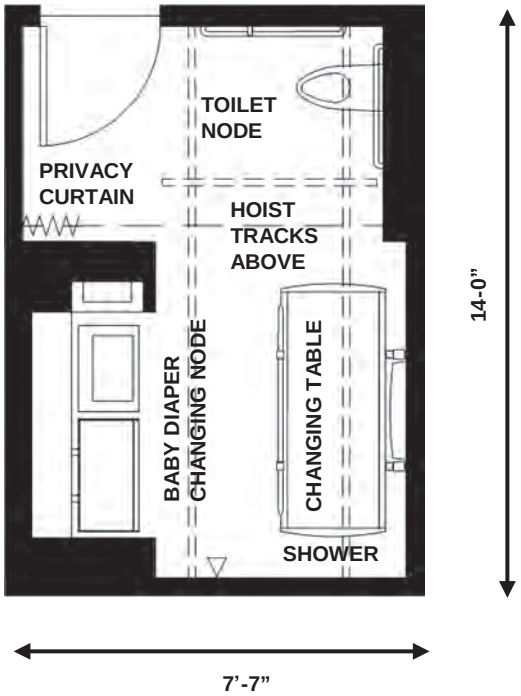


Figure 2-30. Changing table restroom prototype.

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accompanied by one or more children or overburdened with bags. Likewise, a companion waiting in a wheelchair appreciates an alcove to sit in out of the stream of traffic. See Section 2.5.2 for recommendations on appointing this space.

Children's Play Area

Space permitting, an airport may provide small activity areas and/or large areas with elaborate play structures for kids to release their energy while traveling. There also may be an area with tables or carrels for "mental" activities like board or computer games. Seating for adults is recommended (this may be accommodated by adjacent holdroom seating) with charging outlets. TVs are generally not provided here to avoid distractions and discourage lingering. Airports may offer an adjacent TV lounge with options for passengers to use their laptops or tablets. Usually, there is a partial-height barrier surrounding the play area to contain activities. Similar attractions include aquariums, game arcades, and so forth.

Sensory Room

The key feature of a sensory room is the provision of a calming environment. In addition to features noted in the research team's case study example at Gatwick Airport (LGW) (see Appendix C.2), the following design elements are provided in the new sensory room at Pittsburgh International Airport (PIT). PIT's sensory room opened in July 2019 and provides a great example of this new and evolving amenity space. PIT and its design team reached out to the community for input on how best to create such a space. The following suggestions were implemented:

- Located in a transition space in the concourse gate area
- Options for spaces that serve both children and adults
- A variety of comfortable seating
- Soundproof
- Neutral smell
- Neutral and/or cool colors
- Adjustable lighting
- Calming activities and equipment such as fidget boards, tactile objects, and bubble machines
- Cleanliness without the use of toxic cleaners
- Visual connection to flight information
- Outlets to charge devices
- Ability to see into room before entering

While sensory rooms have typically been focused on children, PIT also included a separate soundproof space for adults. An additional unique feature is a realistic airplane cabin experience, designed to acclimate children or adults before they head to the gate. A sensory-friendly bathroom with an adult changing table is located right next door. It does not have automated hand dryers, paper towel dispensers, flushing toilets, or faucets as these features are too noisy for individuals with sensory processing issues.

Meditation/Quiet Area

Airports are busy and loud places. For people who are travel-weary or are sensitive to noise, lights, commotion, and so forth, it can be difficult to find a calming place to wait for their flights. Quiet rooms or areas are intended for reading, resting, and are sometimes dedicated to meditation. They typically feature comfortable seating, no TVs or aromas, dim lighting, and artwork and/or a view beyond the airfield, as well as being kept at a comfortable temperature. Large spaces might be divided into smaller nodes with translucent screens. A variety of comfortable seating, including recliners, couches, or benches, can accommodate people with various

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mobilities and groupings. Some cultures separate men, women, and families, following local protocol. Some airports also provide vending machines that feature sleep kits (water, mask, and so forth).

Yoga Area

Airport-provided yoga spaces, if not provided by a fitness-oriented tenant, are typically a semi-private area for individuals and/or instructed sessions. The space should be dimly lit and temperature controlled. Mats, bands, and balls are sometimes provided, particularly if instructors are available.

Worship

Most of the airports interviewed for this guide stated that they provide some form of worship space, from general space to well-appointed areas. Airports use a variety of names for worship spaces, including chapel, reflection room, meditation room, and prayer room.

These spaces are typically multi-faith with no religious symbols portrayed in the space. Most have a seating area with a pulpit and a separate, open space, typically with prayer rugs and shelves for shoes. Sometimes kneelers are provided. Some spaces offer separate male and female entrances to accommodate Muslim practices. A few offer ablution facilities within the complex (more on this amenity can be found in the next subsection). Most are available for travelers and staff and are open 24/7. Some airports provide wedding services as well.

Larger facilities may have multiple chaplains available and/or volunteers with private offices that offer counseling services. Most have a small reception area with shelves for religious texts and pamphlets. Storage space(s) are typically needed for worship spaces.

Ablution

There are two common configurations of space for ablution. One provides a floor-mounted ablution fixture in a private space. The other has one or more rows of individual seats lining a continuous "trough" that collects the water. A wall-mounted faucet is located facing each seat above the trough. See Section 2.6.3 for additional information.

Smoking

Where smoking is permitted indoors, airports typically provide an enclosed smoking room on each concourse. Smoking rooms are often enclosed with glass walls and provide charging stations. They sometimes provide cocktails, gambling, and a cigar room.

2.6.3 Multicultural Considerations

To accommodate travelers of various cultures and religions, especially Muslim and Asian populations, there are many types of amenities that airports can consider.

For instance, Wudu is an Islamic ritual for cleansing (ablution) parts of the body for purification. It is important for spiritual and physical health. It includes washing the hands, mouth, nostrils, arms, head, and feet. Wudu is typically performed in preparation for formal prayers and prior to handling and reading the Qur'an. Keeping clean and pure is a key tenet of the Muslim faith.

Ablution is sometimes misunderstood as simply foot washing because this aspect of the cleansing cannot easily be done at a sink and therefore is very noticeable. Cleansing is sometimes done in a toilet. This is permitted if the water and bowl are visibly clean. However, it can be hazardous as the bowl is curved and slippery. Furthermore, if the person is standing on the toilet seat, it may break, causing injury. There are commercial porcelain ablution fixtures readily

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available in most countries except for the Americas. Ablution stations are typically located in a space separate from the toilet stalls and, ideally, adjacent to a prayer space.

While Wudu lasts up to one day, ideally, it is performed before each prayer (Salat, which occurs five times a day). The Wudu ritual is nullified by urination, defecation, flatulence, and light menstrual bleeding, and so re-purification with the more comprehensive cleansing of Ghusl, or full ablution, is required after using the toilet. Ideally, a hygienic bidet shower (similar in appearance to a kitchen sprayer) is hung at the right side of each water closet to facilitate this religious cleansing requirement. It is on the right because, in Muslim culture, the left hand is used for cleansing related to elimination. After the spray is used, toilet paper is still used for final cleaning and drying. Muslim travelers visiting Western countries will often bring a "shataff," which is a plastic spray bottle—a modern version of the "lota," the traditional copper, spouted vessel.

The bidet shower has a growing environmental appeal as well. According to *Scientific American*, the average American uses 50 pounds of toilet paper each year ("Wipe or Wash? Do Bidets Save Forest and Water Resources?" December 2009. https://www.rsph.org.uk/uploads/assets/uploaded/e6cda5d7-bc38-4479-9a517703c22b1316.pdf). Every roll is estimated to require 37 gallons of water to produce and an additional 1.6 gallons to flush down the toilet.

In place of a bidet shower, a bidet or toilet-mounted, in-seat bidet can be provided with a nozzle that either sprays from the front toward the back or from the back and straight up. In Japan, an in-seat bidet is often an integral component of toilets. As of this writing, in-seat bidets are only available for Western toilets as add-ons and are typically intended for residential use. Bidets and in-seat bidets both use about one-half gallon of water per use and require an electric receptacle. They do not appear durable enough for the intense commercial use of an airport restroom. In addition, local plumbing codes often require that toilet seats on public toilets have an open front. Washlets have a closed ring for the seat and therefore may not be code compliant for public restrooms.

Alternatively, a bidet could be paired in a larger stall with a toilet, possibly even with its own grab bars (a review with local building code authority would be required). Because a bidet looks like a toilet, those unfamiliar with bidets and those bent on mischief may use the bidet as a toilet. Similar consequences may occur if a bidet is provided within its own stall (this will also reduce the fixture count).

Another restroom consideration is the squat toilet, mostly used in non-Western countries. The squat toilet, once a primitive hole in the floor, has been modernized into a porcelain trough in the floor with a water seal and flush valve. Squatting is the natural position for elimination and is commonly considered more hygienic and healthy than sitting on a Western toilet. In addition to porcelain, stainless steel and fiberglass are used for making squat toilets. Squat toilets are either level with the floor or on a raised platform (about a foot high). There is typically an integral textured, but drainable, area for each foot to prevent slipping. Some Western airports provide one stall in each men's and women's restroom with this type of toilet fixture as an accommodation. On the downside, squat toilets increase the risk of soiling clothes, and they are not accessible or easy to use for persons who are elderly, who may have trouble squatting. In countries where squat toilets are common, a single Western toilet is provided to accommodate those who are physically unable to squat, including persons using wheelchairs.

Provisions for the Wudu ritual are an important traveler and employee consideration in locations where Muslims are a significant percentage of airport customers and/or staff. Airports should consider accommodating the customs of travelers/employees from other countries by providing restroom fixtures they are used to. Installing at least one of each familiar fixture type, if a significant percentage of travelers/employees would use it, could help provide safer and cleaner restrooms and amenities. At the same time, providing restroom fixtures that may be

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unfamiliar to many Westerners may bring the risk of accidents or even misuse. Their provision will require considerable evaluation by an airport's restroom and amenities team and, importantly, input from users of these spaces.

2.6.4 Layover-Related Amenity Space Prototypes

Layover-related amenity spaces are commonly provided by tenants and so providing a prototype isn't practical for this guide. Instead, the typical features of each space are described. These spaces are commonly located at hub airports with a significant percentage of long-haul, international flights.

Sleeping

With the compact efficiency of a mobile home, spaces for sleeping usually offer a small desk, chair, and a slide-out bed (from a couch). A larger accessible room is typically provided. Pairs of rooms may have the ability to be joined with an interlocking door for families. Separate restroom and shower facilities are available at some locations as are various sundries at the check-in desk. Most of these facilities provide both overnight rates and hourly rates for those needing a quick nap between flights.

Fitness

Like most gyms, compact airport fitness areas provide equipment and workout spaces, some with scheduled sessions. Restroom, shower, and changing facilities are usually also provided. Some locations may allow travelers to shower and/or change for a small fee even if they are not using the equipment. Airport staff discounts are often available to boost usage.

Public Showers

Public showers usually provide individual shower stalls for privacy. Typically, these areas have lockers and sometimes a changing room for each shower. These most likely also feature a restroom.

Clothes Changing

Most travelers go into a stall or, if available, into a companion care restroom to change clothes after a long flight or if they have spilled on their clothing. If the restroom has not been cleaned recently, this is not pleasant. A person traveling with an ostomy bag or a person with incontinence may accidentally soil themselves, so a private place to change offers them a more dignified experience. More and more, airports, particularly hubs, are providing options for changing clothes and sometimes freshening up.

The research for this guidebook showed that such spaces are commonly found in tenant spaces, such as airline lounges, hotels, and fitness clubs. Non-members can typically pay a fee for use, or the airport may provide a voucher for personal accidents. As with public showers, larger hubs might consider providing clothes changing stalls collocated with the restrooms.

Business Center

With the advent of charging stations in holdrooms and elsewhere in airports, the need for business centers is waning. Travelers are likely to work on their laptops at the gate or in restaurants. Instead of a room with carrels, concession providers offer a comprehensive business suite with conference nodes and private rooms, snack areas, workstations, printing facilities, and so forth. Onsite hotels often offer these services as well. Use is typically fee-based depending on needs, time, and so forth. Some also provide shipping and mailing services.

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Health/Urgent Care

Because an airport is like a small city, this amenity type is becoming more common, especially at larger airports that have the number of travelers and staff to justify the need. Health/urgent care centers are typically provided by a concessionaire or a local provider. Usually, clinic services are provided, but some offer a pharmacy as well. Some are open 24 hours a day. A rare few also provide dental services.

2.6.5 Other Amenity Spaces

A few other amenity types were occasionally encountered in this research. These were typically provided by the airport but were not noticeably commonplace.

Assistance Areas

These are dedicated areas within the concourses where people can get a wheelchair, wait for a cart to the gate, get assistance through security, or get an interpreter. These areas usually have multiple types of seating for different kinds of mobility, open spaces for wheelchairs, and are set somewhat apart from the bustle of the concourse. Some are called "Special Assistance" but others are shortening the name to the more appropriate "Assistance." These services are typically free to travelers.

Childcare

These spaces provide a fee-based, short-term drop-off for children while parents/guardians take care of other child-unfriendly tasks. Like most childcare areas, these offer a combination of physical play areas, game tables, crafts, movies, and so forth. Typically, these also feature a restroom and a check-in desk.

Gaming

These spaces are newer, tenant-based destination areas with rows of gaming chairs and screens. Blocks of times can be purchased along with a variety of drinks (adult beverages included) and snacks. These can be geared toward adults as well as children. One airport noted that travelers will come a few hours before their flight just to play.

Arts

Typically, art installments are a joint venture between an airport and one or more local art institutions. Some airports have satellite installations of art museums, short film features, or exhibits of local organizations. These can be found in either the landside or the airside area.

Mega-Amenities

Some airports, especially those with a major natural or manmade attraction, have made their airport an extension of those attractions. These include large indoor and/or outdoor parks, zoos, shopping malls, amusement rides, gambling, and/or concert stages.

2.7 Restroom and Amenity Space Locations

A master plan can be developed once the number of existing restroom fixtures and locations and the number and locations of required restroom fixtures (for both current passenger loads and future projections) are determined (see Section 2.8). The master plan document is an effective communication tool for a variety of audiences from airport executives and authorities to the public. In addition to documenting the number and locations of restrooms, the master

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plan includes locations for the amenities discussed in this guidebook. The master plan is a living document that is expected to be modified over the years, as air travel is an ever-changing industry. Airport master plans are typically updated every 5 to 10 years or as needs warrant. The core of the master plan is the distribution of restrooms and amenity spaces.

2.7.1 Restroom Distribution

When a master plan is being developed, it is important to understand the zone each restroom serves. Figure 2-31 shows the exterior catchment zone that each restroom location (represented by a red star) serves. Exterior catchment zones consist of the number of gates for each location. This relates directly to the airside calculations presented in Section 2.4.2. These zones will likely change in 10 years (or more) as the fleet mix changes or when a concourse is modified. Both scenarios should be understood from the airport's master plan, and the restroom locations should be planned to ultimately serve the projected conditions.

The interior catchment zones are equally important to delineate. This is more complex because other amenities are added to the mix such as stores, food courts, and rental car desks. Interior zones occur in both the airside and landside areas as seen in Figure 2-32. In this example, the zones and corresponding restrooms are given a number. Zones served by multiple restrooms use a letter suffix to indicate this. Because some areas of airports have convoluted passenger flows, periods of onsite observation may be required to see which restrooms travelers typically access from these points.

An important consideration is the provision of restrooms at remote airport locations such as shuttle drop-offs and parking garages. Travelers and/or their non-traveling companions

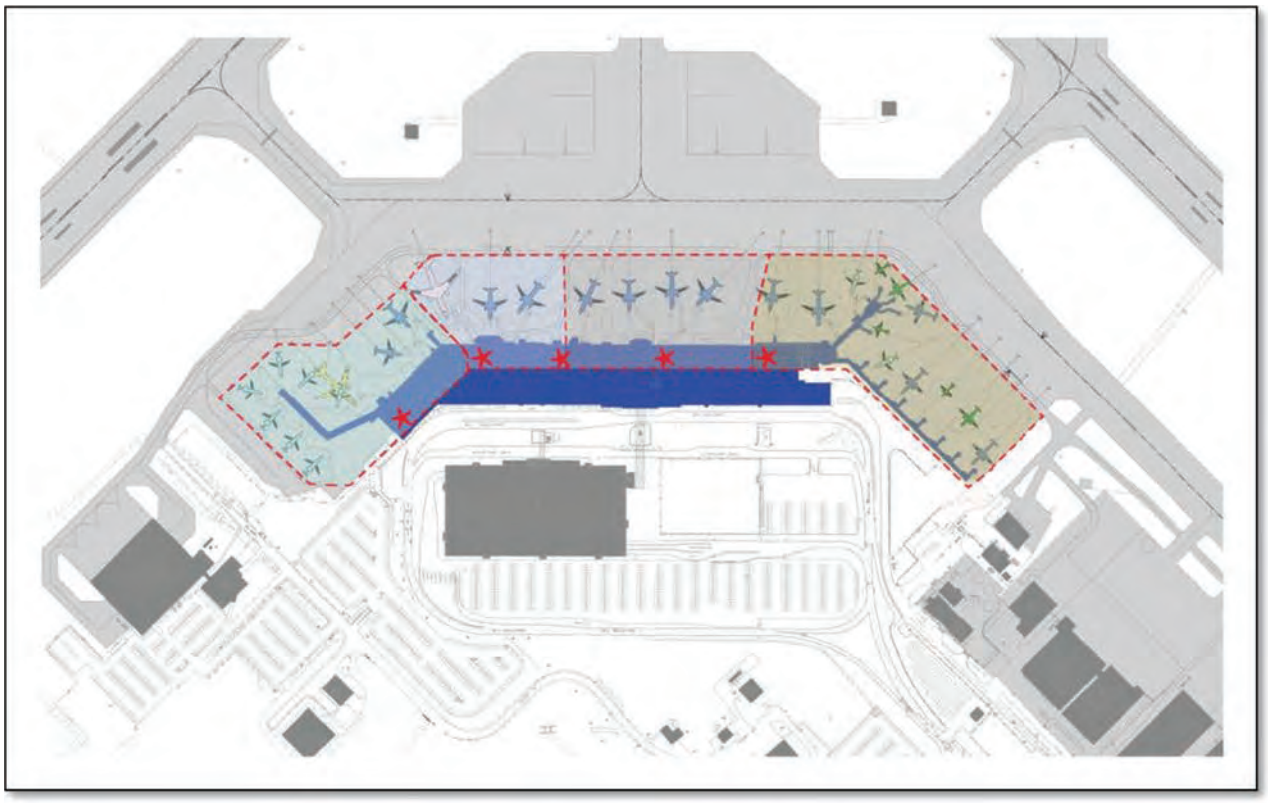


Figure 2-31. Exterior catchment zones (red stars represent restroom locations).

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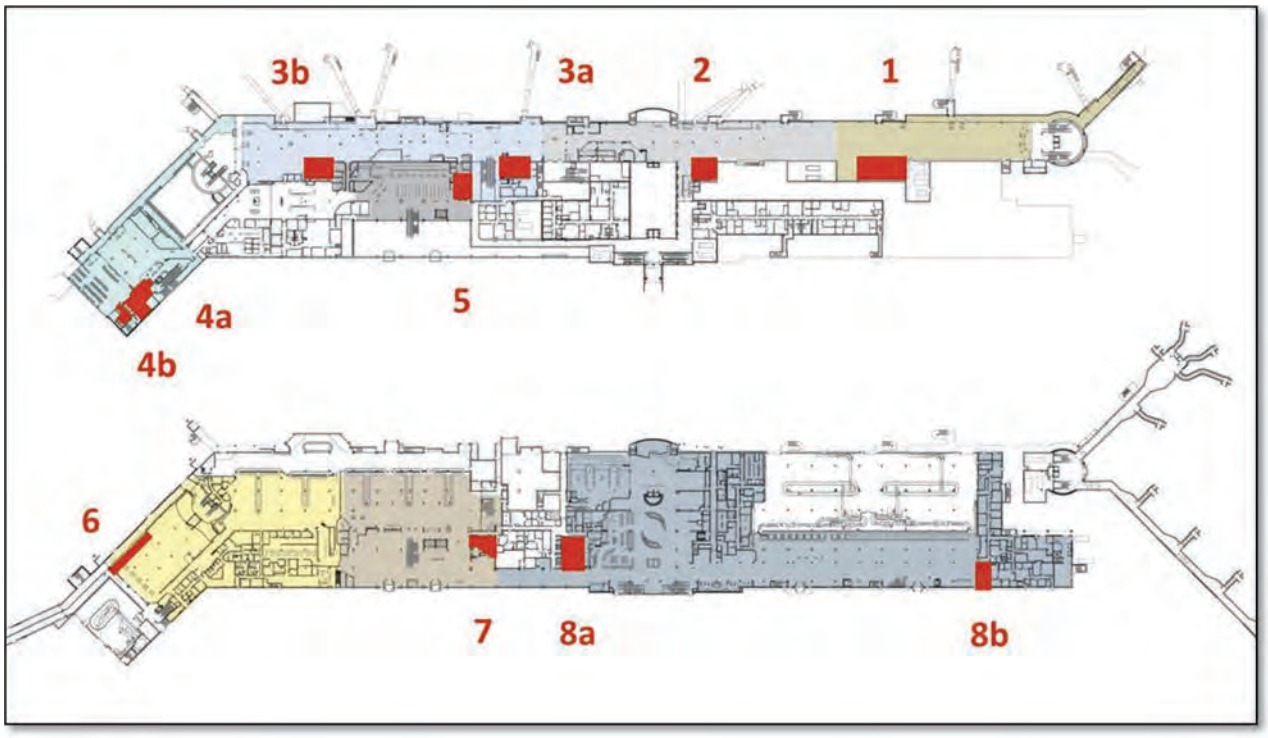


Figure 2-32. Interior catchment zones (restroom sets shown as red blocks).

with mobility or health issues may need to use a restroom frequently and may not be able to wait to get to a terminal. Travelers arriving after a long drive may need to use the restroom. It is important to analyze the travel path for these airport visitors and the convenience of remote restrooms. *ACRP Report 210: Innovative Solutions to Facilitate Accessibility for Airport Travelers with Disabilities* provides additional information for these considerations.

2.7.2 Distribution of Regulated/Essential Amenity Spaces

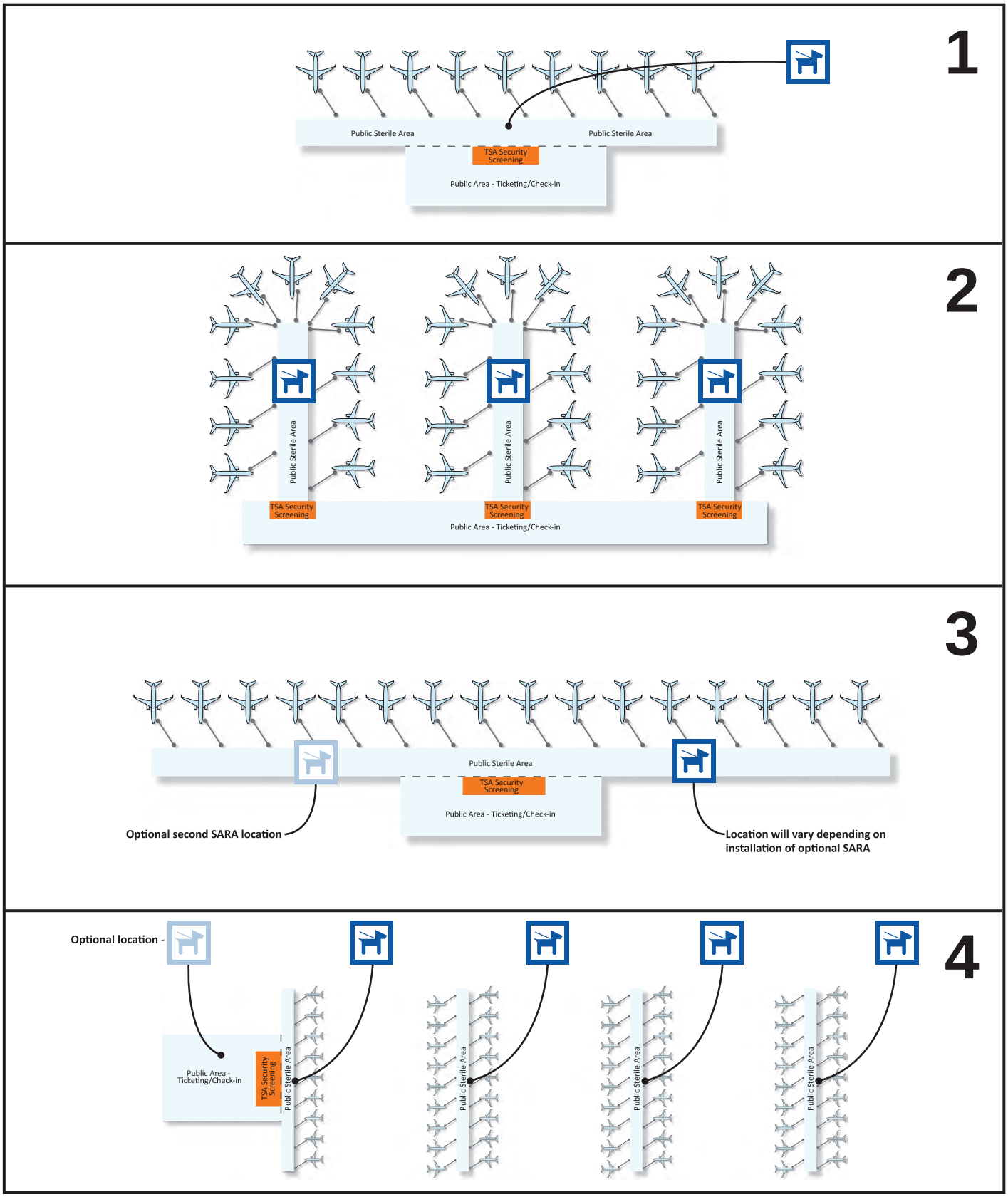
It is recommended that spaces for lactation/nursing mothers and changing table restrooms have the same distribution throughout an airport as SARAs. Figure 2-33 shows example SARA locations provided by the FAA in Advisory Circular 150/5360-14A—Access to Airports by Individuals with Disabilities. The basic rule of thumb is to provide one location on the secure side of each concourse. However, travel distances between like amenities in long concourses and large terminals should be considered as well. It is recommended that airports have a SARA, a room for lactation/nursing mothers, and a changing table restroom every 900 ft. The research for this guide suggests that this distance is a good target because it is generally manageable for those with limited mobility. Longer distances to disability-related amenities also impact the time needed and cost of assistance that airlines and their subcontractors are mandated to provide under the Air Carrier Access Act (ACAA).

While regulations vary outside North America, most major airports and airlines worldwide are members of and try to meet the standards of international organizations such as ACI and IATA. Thus, they too are developing programs for at least some of these amenities.

Service Animal Relief Area Distribution

In addition to the locations listed previously, a location on the non-secure side(s) is recommended, but not required. An outdoor SARA is typically located near an entrance. It is important

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Graphic courtesy of the FAA.

Figure 2-33. Examples of SARA locations in various airport configurations.

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to consider the travel distance for persons with limited mobility to get from any point to the outdoor SARA locations, especially in inclement weather.

Distribution of Spaces for Lactation/Nursing Mothers

Portable units do not provide adequate discretion, space, or accommodations for these two functions, but they are a relatively inexpensive first step. Furthermore, the use of these spaces can be tracked, so locations throughout the airport can be tested to assess demand. While spaces for lactation/nursing mothers should never be located within a restroom, proximity to a restroom is desirable for the built-in-place spaces to minimize plumbing infrastructure and also to accommodate restroom needs in one location.

Companion Care Restroom Distribution

These spaces are not required, but one or more is typically located adjacent to the primary restroom set. Companion care restrooms were originally intended to provide proper wheelchair accessibility. They were also meant to make it easier if a traveler had children of opposite or multiple genders or if the traveler was traveling with a companion of the opposite gender who required assistance. The movement toward all-gender restrooms has also increased the use of this space. Numerous airports are looking at combining multiple single-user restrooms for more flexibility (see Chapter 3).

Changing Table Restroom Distribution

These spaces are essentially a companion care restroom with the addition of an adult-sized, diaper changing table and related accessories. For efficiency, these spaces can also serve as companion care restrooms. However, the demand for the latter can create a situation similar to that of a single accessible stall in a restroom. Someone who needs the space will likely have to wait. These spaces should be located as part of a restroom set.

2.7.3 Distribution of Waiting-Related Amenity Spaces

As these spaces primarily serve those with a little time to spare between flights or before boarding, they are typically located on the concourse(s), not too far from the holdrooms.

Children's Play Area

Play areas are typically located near holdrooms or food courts for better monitoring by parents/companions. Locating play areas near restrooms is not recommended since companions can have a difficult time monitoring their children from there. Most large airports have several play areas placed throughout the concourse areas to provide minimum paths of travel from the visitor's gate.

Sensory Room

Sensory rooms should be located off a relatively calm circulation path and ideally not by areas with large groups of people and noise such as food courts, checkpoints, and restrooms. As these are new (and relatively costly) amenity spaces, starting with a prototype location to test demand and functionality is recommended.

Meditation/Quiet Area

This space is typically located among the holdrooms (although locating it near a large wall of exterior glass may make it too bright). An oddly shaped room that is difficult to lease may be a perfect location for a quiet area. Semi-private seating alcoves placed in found spaces

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are also valued accommodations. However, it is important to locate these spaces away from noise and traffic.

Yoga Area

Like meditation/quiet areas, spaces for yoga should be in quiet areas, out of common view. It is desirable to have nearby storage space for mats and other equipment. A yoga space is frequently part of a fitness center if such a tenant space is provided at the airport.

Worship

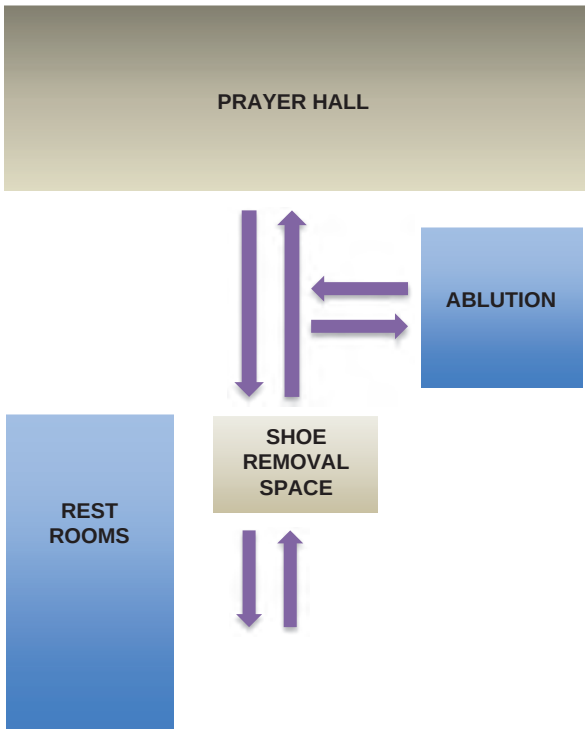
Worship spaces are typically provided on the secure side, but some airports occasionally also have a space on the non-secure side set up to include participants from the local community.

Ablution

Ablution facilities are ideally located adjacent to prayer rooms (see Figure 2-34). Restrooms should be nearby to complete elimination prior to the ablution ritual but should be separate from ablution. Men and women should have separate but identical spaces. See Section 2.6.3 for additional information.

Smoking

The trend is that airports are minimizing or eliminating smoking spaces because smoking is not permitted by law inside public spaces in the United States and some other countries. Airports typically have designated areas for smoking located outdoors on the non-secure side



Source: Mokhtar, A., Design Issues for Muslim Praying Spaces in Shopping Malls. 2008.

Figure 2-34. Recommended sequence and adjacencies for restrooms, ablution, and worship spaces in public places.

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of the airport away from the entrance doors. However, airports with a significant number of travelers from countries where smoking accommodations are common tend to make this amenity more of a destination. Some airports have exterior courtyards or balconies on the concourses. FAA regulations need to be confirmed related to proximity to the airfield. Where permitted, some airports provide an enclosed smoking room on each concourse.

2.7.4 Distribution of Layover-Related Amenity Spaces

Sleeping

Areas for sleeping are typically onsite hotels that are independently operated and sometimes located within security at airport hubs with frequent overnight layovers. More recently, concessionaires have begun offering a bank of small rooms for naps of varying time durations right off of the concourse (ideally in quieter areas, like a mezzanine). Some airline and airport lounges also provide sleeping accommodations for members. Sleeping pods, another concession alternative, are also available but may not have dedicated restroom and showering facilities. There are also accessibility issues with some pod designs, particularly when they are stacked or when users must crawl in.

Fitness

These areas are usually located within one or more concourses if provided. Typically, these have their own restroom, so adjacency is typically not required. Some onsite hotels and airline lounges also provide access to fitness facilities for a fee.

Public Showers

Public showers are typically found in fitness centers, lounges, concession sleeping rooms, and spas. Some onsite hotels provide showers for a fee.

Business Centers

These centers are commonly located in each secure zone when provided.

Health/Urgent Care

Typically, one care center is located on the secure side. Those that also serve the local community (as in large cities) may have an additional location landside.

2.8 The Restroom Master Plan

For renovations, a useful master plan can superimpose proposed restroom locations over the existing distribution of restrooms. Figure 2-35 shows a page from an airport-wide master plan document. In the center, it lists the existing, required, and proposed fixture counts for the entire concourse.

For this section of the concourse, the plan shows a dashed black box around the existing restroom locations along with the room numbers, which ties into the existing restroom evaluation forms described in Section 2.3 of this guidebook. The red circles indicate the proposed restroom locations with the fixture counts in red. Most of the locations in the example are renovations, but the plan also shows the location of one new restroom and one that is expected to be eliminated. The plan should indicate the proposed timing of the restroom projects in this area.

The master plan also captures issues of interest for the airport, including restroom locations that typically have long lines (there is one in this example) and the frequency of cleaning

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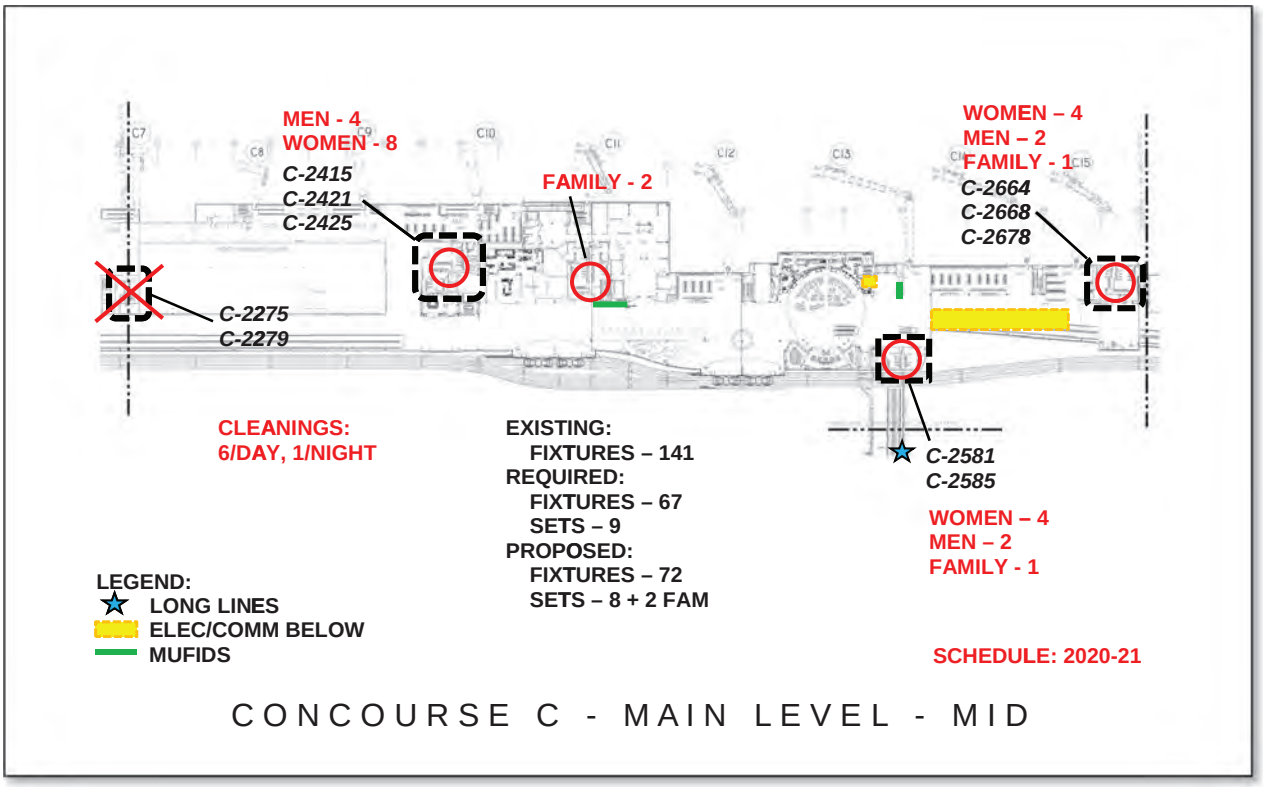


Figure 2-35. Partial concourse master plan.

throughout the day. Compared to other areas of the airport, this number is high, so this is a comparatively busy concourse. This example also demonstrates that restrooms should not be located over electrical and communication rooms in the event of leaks in the restrooms. This plan indicates the location of these spaces on the level below.

The restroom team at this airport designed an information hub at each restroom location so over time travelers will know that these components are co-located at this hub. This hub also installed its multi-user FIDS (MUFIDS) in an area where travelers can peruse it while waiting for companion(s) using the restrooms. This plan indicates where existing MUFIDSs will ultimately be relocated to new restroom locations.

A planning strategy to consider for larger restroom blocks is to divide the restroom into two halves so one-half can be closed at a time. This eliminates the loss of an entire restroom for men or women in a particular area during maintenance. It can be difficult, however, for maintenance staff to shut down one side of a restroom without entering the restroom. This can be an issue if the maintenance staff member is a man for a women's restroom or vice versa. Ribbon barriers, cones, and pull down (or out) grilles are strategies often used to shut down half of the restroom.

2.9 Restroom Renovation versus Building a New Restroom

An important part of the master planning process is evaluating the pros and cons of renovating a restroom versus relocating it or building a new space. The last option is relatively easy. If a new concourse or terminal is planned, restrooms will be constructed with it. The benefit of

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new construction is that disruptions to existing restrooms and the adjacent spaces are avoided. There is also more planning freedom, so the restrooms can meet most, if not all, of the restroom team's design criteria. If new restrooms are constructed, bringing existing restrooms up to the same or a similar standard should be considered. If this cannot be accomplished immediately, then it should be scheduled over a master-planned timeline. This ensures consistency throughout the airport, which is important for travelers from a wayfinding and LoS perspective and for maintenance by keeping attic stock to a minimum.

Renovating an existing restroom is easier by far than relocating a restroom, especially if it simply involves updating fixtures and/or materials. Renovation can be phased over time to minimize disruption. If renovation is phased, the cost of the extended construction period should be analyzed because piecemeal construction can cost considerably more than a more aggressive schedule (that renovates more restroom sets at one time) due to the repeated start-up and close-out costs for each project.

Sometimes, however, there are issues that a simple renovation cannot fix. Perhaps the restroom would be more effective in a different area of the concourse, or maybe the space is too small in its current location for the number of fixtures required to accommodate increased deplanements. Relocating a restroom is a logistical puzzle that everyone on the restroom and amenities team should be involved in solving. The biggest challenges to consider include the following:

- How long will construction take?
- What utility shutdowns will be required and for how long?
- Can the existing restroom remain operational until the new location is complete?
- Is there a nearby restroom(s) that can absorb some of the overflow?
- What other spaces will be affected at the new location?
- Will the overall project cover the cost of modifications to the affected spaces?
- If the affected spaces are those of a tenant, will the timing be dependent on the tenant's lease?

These are just some of the questions requiring consideration and discussion among the restroom team in order to make the most effective decisions for the long term.

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CHAPTER 3

Airport Restroom of the Future

3.1 Introduction

This chapter makes a case for the possibility of creating airport restrooms in the future that are free of barriers for most travelers by

- Using small, flexible planning modules
- Adding right-sized restrooms easily
- Prefabricating major nodes
- Minimizing maintenance shutdowns
- Accommodating all abilities
- Helping to calm the harried traveler

3.2 The Issues

Many airports were built with restrooms squeezed into inferior, leftover spaces. Now, especially in the wake of 9/11, it is difficult to enlarge the typical restroom to accommodate travelers' carry-ons as well as make it accessible for travelers with disabilities without decreasing the fixture count (which keeps increasing with larger aircraft). Some of the issues with restrooms for airports are described in the following.

3.2.1 Construction

Renovating restrooms in airports is disruptive and expensive because it requires 3RD PAGES varying vintages of layouts, infrastructure, and finishes. With varying space constraints, it can be difficult to standardize layouts.

3.2.2 Maintenance

In a typical restroom layout, containing a men's and a women's restroom, the entire men's side or the entire women's side becomes unavailable during cleaning or repair. Even when restrooms are designed so that only a part of a men's or women's space needs to be closed, it can be difficult for maintenance staff to shut down one side of a restroom without entering the restroom. This can be an issue if the maintenance staff member is a man for a women's restroom or vice versa.

3.2.3 Accessibility

The ADA requires one accessible stall in a typical airport restroom. As travelers with disabilities are a small (but growing) percentage of the public, those in need often find the

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accessible stall occupied by an able-bodied person. More of these types of stalls would alleviate this issue.

3.2.4 Gender

To describe the issue of gender requires a brief history of the public restroom (synopsized from *The Bathroom* by Alexander Kira and *Toilet* edited by Harvey Molotch and Laura Norén). The earliest evidence of public restrooms is from approximately 1700 BCE at Knossos on Crete. The then technologically sophisticated facilities were not unlike contemporary restrooms, with places for elimination and washing. They were typically available for townsfolk as well as travelers. Most of the great cities of the ancient world up through the time of the Roman Empire had similar accommodations for the public, although it is not known if genders were separated.

For the next thousand years, public restrooms fell out of fashion to the point where public dung heaps were the norm. Even during the opulent times of Louis XIV, it was common to use streets, cellars, and yards when the need arose. In his proposals for public housing in new towns, Leonardo da Vinci noted that all stairways should be spiral to prevent sanitary misuse of landings.

The nineteenth century saw a renewed interest in public facilities because of disease epidemics like the cholera outbreak during the Civil War in the United States, large public events such as the Crystal Palace Exhibition in London that drew 800,000 visitors, and, more significantly, the rise of rail travel and the new terminals to accommodate the many travelers. Today, public restrooms are generally easy to find in malls, gas stations, parks, and so forth with separate facilities for men and women and even restrooms for families and those traveling with a companion for assistance. Separate public restrooms for men and women, however, are a relatively recent provision.

The first law in the United States to require restrooms for women in addition to men was enacted in 1887 in Massachusetts and was aimed at factories and workshops. By 1920, only 43 of the 48 states had adopted similar legislation. The reason for such a law in the first place came into being in the early nineteenth century when the Industrial Revolution saw men leaving their homes daily to work at the factories, etc. The rapid growth in technology and industrialization invaded life with new dangers and filth at every turn. This, coupled with the Victorian assumption that females were endowed with greater moral sensibility and religious inclinations than men, effectively made the workplace the domain of men and the home that of women. These intrusions of urban life also increased the desire for personal privacy, which in turn led to increased concerns about modesty, especially regarding the human body and bodily functions.

Society's anxiety about women emerging from the home to work, shop, and socialize led to the notion of creating separate, home-like restroom spaces in public places like libraries, department stores, hotels, restaurants, and railroad cars. But in the industrial workplace, men and women took turns using the meager facilities provided by employers.

Women's "physical vulnerability" to unsanitary conditions was a primary justification behind the creation of separate restrooms as was the protection of their virtue and morality. Also included in the "Ladies Room" was a space for emergencies resulting from dizziness, fainting, and other symptoms of illness. In summary, the first sex-based separation laws for public restrooms were not based on the anatomical differences of men and women but were enacted to vindicate Victorian-era moral ideologies concerning the appropriate role and place for women in society.

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So how is this history relevant to today's public restrooms, particularly restrooms in airports? For many people, it seems that the commonly found equality in the number of restroom fixtures for males and females has functioned just fine for decades. But the "toilet laws" have impacts that remain quite influential and detrimental. For starters, the typical 50/50 ratio of fixtures in men's and women's restrooms has created a barrier for women in terms of time. Most building codes allow urinals to substitute for up to two-thirds of the toilets in a men's restroom. The quick-zip design of men's clothing and the ease of maneuvering at urinals shorten men's restroom visits dramatically compared to women's. Women also have needs due to pregnancy, menstruation, and higher levels of incontinence in old age. Women are more likely to be traveling with babies, small children, or elderly relatives who need to use the restroom often, take longer, and require more space within the stall to allow for the number of persons and appurtenances. Last, women make up a larger proportion of people with disabilities, which adds to the need for more fixtures.

Even though current building codes have been revised to provide more fixtures for females in certain building types, the ratio of fixtures in men's and women's restrooms remains 50/50 in airport terminals as specified in the current International Building Code. Another issue with restrooms at airports that affects both women and men is the constantly varying gender mix of travelers throughout the day. At any point in time, there may be a line at either a men's or a women's restroom, for example, if a sports team of men or women flies in. Gender segregation also creates difficulties for parents with opposite-sex children or traveling companions in need of assistance such as an elderly relative or someone with a physical disability.

3.2.5 Wayfinding

A more subtle side effect that persists is the symbolic message in restroom signage that there are only two sexes, which impedes the acceptance of those persons who are transsexual, transgender, or intersexual. Modern society has taken steps to remedy some of these discriminations with the provision of a unisex restroom. However, the common names of "family room," "companion care room," and "assisted care room" have their own exclusionary connotations.

As attitudes have evolved about restrooms and gender, so have ways of identifying them. Care must be taken to develop signing that does not reflect an unconscious bias.

A new sign would focus on function rather than persons. The toilet symbol shown in Figure 3-1 is becoming common in Europe and is slowly gaining acceptance elsewhere for its functionality and its blatant neutrality. The wheelchair graphic remains a predominant building code requirement to indicate that a restroom is accessible. Ideally, all restrooms, particularly in airports, where diverse peoples gather, would be accessible. Thus, the simple toilet symbol would be universal and suffice.

3.3 The Restroom of the Future Prototype

The next step in the evolution of public restrooms is to provide a private space with provisions for personal hygiene that works equally well regardless of gender and needs—the restroom of the future.

The restroom of the future prototype layout shown in Figure 3-2 utilizes private, accessible compartments along a common corridor. This would provide gender flexibility and, as will be seen, a variety of accessibility options that would also accommodate traveling families.

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Figure 3-1. Prototype signage for the airport restroom of the future.

In addition, there is a companion care/changing table restroom, a diaper changing alcove, and a waiting area near the entrance to minimize traffic congestion in the corridors.

At one end of the corridor is a janitor's closet/storage room for cleaning and paper supplies. This would allow the cleaners and maintenance staff to access the restrooms without pushing their carts through the main entry area. An important feature of this layout is that pipe chases would not be needed, saving a significant amount of space. Maintenance could be done within each compartment via cleanouts and access panels, allowing the remaining compartments to stay open. Similarly, cleaning could be done one compartment at a time, ultimately providing cleaner restrooms.

Each typical compartment (see Figure 3-3) has a toilet node, urinal node, and sink node. It would be accessible per the current ANSI A117.1 because there is space for the 30-inch by 52-inch clear space (recently increased from 30 inches by 48 inches to accommodate the battery on electric wheelchairs) within the room without overlapping the door swing. There is an "L" grab bar layout in both left- and right-hand versions. The accessible urinal is strategically placed so it would not be directly visible upon walking into the compartment or when sitting on the toilet.

Two ambulatory stalls are also provided in the prototype layout (see Figure 3-4) with parallel grab bars in the toilet node. Here, too, a sink node is provided, and an accessible urinal node is also located out of direct view.

Additional signage is recommended in the plan for the restroom of the future. With the corridor lined on both sides with a series of identical doors, how would a user know what type of restroom is behind the door? Is it ambulatory with parallel grab bars? Or the traditional model with a left-handed, L-shaped grab bar configuration? A right-handed version? There would be a need for a new type of sign to prevent people from having to open door after door in search of the accessibility configuration that they need.

Figure 3-5 shows a concept that takes the accessible symbol one step further by adding the grab bar configuration. This sign would also help identify in existing restrooms ambulatory stall locations within the mix of standard stalls. This could be a small sign applied to or etched into the outward face of the stall door. The International Organization for Standardization (ISO) could develop and adopt a sign such as this.

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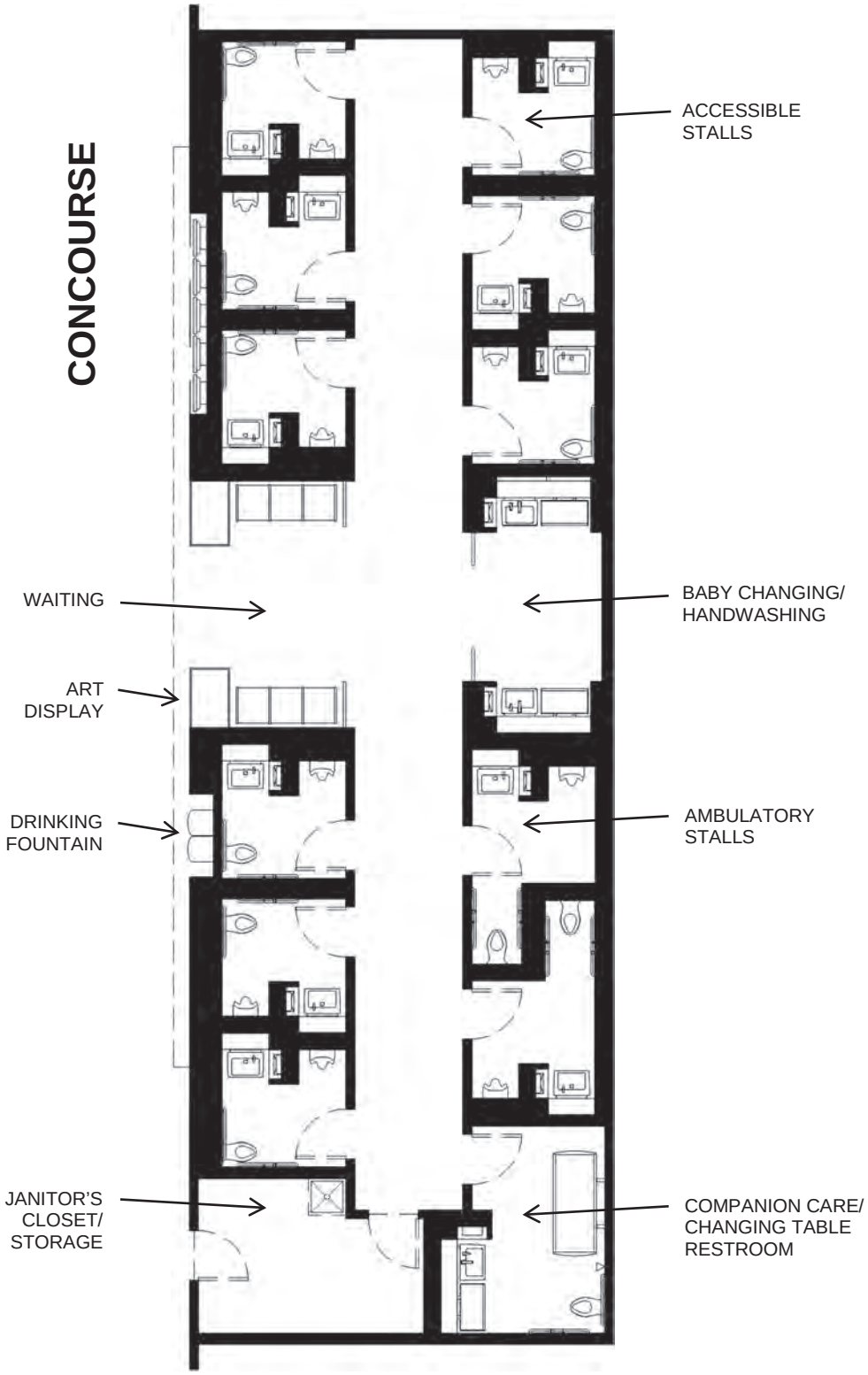


Figure 3-2. Prototype airport restroom of the future layout.

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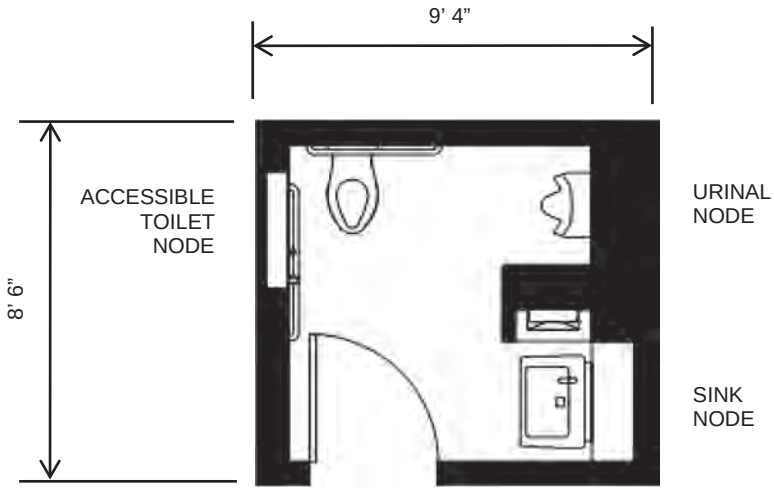


Figure 3-3. Accessible compartment layout.

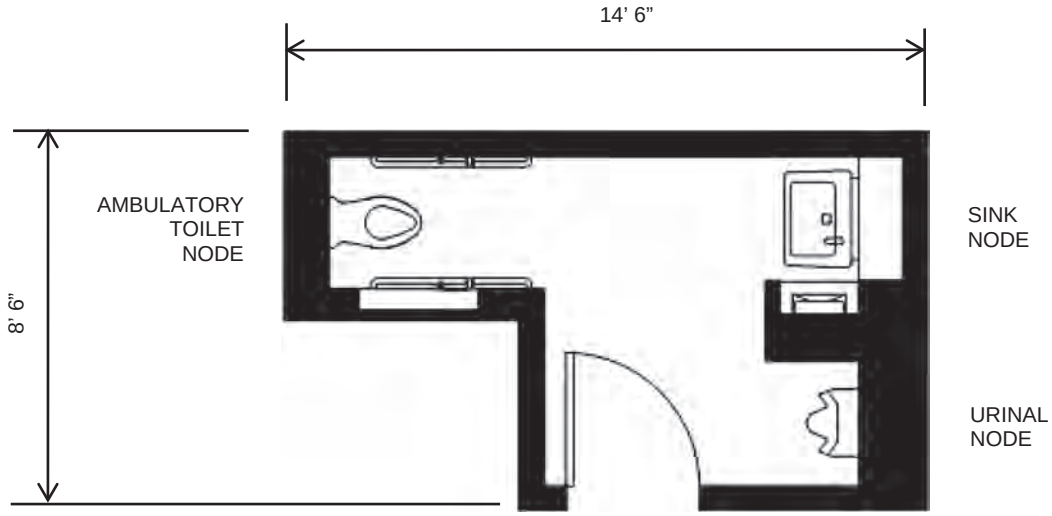


Figure 3-4. Ambulatory compartment layout.



Figure 3-5. Stall sign types—accessible left hand, accessible right hand, and ambulatory.

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3.4 What Would Be Solved

The restroom of the future prototype would solve issues in the following areas:

- **Planning.** The compact module would maximize the number of fixtures by being able to fit the compartments around obstructions (such as utility shafts) and would be more flexible to layout in odd-shaped spaces.
- **Construction.** The modules are ideal for prefabricated assemblies. The entire module or certain nodes, like the sink node, could be built offsite and then be brought in and fit together with other modules. This would speed up construction and minimize disruption from renovations within the airport.
- **Maintenance.** One restroom space could be cleaned or repaired at a time, taking only one fixture out of service instead of an entire restroom.
- **Barrier free.** With the prototype shown in Figure 3-2, every fixture would be accessible to all genders and most people with disabilities.
- **Personal space.** The compartment would provide privacy and room enough to change clothes, shave, put on make-up, and so forth.

3.5 What Would Remain to Be Solved

What would remain to be solved by the restroom of the future prototype are issues in the following areas:

- **Security.** Airport police would likely have concerns about illicit behavior (such as drug use and sex) and discouraged behavior (such as airport staff napping) in the closed rooms. These rooms, however, would be no different from current companion care restrooms. In addition, having more people circulating in the restroom area would likely deter undesired behaviors.
- **Gender separation.** People expect privacy when they go to a restroom, which includes separation from people of other genders. There may be discomfort about commingling in the common corridor and likely also about using a restroom space right after someone of a different gender has used it. There is a precedent for sharing restrooms, however, with companion care restrooms, airplane restrooms, and porta-potties, although the latter two are universally considered to be unpleasant experiences.
- **Cleanliness.** Probably the biggest obstacle to the acceptance of this restroom concept is the perception of differing cleanliness practices among men and women. It's easy to say that people just need to "shape up and be considerate," but changing behavior is difficult, especially as it relates to a transient public space in which users have no vested interest.
- **Multiple accessibility accommodations.** Some disabilities require getting onto the toilet seat from a front approach while others require a side transfer, which may require that the side bar be on the left for some and the right for others. Some disabilities require a side and rear grab bar; others need parallel side grab bars. These two configurations conflict, which is why the current concept plan has left and right as well as ambulatory versions. Ideally, a grab bar system would be created that could accommodate both scenarios. There are swing-down grab bars available that are allowed in housing and public restrooms in some other countries. Swing-down grab bars are not allowed in U.S. public restrooms because this product conflicts with the location of the rear grab bar in an accessible stall. Until these two configurations can be combined, two types of restrooms will be required, and therefore the module will not be completely equitable.
- **Companion care restroom privacy.** Ideally, a companion care restroom would be configured to provide visual privacy from anyone else in the space for the person using the toilet. The

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accessible compartment in this concept is purposefully compact for efficiency. However, with the required clearances for accessibility, there is no privacy from another person.

- **Perception of cleanliness.** It was noted by some of the airport restroom case study participants that having a full-time cleaner in a restroom creates the perception that the space is clean because a person is seen to be cleaning. Even if some part is dirty or a toilet paper dispenser is empty, people tend to be more forgiving because they see the cleaner will get to it soon. The configuration of the airport restroom of the future concept is ideal for being staffed by a full-time cleaner who can move continuously from one unoccupied room to the next. Developing technologies such as motion detectors used to monitor the number of users in each module could also inform cleaners which rooms need to be cleaned. This person could also be available to help people who need assistance.
- **Building code requirements.** Current building codes require separate restrooms for men and women. Unisex restrooms, such as companion care restrooms, are an optional extra that the latest International Building Code does allow to be counted in the fixture count, but not for the entire count. A significant change in thinking and legislation would be required to universally allow the airport restroom of the future concept.

3.6 What Next?

Ideally, the restroom of the future prototype will lead to discussion within restroom and amenities teams and inspire airports and their designers to test the waters with one or more aspects. Current society and culture may not be ready yet for such a drastic change. Yet, among air travelers, who have become increasingly vocal about their expectations, there is a strong demand for improvement.

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CHAPTER 4

Design

4.1 Process

Once an airport determines restroom and amenity locations and the first project(s) is (are) identified, the next step is to develop the design. The prototype plan has presumably been massaged and shaped to fit the location, but may require fine-tuning as the components are determined and finalized. The design stage is the ultimate tug of war among functionality, space requirements, materials/products, initial costs, and long-term costs (such as maintenance and resource usage). Everything that goes into and around the restroom and amenities needs to be researched, discussed, prioritized, priced, selected, and located within the space.

This chapter walks the reader through this process. The intent of this guide is not to recommend design solutions, but rather to offer suggestions and considerations as the design is pulled together by the restroom and amenities team. Section 4.3 looks at each of the primary components in the restroom spaces including the following:

- Signage
- Surfaces
- Restroom Accessories
- Plumbing
- HVAC
- Electrical
- Lighting
- Technology

For restroom components that have multiple attributes to compare, a matrix is provided in Appendix A that compares

- Initial cost
- Life-cycle cost
- Warranty
- Maintenance
- Sustainability
- Pros
- Cons

Similarly, the regulated/essential amenity spaces have primary components that are identified in Section 4.4, and Section 4.5 offers considerations for the waiting- and layover-related amenity spaces. Prioritization worksheets are provided at the end of this chapter to help the restroom and amenities team evaluate and prioritize which components the space and budget constraints can accommodate. However, before jumping into this level of detail, it is important to consider the impacts of maintenance, sustainability, and universal design.

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4.1.1 Maintenance

Every material and product will require frequent cleaning and occasional repair or replacement. Components (such as paper towel dispensers) installed in the restroom of an office building may last 10 to 15 years. However, in a large-hub airport restroom, components may last an average of 2 to 5 years due to 24/7 operational use; inadvertent damage from luggage; and, unfortunately, vandalism. Durability is an important consideration for every item that is included in an airport restroom, and this feature typically comes with a higher price tag than other options. The high price is not necessarily extravagant. A material that will hold up for 20 years instead of 5 may be worth even double the cost.

Key maintenance attributes of amenity surfaces are that they are durable and easy to clean. One way to achieve both is by using minimal joints. Innovations in manufacturing are large-format tiles and wall panels that can be up to 4 ft by 10 ft, with butt joints sealed with epoxy adhesive. These panels eliminate grout joints, which over time discolor—often to the shade of mop-water gray—and can develop cracks that let moisture seep into the wall or floor cavities. The elimination of grout joints can help keep restrooms looking fresh and clean.

Of course, new issues come with innovation. In the example of the large-format wall panels, airports must understand wall and floor movement to avoid stress cracks in panels. Such deflection can be significant in older airports. However, it may be difficult to find examples of tried and tested panel installations or even multiple, competitive manufacturers to compare products. It is important that the entire restroom team participate in researching and evaluating products, which should be an ongoing process. New materials, products, and technologies are introduced every day. Most will not pass muster, but occasionally one will fulfill a need or provide an enhancement that will greatly benefit customer service or operations. For this reason, it is important for the entire team to perform diligent research.

Unintentional or intentional damage is also an ongoing maintenance issue at airports. For example, airports with international flights are visited by travelers that may never have previously encountered a Western restroom. In their attempts to figure out a stall full of unfamiliar fixtures and devices, they may cause damage. There is also intentional damage from disgruntled travelers, unruly youth, and others—every airport has its share of stories: graffiti on mirrors, dented partitions, and so forth. While some airports make efforts to minimize damage such as providing instructional signage or frequent monitoring by staff, at the end of the day, replacing and repairing broken fixtures and extra cleaning are part of operating an airport. Therefore, it is imperative that selected materials and products can be easily repaired or replaced. While a light fixture from Italy may provide gorgeous lighting at the sink mirrors, when the replacement takes 4 weeks to arrive, it has a significant impact on a traveler's experience and, ultimately, the airport's reputation. It is best to select products that have locally available stock or at least an attentive local representative. For those products that do have long lead times, it is helpful to specify a supply of attic stock with the project.

4.1.2 Sustainability



Any opportunity an airport has to reduce the waste of natural resources should be explored. Some opportunities may not be economically feasible, but airports have employed several solutions in restrooms that can have a significant impact on resource waste management with a relatively low impact on a project's long-term costs. The most obvious opportunities involve water usage. Low-flow fixtures can substantially conserve the amount of water consumed. Toilets, urinals, sink faucets, and drinking fountains all have sustainable features available (see the component comparison matrix in Appendix A). A criterion that should be

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considered for every product is the LEED Program's 500-mile maximum radius from extraction and manufacture to the project site.



Leading-edge products and technologies are often still in development, and some end up having adverse effects. For several years, the sensors on some automatic flush valves were activated by the slightest movement of a person in a stall. This caused the toilet to flush several times during one visit. Reflective stall materials can also confuse the sensors. Recent developments have corrected this annoyance in newer models. However, legacy products will likely remain for years to come. Similarly, some waterless urinals have issues with odors depending on the model and maintenance procedures. Several of the case study airports visited by the research team had tried waterless options with mixed results. Some had great success with them, while others replaced the fixture not long after the project was completed.



Some products have sustainable effects in collateral areas. Drinking fountains can have a water-bottle filler attachment that allows travelers to bring their own (empty) water bottle through security and fill it before boarding. This reduces the number of plastic bottles that need disposal. A further improvement in this vein is bottle-emptying receptacles located pre-security that reduce recycling costs for the airport. (This also reduces income for retail tenants that sell bottled water. Thus, tenants may oppose such a development.) Another innovation is faucets and flush valves with small turbines that power the fixtures' automatic sensors via water flowing through the spout. This eliminates the need for batteries or a connection to building power. Note that automated toilets are required to have emergency backup power, which may also apply to self-powered and battery-powered models in certain jurisdictions.

4.1.3 Universal Design



The concept of universal design has been around for decades. It goes beyond providing accessibility as a civil right for persons with disabilities via the ADA and translated to architecture through ANSI A117.1. The intent of universal design is to provide equitable access to spaces and components for everyone, regardless of mobility; age; status in life; and, most recently, gender identification. So far, in the United States at least, cultural diversity has not been a major thrust of universal design (e.g., provision of squat toilets for Asian and African travelers), but it is gaining traction.

In our focus group for travelers with disabilities (see Appendix D.1), the comment was made by a person with limited vision that universal design should also mean that fixtures are always in the same relative location. For example, the toilet paper dispenser would always be on the right side of a stall. For this person, every visit to a restroom was a searching expedition.

Responding to Disability

As members of the restroom team interact with people with disabilities, they may feel tongue-tied as they try to communicate without using words that may sound insensitive or offensive. The Minnesota State Council on Disability has a helpful booklet, "Responding to Disability: A Question of Attitude," that provides guidelines for navigating these interactions for those who haven't had the opportunity to get to know people with disabilities. The booklet is also available online at www.reachable.org/content/responding-disability-question-attitude-9.

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However, in the case of accessible stalls—commonly used by guide dog handlers—standardizing fixture locations would conflict with the needs of individuals transferring from a wheelchair to a toilet, some of whom need a left transfer, others a right one. Thus, one should always be careful when considering an accommodation for one type of disability to ensure that it does not disadvantage another. (See Chapter 3.)

Given that the traveling public comes from all walks of life and all corners of the globe, it is important to consider universal design, especially from the perspective of providing excellent customer service. With our increasing elderly population, for example, accommodation for limited mobility, visual and hearing impairments, and slower movement no longer pertains just to the realm of the accessible stall. In addition, small children and persons of short stature need to reach the faucet, and transgender persons need restrooms where they can feel comfortable. Comfort is the key. When a person is uncomfortable, for whatever reason, it is a distraction that degrades their experience; this is exactly the situation this guidebook endeavors to eliminate.



Accommodating universal design, however, is not an easy task. Almost every accommodation conflicts with another need. For example, the accessible, 18-inch-high toilet seat works well for transferring from and to a wheelchair. This height, however, can create problems with bowel movements. The elderly, for example, often need to assume a squat position. An expert on disabilities of all types is an invaluable individual to have as part of the restroom team.

4.2 Standardization

It is in the best interests of an airport to develop standards for its restroom design. This will provide consistency throughout the airport, which will help with wayfinding, streamline the design and construction process, and make the maintenance process more efficient. Standards should be determined for the following:

- Prototype layouts for the restroom and regulated/essential amenities
- Products
- Finishes and colors

A sample standards document from one of the airports that participated in the Airport Managers Survey and the case studies conducted for this research is included in Appendix F. Like the master plan, the standards document should be a living document. Products are discontinued, innovations come to market, and so forth. The standards document should be updated with each project and reviewed holistically every 5 years.

4.3 Restroom Components

The restroom components—the materials and products that fit out the space—are the heart of the design. This section provides information on the initial selection of each of these. Research needs to be undertaken by the restroom team to understand what components fit best within the airport's customer service, operations, and fiscal philosophies.

Because of the extreme wear and tear on components in airport restrooms, it can be difficult to find products that are durable while not looking utilitarian. Given the quantities needed in an airport, customization of existing products is a viable option as is engaging manufacturers to develop new products to fulfill a specific need. See the summary of the product development focus group in Appendix D.1 for more information on this process.

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Figure 4-1. Wayfinding sign.

4.3.1 Signage

Concourse Wayfinding

When travelers deplane, often their priority is finding a restroom. Likewise, after a long drive to the airport and facing lines at both the ticket counter and security checkpoint, many travelers make a quick stop at a restroom. If restrooms are not readily visible from gates and near checkpoints, clear directions to the nearest restrooms are critical (see Figure 4-1).

Wayfinding signage for restrooms often consists of the international men/women symbol on an overhead sign with an arrow. It can be useful for travelers unfamiliar with the airport to see the distance in feet to the nearest facility, especially if there are restrooms in multiple directions. While locations are typically shown on directories, providing a phone app with restrooms highlighted, especially in relation to a person's current location, is a great customer service feature. *ACRP Research Report 177: Enhancing Airport Wayfinding for Aging Travelers and Persons with Disabilities* notes the importance of also providing signage as travelers leave restrooms to reorient them to gates, baggage claim, and so forth.

Room Location Sign

Airports have become as cluttered as highways with signs of every size, shape, and color encumbering the visual field, making it easy for travelers to miss the restroom or amenity sign. Therefore, it is the recommendation of this guidebook to forgo the ubiquitous blade sign and let the restroom speak for itself. Using a consistent, iconic façade and entry on restrooms

Phone Apps and Directories

There is tremendous inconsistency in functionality and graphics between and within airports regarding their apps and physical directories. The consumer aviation industry needs to agree on symbols, particularly for regulated/essential amenities. Also, the naming of the spaces for directory listings needs to be consistent. Several industry organizations could make this task a priority and develop a standard.

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throughout the airport helps travelers to recognize them and eliminates the need for identifying signs. (An exception to this recommendation is high-ceilinged spaces. A blade sign above the restroom can aid in locating the space when a ceiling-mounted wayfinding sign is not desirable.) That said, some indication will be needed, from about 50 ft away, to identify which restroom entry is for which gender. Airports are beginning to use bold colors, text, or graphics to draw attention, sometimes with a playful sense of humor.

Room Name and Number

Accessibility and building codes typically require that a sign be provided that identifies the name and the number of the room. Specific requirements for the size and location of such signs should be verified. Room signs allow the facilities staff to locate doors or openings for maintenance and emergencies. The aesthetic of the room sign is often an afterthought and the airport's standard room signage—often a basic, primary-colored plastic plate—is applied to the wall. Consideration should be given to integrating the sign into the overall design by, at a minimum, using colors and/or material that complement the design palette.

Room names should be brief so they can be quickly read. There is no need to use the word "room" on the sign because it is commonly understood that the sign is for a room (e.g., "Women" instead of "Women's Room").

Accessible Stall



In larger restrooms, it can be beneficial to provide a symbol graphic identifying the accessible stall(s). This graphic can be on the stall door or the wall next to the door; however, it should be clear which door is to the accessible stall. The graphic can be integrated into the surface material rather than being applied as a sticker or other applied signage material, which is vulnerable to damage or cleaning wear. Stainless steel, for example, can have the symbol etched into the surface. (It is important to ensure there is enough contrast for people with visual impairments to see the symbol).

The FAA Office of Civil Rights requires indicating on the exterior of a restroom if it has one or more ambulatory stalls. It is also helpful to identify the ambulatory stall within the restroom. In a typical restroom, the ambulatory stall is the one with the door that is slightly wider than the doors on typical stalls. That is a subtle difference for wayfinding. Unfortunately, there is not a standard symbol for ambulatory stalls. This guidebook suggests that all typical stalls should be ambulatory, although this would not be known until the doors of the stalls are open. The FAA does waive the external indication requirement if all the typical stalls are ambulatory.

Informational Signs

Informational signs, such as a sign indicating that employees must wash their hands, are often an afterthought. Once a project is completed, these and other types of informational signs are attached to the walls with double-sided tape or screws, which then become a cleaning obstacle and take away from the otherwise well-thought-out placement of wall elements. If possible, informational signs should be recessed within the wall material so the surface remains flush and smooth.

A simple technique to recess signage is to print the sign graphic on the back surface of a ¼-inch sheet of tempered glass. The opening in the finished wall surface should be ⅛ inch larger than the glass on all sides and the gap should be sealed with clear silicone for future removal if a change to the sign is needed.

Like all toilet accessories, signs within the restroom should be deliberately organized on the wall to create an orderly composition.

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Advertising

Although placing an ad or flat screen above a urinal or on the inside of the stall door seems ideal for generating extra revenue with a captive audience, it is probably not the best idea. As mentioned previously, travelers are already overstimulated moving through the airport and tired from the processes of air travel. "Rest" is an important part of the term "restroom." Optimize customer service by keeping the space calm and free of unnecessary distractions that encourage visitors to linger.

Severe Weather Shelter

Intuitively, a restroom block seems like a good place to shelter in severe weather such as tornadoes, hurricanes, and so forth. Restroom blocks are rigid, with a network of pipes and conduit, and are often constructed with concrete block partitions. The terminology used to indicate that a restroom is a severe weather shelter is important. Spaces labeled "safe rooms" and "storm shelters" must meet specific requirements set by the Federal Emergency Management Agency (FEMA) and the International Code Council (ICC).

The construction elements surrounding a restroom (roof structure, exterior glazing, and so forth) impact the labeling of the restroom. If a restroom has windows, security film can be installed to protect against airborne projectiles to meet FEMA or ICC requirements. However, this film often compromises the warranty of insulated glass. One heavy-duty approach is to provide an exterior, specialty, insulated-glass unit with a sandwich of laminated glass and high-strength film. These window units typically are thicker and require a deeper curtain wall system, which may not match the adjacent existing curtain wall system.

It is also important to check with the airport's legal department to assess potential liability. The Metropolitan Emergency Managers Association has put together a useful checklist for determining the best spaces for shelters within a facility. Again, signage should ideally be incorporated into the overall material palette (see Figure 4-2).



Figure 4-2. Severe weather shelter sign.

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Signage for Persons with Vision Loss

Review with your airport's or community's accessibility advisory committee the best strategies for communicating information on signs for both those who are blind and low vision, as the needs of each differ. *ACRP Report 52: Wayfinding and Signing Guidelines for Airport Terminals and Landside* is a recommended resource as well as *ACRP Research Report 177*, where this is covered in greater depth.

ADA standards require braille and raised characters for signs that identify permanent spaces such as restrooms or other ancillary spaces such as lactation rooms or SARAs, but braille and raised characters are not required for directional or informational signs. An airport may, however, decide to exceed code requirements where it feels that the information is critical for individuals with vision loss. London Heathrow Airport (LHR), for example, includes tactile characters and pictograms for its floor directory outside elevators. Persons with low vision, as well as those of short stature, and those who use wheelchairs benefit from placement of signs, including a FIDS, at a lower level for close approach. This enables them to use apps such as Seeing AI to read the text viewed by the smartphone camera. See *ACRP Research Report 177* for detailed information on making signage accessible for persons with vision loss, including which fonts best serve their needs. Beacons, phones, apps, and directories with audible instructions are becoming common for wayfinding assistance.



Instructional signs such as TRASH are another issue. Using raised letters on the face of a waste disposal bin is the most common approach. Most people with vision loss lost their vision later in life, so they recognize the shape of the letters with their fingertips, which is an easier adaptation than learning a new language (braille). However, this accommodation doesn't help those who only know braille or who speak a different language. Unfortunately, many people who are visually impaired have to feel their way around a restroom, recognizing fixtures and accessories from experience. A visually impaired participant in one of the accessibility stakeholder outreach sessions for this research lamented that he had inadvertently put his hand in many a urinal.

4.3.2 Surfaces

Wall versus Partition

See Appendix A for component comparisons.

The floors and ceiling/roof structure of restrooms are non-negotiable. The restrooms are constructed between the concrete and steel that supports the entire building. There are, however, options for perimeter interior partitions. "Partitions" do not support weight from above as opposed to "walls," which are part of the building structure.

There are two basic partition types: concrete masonry units (CMUs) and metal studs. CMU walls are more durable and should always be used for plumbing walls, as metal studs will rust from leaks. When CMUs are combined with steel reinforcement, they provide more durable support for plumbing fixtures. CMU walls are heavy, and some older airports do not have the structural capacity in the deck to carry such a concentration of heavy walls. If metals studs need to be used for structural or budgetary reasons, a concrete curb provided under the studs will keep them off the floor and protect them from corrosion in case of large spills. This is especially critical at plumbing walls and in pipe chases.

Floor

See Appendix A for component comparisons.

The use of large-format porcelain tiles with thin 1/16-inch maximum grout joints or seamless terrazzo is recommended for restroom floors. Seamless floors avoid the mold and bacteria

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Effects of Uric Acid

Urine splashed on floors and walls is a prevalent maintenance issue in high-use restrooms. Over time, the surfaces become stained and then break down while a urine odor pervades the space. The key to mitigation is how well a material is sealed. It is not enough to have seamless flooring or wall finishes. If there are tiny holes in the surface, uric acid–borne bacteria will find its way under the sealer and propagate, turning into mold and mildew and creating associated stains and smells. There are numerous commercial cleaners available to tackle urine spills, but the first line of defense is installing completely sealed antimicrobial flooring.

build-up that can happen with grout. A seamless surface also eliminates the clicking sound of rolling luggage, which adds to the din already common in restrooms. However, comments from the research team's focus groups indicated that for people with visual disabilities, the clicking sound helps them discern if others are around.

The slip resistance of floor materials should conform to building code requirements to avoid creating a hazardous condition when the floor is wet. It is important to point out that glossy floors are not necessarily more slippery and that water can collect in textured surfaces, creating a highly perilous surface.

Grout, if used, should be a medium gray color as both stains and the cleaning process over time tend to turn grout to a mop-water gray color anyway. Another strategy (used at LAX) is to match the dust color found in an airport's location with the grout color. The color of dirt varies in different regions of the country, so it is important to discover the correct color.

It is recommended that flooring in pipe chases and janitors' closets have a waterproof traffic coating that continues up the wall four inches to contain leaks in the space. Storage rooms without plumbing fixtures can simply use sealed concrete flooring.

Wall Base

See Appendix A for component comparisons.

Wall base is often the same material as the wall surface. To help low-vision customers trace their way through the space, it is recommended that the base color be different from the wall and the floor. The National Institute of Building Sciences recommends that the value contrast of floors/doors to walls have a light reflectance value difference of 30 to define space and passageways.

Design trends are moving away from a cove base to a right-angle base. Cleaning crews find the right-angle transition easier to clean because the equipment can get tight to the walls. Cove bases can also be expensive if there are many corners (e.g., at urinal alcoves) and are less flexible for future expansions or modifications.

Wall

See Appendix A for component comparisons.

As with floors, large-format porcelain tiles with 1/16-inch grout joints are the current trend for wall surfaces in airports; however, new wall materials in much larger sizes have been developed and should be considered. Wide, floor-to-ceiling quartz or stone wall panels (3/8-inch to

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½-inch thick) create an elegant, monolithic appearance. Edges are butted together and sealed with hairline-thin epoxy resin sealant.

Another design feature of these monolithic panels is the ease of inserting flush graphics and text with contrasting colored material using water jet cutting. This makes the graphics easy to clean. Durable, metallic braille can also be countersunk into the wall panels.

Accent wall tiles of various types and sizes offer design opportunities to add punches of color, reflectance, and/or texture. However, it is important to avoid placing less durable tiles in locations that are susceptible to wear and tear from stored personal belongings and long-term maintenance. Again, grout color and cleaning impacts should be considered. Also, be aware that most tiles bow slightly in the long direction, depending on the proportions. A running bond pattern can show this slight difference between the tile faces. Instead, use a stack bond or one-third running bond pattern.

Ceiling

See Appendix A for component comparisons.

Ceilings in restrooms serve multiple utilitarian functions. They reflect light, can dampen sound (especially in restrooms that are open to concourse and terminal spaces), and provide access to mechanical and electrical equipment above. From an aesthetic perspective, ceilings help organize the room and influence how open a space will feel.

Lighter colors, usually white, provide the greatest light reflectance and feel the most expansive. Higher ceilings in the traffic areas can make a crowded restroom feel less claustrophobic, while low soffits or ceilings over the stalls add to the sense of privacy. In the room prototypes described in Chapter 2, articulated ceilings and soffits can be used to visually group the stalls or sinks together to reinforce the "room" concept.

With all the hard surfaces on the floors, walls, stall partitions, and fixtures, the ceiling is really the only opportunity for noise mitigation. Materials should be carefully considered as some airports have periodic hose-downs of the restrooms in their cleaning cycle. Perforated metal ceiling panels have essentially replaced the linear metal ceilings commonly used in airports in the past. Perforated metal ceiling panels are more durable and perform very well acoustically. They also come in larger sizes, which minimize joints. A variety of hinge and sliding mechanisms make even large panels (up to 10 ft long) easy to access without damaging the panels.

Painted gypsum board is among the most prevalently used materials, especially on soffits, due to its simple aesthetic, ease of repair, relatively low cost, and privacy benefits. However, gypsum board does not provide good acoustics, and, if access is required above the ceiling, access panels will need to be provided. These panels clutter the ceiling plane, and, over time, the finish tends to get chipped and fingerprinted.

It is best to avoid locating infrastructure that requires access, e.g., variable air volume (VAV) boxes, over restrooms because maintenance likely requires shutting down the restroom. Also, some airports have experienced "peeping Toms" among those who have access to lifting ceiling panels in the restroom. If possible, equipment requiring access in pipe chases or ceilings should be located outside the restroom.

Stall Partitions

See Appendix A for component comparisons.

Stall partitions are often the largest surface area in a restroom, so their durability, design detailing, aesthetics, and ease of maintenance are important considerations. A wide range of

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materials is available for different intensities of use. Softer stones, such as marble and limestone, are not recommended because of their inherent tendency to crack, chip, break, and absorb moisture and stains. Lower- and moderately-priced partition systems should be considered for more private airport areas (e.g., employee areas) and moderately to higher-priced partition systems should be considered for greater performance and aesthetics in the most public restrooms.

Most standard partition panels are manufactured to be 58 inches high. Per standard recommendations, this puts the top of the panel at only 70 inches above the floor, which can cause privacy concerns. Fortunately, in recent years, manufacturers have begun offering a broader range of panel sizes as they recognize the demand for stall privacy from all industries. Increasing the panel height and lowering the mounting height of the panel can provide better security and privacy.

Also related to traveler security and privacy is the amount of space provided below partition panels, which can vary widely. It is important to verify local codes for clearance requirements below stall partitions. For the best floor cleaning access below partitions, clearance of not less than 6 inches up from the finished floor is recommended. The 2017 ANSI A117.1 requires a 12-inch-high toe clearance in accessible stalls for wheelchair maneuvering. This requirement can be lowered to 8 inches if certain exceptions are provided, primarily providing additional maneuvering clearance with the walls. Manufacturers often recommend a space at the bottom of the partitions of 12 to 14 inches. However, this makes personal belongings on the floor extremely vulnerable to theft and can also compromise privacy.

There are several stall installation systems for consideration. Cleaners prefer ceiling-hung systems, but their primary drawback is that they offer no lateral stiffness. When this type of stall door swings open, the stall system moves; this movement can sometimes disengage the door latch in adjacent stalls, which makes the doors pop open. If a ceiling-hung system is chosen, it is recommended by manufacturers that a floor-to-ceiling pilaster be dropped at a maximum interval of every three stalls, which means that additional structural support is also required above the ceiling. It is strongly recommended that the pilasters are bolted to red-iron, structural bracing that in turn is secured to the building structure above. When designed properly, the partition becomes as sturdy as a full-height wall. Also, access to the bolts that secure the partition to the structure should be provided, as the bolts tend to loosen over time with the constant swing of the stall door. Alternatively, a permanent cement can be used to "lock" the nuts and bolts in place. It is important to note that a ceiling-hung partition system is not feasible with heavier partition materials.

Floor-and-ceiling-mounted systems are recommended by manufacturers for busy restrooms (e.g., those in airports) that are susceptible to damage from both luggage and vandalism. These systems also require structural support above the ceiling but are far more rigid and durable because of the floor connection.

Two other system options include floor-mounted and floor-mounted/overhead-braced systems. The floor-mounted system is the reverse of the ceiling-hung system and requires a 2-inch penetration into the concrete deck/slab for anchoring. Water can seep under the pilaster and cause the bolts to rust. It is also less rigid than floor-and-ceiling-mounted systems. The floor-mounted/overhead-braced system features a continuous overhead metal brace at approximately 7 ft off the ground. While this system is also quite rigid, a common issue with it is vandalism, occurring when people hang from its metal bar. Anti-grip profiles are available to address this issue.

Urinal Screens



Paruresis (shy bladder syndrome) can impede the quick turnover of restroom occupants. To help alleviate this issue, urinal screens should be installed that are the same height as stall

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partition panels to create a sense of privacy as well as provide visual continuity to the rest of the space. If the recommended floor-and-ceiling-mounted stall partition system is being used, the same pilasters should be used at the urinal screens for extra strength and design consistency.

Stall Doors

Stall doors come in a variety of materials for different intensities of use. Stall doors are typically made of the same materials as stall partition panels. However, partition materials like quartz and granite should not be used for stall doors because they are too heavy for commercially available hardware, and a swinging door made from these materials could cause bodily harm to a person hit by it. For stone partition materials, stainless-steel doors are recommended. Stainless steel is available in a variety of finishes and textures. It is important that the restroom team provide a careful assessment of these alternatives as some finishes hide fingerprints well but may be more time consuming to clean.

Stall partition hardware often comes as part of a manufacturer's package. This hardware is often the weak link in stall systems in terms of aesthetics, privacy, and ease of use. Oftentimes, manufacturer hardware causes a gap to occur between panels, doors, and walls that can be up to an inch wide. This essentially leaves the stall occupant on display from the front and the sides. Continuous brackets are available that can cover the length of the gap, but they tend to look industrial, with exposed fastener heads. Latches typically have small knobs that are difficult to operate for those with limited manual dexterity. Ideally, a person would be able to unlatch the door with their wrist in a single motion.

Occupancy indicators for toilets are increasing in popularity. Traditionally, these have been part of the door latch system and have come in versions that say "open"/"closed" and/or include red and green panels. More recently, a red/green light mounted above a stall more visibly serves the same function. This system operates using wireless technology that allows a sensor to activate the light when the latch is engaged. This alternative is still developing and seems to have promise. However, this system shares a drawback with the manual red/green indicators: those who are color blind cannot distinguish between red and green. While color-blind users make up a small percentage of users (primarily male), a system that excludes people who are color blind does not align with the principle of universal design.

One low-tech solution to indicate stall occupancy is to install out-swinging doors with adjustable hinges that, when not latched, close the door to a position a few inches from completely shut. This allows potential users to see that the stall is unoccupied. Additionally, doors should always swing out of a stall to prevent occupants from having to climb over luggage to open or close the door.

Counters/Backsplash

See Appendix A for component comparisons.



Only a few surface materials are appropriate for counters and backsplashes due to the heavy use of these components, potential graffiti, and chemical reactions with water and other liquids such as fingernail polish and remover. Stone, quartz, and solid surface materials are the most durable for these restroom conditions. Softer stones, such as marble and limestone, are not recommended due to their tendency to crack, chip, break, and absorb moisture and stains.

Historically, restrooms have had countertops with cutouts for the sinks below. A common traveler complaint is that countertops are always wet from water that is dripped from the sink to the hand-drying station (See Chapter 2), leaving no dry place for belongings. A better configuration is to eliminate the counter (one less surface to clean) and provide wall-mounted sinks. A raised shelf behind the sink can provide a dry place for belongings.

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Room Doors

See Appendix A for component comparisons.

Airport restrooms rarely have entrance doors anymore. The constant opening and closing wear out both the doors and hardware quickly and impede traffic flow, especially for those travelers with luggage in tow. An open, labyrinth-type configuration with efficient two-way traffic is more common now.

Doors are still recommended for smaller rooms like companion care restrooms and pipe chases, but finishes for these doors should be durable and, to prevent collisions, these doors should not swing into public corridors. An automatic door operator is recommended on single-use restrooms, like the companion care restroom, to provide a more accessible and touchless experience. The automatic door operator also eliminates some of the ADA clearances at doors, since the door operator is placed out of the space of the swinging door. This is helpful when space is limited. A suggested hardware configuration would include the following:

- An automatic door opener and closer on both sides
- A wall-mounted, automatic, locking button on the inside (the interior door opener unlocks the door as does manually turning the lever)
- Two wall-mounted, light-emitting-diode (LED) indicators ("open" and "in use") on both sides of the door

Stainless-steel doors and frames hold up well for access to these locations. This finish also ties in well with the stainless-steel accessories in the space and stalls/doors (if used). Non-public doors such as doors to utility spaces are typically painted steel to blend with the adjacent wall finish. However, a kick plate should be provided to avoid marking and chipping door paint (which in turn requires refinishing) from cart traffic. Installing a durable plastic kick plate that blends with the door color is recommended as opposed to stainless steel, which can mar the aesthetic of public spaces (e.g., concourses) with its industrial appearance.

Wall Protection

In airports, corners are vulnerable to damage from traveler's belongings and maintenance carts and equipment, especially in restrooms where it is crowded, and space is limited for maneuvering. Even the hardest stone materials are susceptible to damage along the edges. Manufactured stainless-steel profiles are available for most thicknesses of tile; however, once materials get thicker than ⅜-inch, these trim pieces can also be easily dented. For thicker materials, consider custom corner guards made of a thicker gauge stainless-steel tubing with tabs welded to the non-visible side for securing to the substrate.

4.3.3 Restroom Accessories

Most restroom accessories include dispensers and receptacles. Larger manufacturers of restroom accessories offer models for most of the functions discussed in this section. Ideally, accessories should be procured from the same manufacturer, and all accessories should feature the same style (see Section 4.2). This provides continuity of design throughout the space and simplifies long-term maintenance, especially when all the dispenser locks can be accessed via a single key.

With so many separate accessories scattered throughout a restroom, the room risks appearing disorganized and cluttered. Careful consideration should be given to organizing components on the walls. Grouping accessories within the stalls on one wall is a strategy that has the added benefit of through-bolting the same accessories in the adjacent stall on the opposite side of the partition, providing a more durable installation. It also creates left-handed and right-handed

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configurations to provide equitable access. In addition, accessories in restrooms should be located consistently throughout the entire airport. This is particularly helpful for the visually impaired. With consistent placement of accessories, they will know, for example, that the toilet paper dispenser is always in the same place within each stall.

Mirrors

To reduce dwell time, some airports have opted not to place mirrors at the sinks and instead provide a mirror or two (including a full-length mirror) on a blank wall for grooming and last-minute clothing checks. Availability and placement of mirrors are also important for travelers with stomas so that they can see to replace the collection device. A mirror is recommended in a "stoma-friendly stall," even if a sink is not available, to allow the person to see their abdomen. As noted in Chapter 2, full-height mirrors can be disorienting for persons with dementia as they think the mirror is a hallway and walk into it. Some airports provide a sign above the mirror that says NOT AN EXIT.



Regardless of their location, mirrors should be tempered so that if they are broken they do not result in sharp shards that can be used as weapons. The downside to this is that the current manufacturing process for tempering distorts the clarity of larger-size mirrors. This issue will likely disappear in the coming years as the manufacturing process improves. The silvering behind mirrors is also vulnerable to damage by cleaning solvents that contain ammonia, so the edges and back should be sealed and protected from any moisture penetration.



The current ANSI A117.1 requires that the bottom of a reflective surface must be no higher than 40 inches above the finished floor or sinks and counters and no higher than 35 inches in other locations. However, it is important to check local codes for height requirements as well. A frequent design-to-construction error is that the opening height for the bottom of the mirror (the backsplash typically) is at 40 inches, thus pushing the mirror up a fraction of an inch into non-compliance. See Section 4.3.7 for information on mirrors with integral lighting.

Paper Towel Dispenser

See Appendix A for component comparisons.

Airports have been gravitating away from folded paper towels toward more economical and eco-friendly paper towel rolls. Touchless models are the most sanitary. Note that touchless does not necessarily mean powered; if only the paper pulled out is touched, the model is hygienically sound. Sensor-operated dispensers, however, are usable by persons with almost any mobility or dexterity limitation.

More models are being offered with a stainless-steel finish, which is more durable than other materials and blends well with other accessories in the restroom that are typically stainless-steel. Stainless-steel enclosures that cover plastic dispensers are also available; however, they may cover the sensor on automatic models.

Sink Node Waste Receptacle

See Appendix A for component comparisons.



The waste receptacle is one of the primary contributors to travelers' perceptions of a restroom as "unclean." This perception arises from both overflowing trash—typically paper towels, but also food, diapers, etc.—and the large size of trash cans that visually dominate the restroom space. To eliminate both conditions, a common waste disposal method in airport restrooms today consists of an opening in the counter with a trash container underneath in a secured enclosure. Typically, there is one waste opening between each pair of sinks. Locating

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Paper

Curiously, one of the big drivers in airport restroom design is paper, primarily the stock for paper towel and toilet paper dispensers. There are many variables to consider: paper thickness, number of plies, folded or rolls, regular or jumbo. There can be significant long-term cost savings with adjustments to any of these variables. Focusing on the cost factor alone, however, can ultimately compromise customer service and maintenance.

The vendors that supply the paper typically provide dispensers at no or low cost; these dispensers are often plastic and not the most durable product. While the vendor will replace broken dispensers, until that happens, travelers encounter a dispenser that does not work or perhaps has a big crack on its face. That experience, within the short time that travelers are in the airport, will leave a long-lasting impression that may affect their future choices in layovers and destinations.

The governing authorities of some airports require that bidding occur as frequently as every year for vendors. This can create a maintenance nightmare for the airport if the vendors change every year or two. One airport interviewed in the case studies, whose vendors did not provide dispensers, complained that every new vendor had a "new and improved" paper delivery system resulting in a half-dozen dispenser models throughout the airport that required their own sets of spare parts and more space to stock the different types of paper.

Restrooms are among the most expensive spaces per square foot. The restroom team spends months planning and designing a space that is welcoming, calming, and memorable.

So why leave the intended experience to chance by waiting to see which vendor is awarded the annual bid and what dispenser they will provide? What are the odds that the red plastic cover on the paper towel dispensers will complement the elegant restroom interior?

One-ply toilet paper versus two-ply is another ongoing, sometimes heated, debate in the industry. Some say that people will use less toilet paper with more plies. Others counter that people pull out their preferred amount regardless of waste so the thinner product may as well be provided. The answer is elusive, but one clue lies with an airport interviewed for this guidebook. It provides staff restrooms with two-ply paper and public restrooms with one-ply.

the trash under the counter efficiently takes advantage of otherwise unused space. It also creates an opportunity for additional storage for soap refills, cleaning supplies, paper products, and so forth.

Traditional surface-mounted and recessed waste units can be combined with a paper towel dispenser directly above them, which saves wall space and eliminates wall clutter. Stainless-steel waste containers should be lined to make cleaning easy and prevent corrosion. While a flipper door helps conceal the trash inside, the door becomes a germ farm, and smears on the door can be quite unsightly and repellent.

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Hand Dryer

See Appendix A for component comparisons.

Hand dryers have evolved significantly in recent years. Attention to the disposition of water blown off hands, energy use, accessibility, noise, and hygiene has fostered design innovation in mounting methods, air movement, and refinements to improve the user's experience. All hand dryer styles are available touchless, and these are recommended. Most hand dryers are wall units although there are faucets available with integrated hand dryers.

Issues remain regarding hand dryer noise levels, which can be especially troublesome for some hearing aid users. One participant in the research team's focus groups commented that the noise frequency can leave her "deaf" for several minutes. Noise from hand dryers is also upsetting to persons with sensory processing disorders, autism, and dementia. In companion care restrooms, one should opt for a quieter solution: paper towels. See the "Paper Towels vs. Hand Dryer" text box (Chapter 2) for further information on hand dryers versus paper towels.

Biohazard Disposal

See Appendix A for component comparisons.

The familiar red plastic disposal for needles is a mainstay of airport restrooms. For those requiring regular injections, such as insulin, the receptacles are vital for safe disposal. In some airports, however, primarily on the landside (non-secure) side of the terminal, these disposals are also a target of vandalism as needles are stolen for further use. Thus, durability is as important as visibility, location, and aesthetics, with stainless-steel enclosures being preferred to plastic ones. The full container is typically replaced by maintenance with an empty one and collected by the airport's biohazard vendor.

Toilet Paper Dispenser

See Appendix A for component comparisons.

Like paper towel dispensers, toilet paper dispensers are often provided by the paper vendor. Within the stall, however, plastic models are especially vulnerable to damage from carry-on luggage. Stainless-steel models are available for both standard and jumbo rolls.



Larger rolls of toilet paper reduce refill frequency but take up more space. To alleviate this, the rolls are often turned sideways. This arrangement, however, can be difficult to access for those with limited motion in their hands or arms. Also, when the dispenser is mounted below grab bars, the toilet paper comes out at an uncomfortably low level. The location of the paper outlet is narrowly defined in ANSI A117.1 as are other accessories found in combination units. It is important to check for modifications in the local code.



Multiple-roll models are commonplace and highly recommended. Coreless rolls are also available that reduce waste but can be difficult to install correctly.

Combination units that combine the toilet paper dispenser with a waste receptacle or a seat paper dispenser help reduce the clutter of accessories within the stall. Through-partition dispensers that have access from adjacent stalls reduce the projection of the dispenser into the stall. Recessed models protrude even less but require thickened partitions instead of panels.

Stall Waste Receptacle

See Appendix A for component comparisons.



Traditionally, waste receptacles have been provided only in women's stalls for sanitary products. With the increase of elderly travelers and people with disabilities, providing a waste

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receptacle for adult diapers or stoma collection devices supports the dignity of the user and provides a receptacle for general trash instead of the toilet. Receptacles that are big enough to accommodate an extra-large, fully absorbed men's diaper should be used.

Seat Paper Dispenser

See Appendix A for component comparisons.

No matter how well a restroom is maintained, some people are simply wary of unseen germs and are uncomfortable sitting exposed on a surface where only a minute before someone else sat. Sometimes, the toilet seat is visibly soiled, and no other stalls are available. Seat paper dispensers provide a flushable cover for users to sit on rather than hover. The dispenser should be installed either behind or to the side of the toilet. This unit is also available combined with a toilet paper dispenser. Again, a stainless-steel enclosure is more durable than plastic.

There are also automatic models available. When activated, a continuous clear plastic covering slides around over the toilet seat. The used portion is automatically spooled into a container for later disposal.

Shelf

See Appendix A for component comparisons.



Travelers are always carrying something—a bag, jacket, and/or food. Even airport employees often have a planner, folders, or another item in hand. There is always a need for a clean, dry, stoma-friendly place to put belongings in restrooms. Counters in the sink node are typically wet from splashing and from drips to the hand-drying station, and floors are usually equally suspect. Shelves at the sink, in the stalls, and at the urinals have become a mandatory customer expectation.

Preferred are built-in shelves integrated with the wall materials. These may be a recessed niche at the toilets and urinals or a continuous shelf behind the lavatories. The shelves should be 8 to 12 inches deep and above wet surfaces. Accessibility is important, even for the agile. Shelves behind sinks should be within easy reach (see Chapter 2 for recommended counter depths). Shelves in stalls at and near urinals should not be placed behind toilets and urinals. Such locations are typically awkward to reach (and not accessible), and there is the hazard of dropping something into the fixture.

Where budget is limited, shelves can be surface mounted, but not where they would create a projection that someone could bump into. For space-constrained locations, stainless-steel fold-down shelves are an option.

Hooks



Hooks are useful in the toilet stall and at urinals and sinks. Travelers often have a coat, bag, or purse they would rather not set on the floor due to concerns about cleanliness, theft, or both. Typically, two hooks are provided, one above the other. The lower hook, at 4 ft off the floor, meets accessible reach requirements. A higher one at 5 ft or more will accommodate a long coat.

Within stalls, hooks should never be mounted on the back side of the door, as a common theft technique is to reach over the top while passing by and lift a purse or bag. It is preferred that hooks are located closer to the toilet to access such things as sanitary products or injection materials for diabetics. A hook can also be installed on higher urinal screens, at the sink on a side wall or hand-drying column, and in the baby diaper changing node for diaper bags.

Hooks should be a strong metal with secure mounting. There should be no sharp edges. They should project enough to accommodate a wide bag strap and hook upward to hold a garment.

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Sanitary Products Vendor

This coin-operated dispenser can be set for different denominations. Most airports charge a nominal price (typically 25 cents). Experience has shown that if the products are free, people will stock up. Newer models have a push button rather than a lever, so it is accessible as well as less likely to break. These can be recessed (preferred) or surface mounted. One unit within a restroom is typically enough. More may be needed in larger restrooms or if the restroom is divided into separate areas.

Baby Diaper Changing Table

See Appendix A for component comparisons.

Durability is a key requirement for the changing table to ensure the safety of infants and to hold unintended loads like luggage or sitting adults. Changing tables can be purchased as a manufactured unit that is mounted to the wall and folds down or can be custom made as part of the counters. There are also hybrid models that consist of a manufactured changing platform that is secured to the countertop.

The materials used should be antibacterial and easy to clean. Stone materials tend to be cold to the touch, so using a pad on top is more comfortable for babies. Most users will have a changing pad in their diaper bag, so providing a low enclosure with a strap that will keep the baby and pad in place is recommended.

Child Step

Some airports provide a child step at one of the sink locations, which can also accommodate adults of short stature. While this is a worthy accommodation, it will make that sink location inaccessible to wheelchairs (which is not too much of a problem if all the other sink locations are accessible). They can also be a cleaning nuisance because the typical models sit on a frame of stainless-steel angles that are bolted to the floor and are cumbersome to clean around.

Baby Diaper Kit Vendor

Most people traveling with babies are stocked with diapers, wipes, changing pad, and so forth; however, sometimes a convenient dispenser can save the day if a traveler's resources become depleted. Providing one dispenser at each restroom block, ideally near the companion care restroom, should be adequate, although one can also be installed at each changing table. The dispensers are coin operated and typically provide a diaper, wipes, a paper liner, and a disposable bag. As with sanitary products, nominally charging for the products ensures a more stable supply.

4.3.4 Plumbing

Plumbing design is critical to the restroom. It needs to be well thought out, organized, clean, and functional. Each restroom should have its own set of shut-off valves to isolate the restroom block.

In a new airport, a dedicated tempered water system should be used that can serve all the hand sinks in the building. This system will always allow for warm water at the sinks; the comfortable water temperature for handwashing is 105 degrees Fahrenheit.

When piping within the pipe chase is being designed, the pipe elevations should be drawn on the construction documents and organized to all fit into the chase. Ideally, the waste piping should slope toward the chase door, and a cleanout should be provided above the spill line to allow for easy access and the ability to clean out that piping as needed. If the cleanout is in the

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back, the maintenance staff must take their equipment to the back of the chase. Enough space should be provided at the chase entry to allow the door to close when someone is working at that end. This conceals maintenance personnel from public view.

In existing buildings, the slope of the existing sanitary mains will be critical in selecting fixtures. The slope in older plumbing systems is minimal and thus it is advantageous to have more water running through the piping. Low-flow fixtures should be evaluated before installing (see "Toilet" and "Urinal" sections).

Sink

See Appendix A for component comparisons.

There are two options for hanging a sink: a concealed arm carrier or safety clips. A concealed arm carrier is much sturdier and safer and therefore requires a structural wall behind the sink to secure the carrier and piping. The clip option requires the proper backing to be on the wall to ensure structural integrity. A single cantilevered sink should be designed to carry a 150-lb load and a double should be designed to carry a 300-lb load.

Each sink should have a shroud that can be easily removed to conceal the pipes, valves, and so forth, ideally by simple lifting (screws and fasteners should be avoided because they take extra time and get lost). The shroud should accommodate the required knee- and toe-space clearances. It is recommended to provide a trap and tailpiece composed of 17-gauge piping. If a shroud is not provided, trap and piping guards should be installed per the local accessible building code.

Sink strainers should be made of stainless steel to prevent corrosion and discoloration. Additionally, strainers should have large holes to prevent clogging and be in a flat area large enough to use a small plunger to loosen clogs. Several trough sink basins are designed with the forward surface sharply angled toward a vertical back wall. This creates a condition where there is only a narrow flat section on which to locate the strainer. The strainers in this style of trough sink are often less than an inch in width. The narrow strainers combined with the angled geometry of the trough basins can make it difficult to service the sink with a small plunger. Often, the sink and associated drainage piping will have to be dismantled to be unclogged.

Providing a cleanout above the spill line of the sink will allow the maintenance staff to clean the piping without causing an overflow. This cleanout is only possible when a sink has a chase behind it. Even without a chase, a cleanout should be provided below each sink.

For wall-mounted sinks, it is recommended to have a gap between the sink and the wall. This makes for one less maintenance-prone sealant joint and allows another option for spills to drain off the flat surfaces where users want to place their belongings.

Adjustable-height sinks are available (and more common in other countries). This can be beneficial, especially in companion care and changing table restrooms. A cleanliness and hygiene concern is that these typically use a flexible drain hose, which may not drain properly (and may not meet building code requirements).

Faucet

See Appendix A for component comparisons.

Today's market has ever-changing faucet designs and features to consider. Features that should be considered first and foremost are water capacity and power consumption. Given the usage in airport restrooms, it is important that an adequate water supply, especially hot water, reach the faucet.

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Energy savings are achievable with innovations like solar- and turbine-powered faucet sensors. Touchless faucets are both more hygienic and are preferred for accessibility. Proximity sensors, a recent innovation, turn on the faucet by sensing the heat from a person's hands rather than motion from fingers placed near the faucet. This prevents accidental activation.

The spout shape should allow water to flow high and far enough out over the basin so the user's hands do not touch the sink bowl. If possible, a mock-up of the sinks and faucets being considered should be set up to test for ease of activation, comfort, and unexpected splashing. An existing sink is a great place to test out devices.



Passengers expect warm water right away at the sink. The system design should bring the hot water main as close to the sink as possible. Section 606.4 of ADA 2010 Standards requires that water flow out of faucets for a minimum of 10 seconds.

Braided supply lines used in the installation of faucets can collect dirt and dust over time. Chrome-plated brass supply tubes should be used to avoid build-up. It is also recommended to provide quarter-turn ball valves under sinks instead of the multi-turn type for shut-off directly below a sink. If no shroud is provided on a sink, provide loose-key, quarter-turn ball valves to help prevent users from shutting off the water.

Soap Dispenser

See Appendix A for component comparisons.

One of two types of soap is typically used in restrooms: foam soap or liquid soap. Both are available in bulk and usually bought in bulk through vendors. However, the advantages of foam soap are that smaller amounts are used, and it tends not to clog the sink. Foam soap washes through piping more easily and removes bacteria from hands as effectively as, if not better than, liquid soap.

Maintenance staff are typically stretched too thin to keep up with the busyness of airport restrooms. Utilizing a bulk soap-dispensing system that is accessible for refilling from the plumbing chase is ideal because it extends the period between refills. An advantage of a bulk soap system is that it can feed multiple dispensers. One should use a conduit to run the tubes to make sure they are not inadvertently cut, and the tubes should be run no higher than the dispenser output height to avoid drips.

Traditional bottle dispensers are often stolen from underneath sinks if they are visible, which they usually are. The half-empty plastic bottles of pink liquid hanging below the counters are also an unsightly visual distraction.

Toilet

See Appendix A for component comparisons.

The main difference in toilet designs is the flushing system: siphon jet versus blowout. The siphon jet utilizes a jet of water in the trapway to pull water out of the bowl. The blowout method displaces all the liquid (and its contents) from the toilet bowl. It is a more aggressive means of flushing but is more effective at removing solids, which is important for high-use restrooms like those found in airports. The drawback of the blowout flushing system is that it is startlingly loud, and the repeated sound of flushing can be heard in a holdroom if an open restroom entrance is adjacent to it. Further, a sudden loud sound can be problematic for individuals with autism or sensory processing disorder and those with hearing aids.

Wall-hung toilets are the best choice for airports; however, with the increasing weight of travelers, it is important to be aware of the load limits toilets can carry. In the United States,

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the minimum weight requirement for a wall-hung toilet is 500 pounds. With proper selection, the weight limit on a wall-hung toilet can be up to 1,000 pounds. For this kind of weight, it is critical that the carrier and the associated wall structure have the proper capacity. In the case studies conducted for this research, more than one story was told of a person sitting on a toilet that shared a back-to-back carrier with the toilet on the opposite side of the wall being lifted into the air like a seesaw when a heavier person sat on the toilet on the opposite side of the wall.

It is important to provide cleanouts on the waste piping above the toilet spill line to allow for ease of maintenance. Carriers should be installed as flush to the chase wall as possible to allow for more space in the restroom. Water hammer arrestors should be provided in an accessible place in each restroom. Toilet seats should have all-stainless-steel bolts and springs (no plastic or zinc parts) and should feature a soft close to prevent seats from slamming down and potentially breaking. Plumbing codes typically do not permit solid lids over the seat in commercial restrooms.

Water conservation is a critical function in a smart building. Capturing gray water (water from sinks or rainwater) and reusing it in toilets and urinals is a great way to conserve and reuse relatively clean water. Nonetheless, these systems are complex. The water needs to be properly processed as well as maintained and cleaned to prevent bacteria from traveling through the systems. The system will also require separate drainage for collection and a separate distribution system to supply it to the necessary fixtures. Gray water can be cleaned both naturally and chemically.

Urinal

See Appendix A for component comparisons.

Urinals are a staple in men's restrooms in the United States. They are very functional and expedient in getting users in and out. When a urinal is selected, it is important to consider aesthetics because these fixtures are very visible. Their appearance should complement the design of the room.



Within the urinal, it is important to consider flow. Waterless urinals have not been extremely successful, primarily due to installation and maintenance issues. For pint or other low-flow urinals, a sink or toilet should be upstream on the waste piping of this fixture. This will help keep water flowing and help prevent pipes from deteriorating due to the effects of uric acid.

The trap discharge should also be considered. Many urinals have a lip at the discharge of a trap that can make them difficult to clean. Most urinals today are mounted on factory-provided mounting brackets that seem to work well in hanging the units securely. Adequate space should be provided around a urinal for a person's belongings. A shelf and a coat hook are also welcome provisions. Ensuring 3 ft of clear width between the urinal dividers (as recommended for ambulatory toilet stalls) provides adequate room on the floor for a roller bag as well.



A long overdue development in urinal design is taller urinal options. Floor-mounted urinals that are set in the slab are a thing of the past due to pipe access from the floor level below. However, there is no reason urinals cannot all be low enough to accommodate persons of short stature, persons of tall stature, and persons with disabilities. The typical men's restroom has a row of urinals mounted at "standard" height and one urinal on the end that is mounted at a lower height. From a universal design perspective, they should all be at the same height so that one is not highlighted as a fixture for those with special needs. Furthermore, the easier it is for the user to reach the target, the better.

Some airports are using actual targets in urinals to prevent extra cleaning related to aiming or the lack thereof. These can be playful such as small, printed stickers (like a fly) or a custom

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design. Also available from some manufacturers is an embedded stainless-steel ball or an image applied to the porcelain prior to glazing.

Toilet and Urinal Flush Valves

See Appendix A for component comparisons.

Flush valve selection decisions should consider type, location, and utility savings. Most travelers are looking for a touchless environment, so an automatic flush valve is recommended. Low-flow water solutions as well as solar- and turbine-powered valves are available for water and power savings.

Based on the restroom layout, a decision must be made on whether to use a concealed or exposed flush valve. This decision will likely be driven by chase access for maintenance. If there is a chase behind the fixtures, a concealed flush valve has a much simpler look. Also, it can be accessed without the maintenance staff needing to close the restroom to perform their work. This design is also easier to clean than exposed pipe and fittings.

Drinking Fountain

See Appendix A for component comparisons.

Long days of traveling can be dehydrating, so providing drinking fountains is important for the well-being of travelers. While the International Building Code requires only one drinking fountain per 1,000 occupants in a passenger terminal, it is good practice to provide one at each restroom set, in part, because people generally associate drinking fountains with restrooms when seeking them out. There are numerous aesthetic options for drinking fountains. Look for accessible models (bi-level models) that allow a front approach, are easy to clean, and complement the aesthetics of the restroom façade.

It is recommended that the drinking fountain(s) be in an alcove off of the hallway. This avoids having the person at the fountain bent out into the traffic flow. It also keeps their bags more protected if they are temporarily set on the floor. Drinking fountains are not permitted to project more than 4 inches past the primary corridor wall, per ANSI A117.1. This protects persons with a visual disability who use a cane from walking into the side of the fixture. For locations where space is not available to adequately recess the drinking fountain or for retrofits, some manufacturers make cane guards for newer models. The cane guards are inexpensive and can be easily attached under the high fountain to bring the fixture bottom down to the 27-inch maximum height off of the floor.

Filtration on drinking fountains is recommended to provide clean and safe drinking water. The filters require maintenance in addition to the regular cleaning of the drinking fountain to remove debris collected in the drain.

Bottle-Filling Station

See Appendix A for component comparisons.



Restrictions on the amount of fluid that can be brought through airport security, growing public awareness of recycling, and the cost of buying drinks within an airport have encouraged many travelers to use personal water bottles. Bottle-filling stations mounted on drinking fountains support travelers' use of personal water bottles and have become quite popular.



The 2017 ANSI A117.1 code allows that a single water-bottle filling station can be used on a hi-lo fountain if the station is mounted on the low, wheelchair-accessible fountain. If the high and low fountains are separate units, each will require a water-bottle filling station. Filtration on bottle fillers is recommended to provide clean and safe drinking water; however, filters will

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require ongoing replacement. Automatic bottle fillers need to be cleaned regularly to help keep them dry and free of mildew. The manufacturer's recommendations on mounting for accessibility should be followed.



Some airports now provide a receptacle to dispose of liquids before travelers go through airport security so that they can take the bottle with them. The airport then provides water-bottle filling station(s) post-security. This cuts down on a significant amount of water-bottle recycling.

Mop Sink

Mop sinks are a well-used and integral part of restroom maintenance. These should be placed in easy-to-access locations within a pipe chase, providing one mop sink per set of restrooms. There should be at least 24 inches clear space on two sides to allow for both a mop bucket and a person to fill the bucket.

Given the hard use of mop sinks, they should be made of impact-resistant composite fiber-reinforced plastic with an integrally molded drain along with a factory gasket and a removable stainless-steel strainer. A stainless-steel strainer will help prevent corrosion over time and unwanted material from getting down the piping.

Bumper guards (vinyl) on the edges of the mop sink will help prevent chipping and wear. A splash guard made of stainless steel stretching 18 inches up the back wall of the mop sink will prevent dirty and corrosive water from getting on the wall. In addition, providing a hose, hose hanger, and mop hanger will help keep the area surrounding the mop sink organized and clean.



If a mop sink is at the end of a branch, be sure to provide a cleanout per the local plumbing code dedicated to the mop sink. The trap of the mop sink is not considered a removable trap and thus a cleanout is needed. Ideally, a wall cleanout would be provided on the back wall behind the mop sink. If this is not possible, a floor cleanout should be provided in the same room as the mop sink. Either way, it is important to ensure the cleanout is full size. Consider making the back wall 6 inches thick to allow enough room for the cleanout.

Mop Sink Faucet

While a faucet may seem like a simple, necessary accessory to the mop sink, there are many factors to consider. The faucet should have both a hot-water and cold-water supply. On each of these, there should be check valves (integral or separate). Other must-haves include a support arm, an atmospheric vacuum breaker, integral shutoffs, a pail hook, a chrome finish, and lever handles.

Typically, the faucet is installed approximately 48 inches above the finished floor or as called for in the manufacturer's recommendations.

If a third-party chemical dispenser is used, which is often an aftermarket item, a pressure-bleeding device should be provided on the hose connection, and the dispenser should be tested to ASSE 1052. Otherwise, there is a risk over time of back-leaking soap into the water system.

Hose Bibb

A hose bibb is a necessary convenience for the maintenance of airport restrooms. Whether a pipe needs to be flushed, a janitor needs additional water somewhere, or the cleaning process includes a periodic hose-down of the entire restroom, having a hose bibb in an adjacent pipe chase or janitor's closet is vital. Locate the hose bibb near the chase door and ensure the hose-bib has an atmospheric vacuum breaker (either integral or separate) to prevent backflow into the system.

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4.3.5 Heating, Ventilating, and Air Conditioning

Providing HVAC to the restroom is critical to creating the perception of cleanliness and managing odors. With the right amount of ventilation, restrooms will smell better and users will enjoy the experience more. It will also allow for the elimination of air fresheners, which can often be overpowering, bothersome to people with allergies and sensory processing disorders, and a maintenance hassle.

Radiation

Finned-tube radiation is a passive heating method used along exterior walls for both heating and reducing the formation of condensation on windows and cold surfaces. Radiation covers should complement the overall restroom aesthetic. Stainless steel is a common choice as it is durable, easy to form, and matches other stainless-steel elements in the space. Painted metal is easily scratched, can rust, and is therefore not recommended as a cover choice. In addition, covers with flat tops are to be avoided because people tend to sit, place luggage, or step on them. Consider a pointed or sloped top to provide additional strength and prevent dust collection. Make sure the surface is easy to clean if it is textured.

Air Supply

Most building codes do not require supply air in restrooms because they are not occupied spaces (they are considered transient). Providing supply air, however, is good practice. It makes sense to provide supply air below sinks, given that the area surrounding the sinks is often wet. This will help dry the wet surfaces and keep the area around the sinks at a comfortable temperature.



The space under the counter or sink is conducive to a linear diffuser for this purpose. For a clean look, run the diffuser from wall to wall. This will also allow for an equal distribution of air throughout the space. The ductwork, located in the chase behind the sink, will need to be coordinated with piping to ensure ample space for both. The ductwork will also have to be coordinated with the plan layout as the chase will need space for the elbow from the diffuser to the duct and a clear vertical run (usually a minimum of 8 inches) behind the sink and mirrors for the insulated duct.

If space is limited, at the very least one should provide supply air above the sinks to promote air mixing and eliminate short cycling of the air. Short cycling is when the air comes directly out of the supply diffuser and is sucked right back out through the exhaust without having time to condition the space.

It is also recommended that supply air be provided in single-fixture restrooms, like companion care restrooms. Where supplying air below the sink is not feasible, an overhead diffuser is better than no diffuser.

Exhaust



Without restroom exhaust, restrooms would be unpleasant places with lingering disagreeable odors. Each restroom should have a dedicated restroom exhaust system with diffusers located directly above the toilets. Diffusers can be dedicated to each stall or can be one linear run above all the stalls. At a minimum, the code-required exhaust should be provided in the restroom. Whenever possible, provide an occupied and unoccupied setting by using the spaces' occupancy sensor (see "Technology" section) and modulation of the fan. The code-required minimum should be the unoccupied setting; the occupied setting should be one and one-half times that. If the restroom has a restroom management system (RMS), as described in Section 4.3.8, the exhaust can be connected to provide a cleaning cycle that will be set for another one

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and one-half times the occupied setting. The cleaning cycle will allow additional air changes to remove cleaning chemical smells and help to dry floors that are wet from cleaning more quickly. This can be accomplished with either a standard exhaust fan with a variable frequency drive or an electronically commutated motor.

If the exhaust fan is not located in the pipe chase, be sure to provide an access panel on each side of the fan to allow for duct cleaning because, over time, toilet paper dust builds up substantially in ducts and fans.

Diffusers

The aesthetics and layout of the space should be considered when selecting diffusers. Be sure to verify the diffuser finish with the architect. Placement of diffusers is explained in the exhaust and supply sections. Fan noise is not a major concern in restrooms since there is so much noise that a little white noise is often welcome. Squealing noise, however, should be dampened.

Thermostat

Locate the restroom thermostat in an inconspicuous location to avoid having it tampered with. A ceiling-mounted thermostat or a wall thermostat with no display is the best type to consider. These types can only be installed in spaces with a building automation system, common in larger airports.

4.3.6 Electrical



Except for use in maintenance or cleaning, receptacles for power are often overlooked in restrooms. Outlets for convenience power are a great amenity for travelers. The best locations for outlets are at sink mirrors/shelves and make-up/grooming counters to allow for a quick shave, etc. Utilitarian functions that require power include automatic sensors on plumbing fixtures, soap dispensers, hand dryers, RMS components, and tools in plumbing chases.

All receptacles near moisture need to be the ground-fault circuit interrupter (GFCI) type. There are two methods of ground-fault protection: at the circuit breaker and the individual device. GFCI circuit breakers have better sensors and are less likely to nuisance trip, but they typically cost more and require the breaker to be reset, typically by facility staff. GFCI receptacles are a bit more cost-effective and allow the user to reset the device, but they do trip more often. Refer to the National Electrical Code and local codes for exact ground-fault protection requirements.

4.3.7 Lighting



Of all the building systems, lighting can have the most dramatic effect on a space. In restrooms especially, lighting affects people's perceptions of everything from cleanliness to noise levels. Bright lights can be jarring and make travelers appear garish. Muted lights are calming and can make the room feel "quiet," a welcome benefit considering the whirring of hand dryers and flushing toilets. Lighting that is too dark or a space filled with deep shadows can be potentially hazardous as occupants bump through the space. Lighting that is too dark can also make a person feel unsafe and the space seem dirty.

The color temperature of lighting, measured in kelvin (K), is also a critical consideration. Warm white lights are in the 2,700 K to 3,000 K range. This coloration is like incandescent lamps. Cool white lights in the 3,500 K to 4,100 K range appear bright and are good for detailed tasks. Natural daylight is in the 5,000 K to 6,500 K range and is the coolest colored light. Cool lighting can create a bluish cast in a space. In the mirror, this can make a person appear as if they have

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traveled for days rather than hours. Warmer color temperatures feel welcoming and soft. Colors and finish materials under consideration should always be reviewed under lighting that has the same color temperature as the lighting that will be used in the restroom space.

General (Ambient) Lighting

Ambient lighting fills the space as opposed to task lighting, which highlights areas of specific activity. Ambient light levels should be bright and even in order to make the space feel clean and airy, not dark and dismal. This lighting type should provide approximately 5 foot-candles (FC) in the walk zones for safety, security, and cleaning.

Task Lighting

The following locations have different task lighting needs:

Entry. Lighting at the entries should accent signage that designates wayfinding, art, or other pertinent information. Lighting levels should be set not so bright (or so dim) that users have a hard time visually transitioning from the adjacent (typically bright) area into or out of the restroom.

Sink Node. Lighting at the sink should be oriented to provide even vertical illumination (from the sides) onto the face, but somewhat softly to avoid glare in the eyes. Lighting from above the mirror can supplement the vertical lighting but should not be the sole source as it casts shadows that make a person's face look tired and gaunt. Vertical lighting should be evenly balanced on both sides of the face. Lighting integrated within mirrors works well to keep the walls clear of clutter and surfaces that collect dust or are hard to clean. Recommended light levels at the sink node are 15 FC horizontal and 20 FC vertical.

Toilet Stalls. Lighting at the utility wall or over the toilet stalls will offer a clean and crisp appearance. Providing a linear slot or a light panel along the toilet wall or downlights within each stall will assist in evenly lighting all locations. The recommended light level is 15 FC. Lighting from above and behind the toilet can cast the stall in shadow when a person is finished and stands. Consider providing light from the walk aisle or adding a small downlight between the toilet and door.

Switch/Occupancy Sensor



Occupancy sensors provide energy savings by switching off lighting when no one is present. Occupancy sensors can also be used more creatively, such as tying them into mechanical systems to increase ventilation when the space is occupied.

Care needs to be exercised when selecting types of sensors for restrooms so that the lights do not go off when people are present. Line-of-sight sensors (for example, infrared) should not be used in restrooms with partitions or walls that shield the sensor's view of the space. Ultrasonic sensors placed too close to an entrance without a door may pick up people walking by and turn on lights when not needed. Dual technology sensors cost more but can "see" around walls and partitions, making them the best choice for larger restrooms. The time and sensitivity on devices should be properly set to maximize energy savings and minimize "false offs."

Lamps

See Appendix A for component comparisons.

Lamp types can have a significant impact on maintenance and operational costs for any facility. Longer-life fixtures can lower maintenance and operational costs but typically have a higher initial cost.

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LED lamps have a long lamp life and good color temperatures and color rendering. The technology is changing rapidly, however, so fixtures and lamps selected now may soon be obsolete. Initial costs for LED lamps are higher than fluorescents, and the drivers are likely to fail before the lamp expires. Maintenance costs are lower as relamping is rare.

Fluorescent fixtures also have a long lamp life and good color temperatures and color rendering. However, relamping adds more cost over time, and ballasts for fluorescent fixtures need to be maintained. In addition, although compact fluorescent lamps provide more flexibility for use, they have a slight warm-up time before coming fully on. This may be bothersome in smaller restrooms where they are the only light source and an occupancy sensor is utilized to turn lights on.

Daylight



Daylight can have a positive effect on people's demeanor, especially after they have spent time in parts of the terminal where there is limited natural lighting. Many restrooms are located with at least one exterior wall. Unfortunately, these walls are usually opaque. Even curtain walls have metal panels instead of glass for privacy or a pipe chase. With a little creative planning, daylight can be pulled into a restroom, enhancing the ambiance of a calm oasis.

For instance, daylight can be borrowed from adjacent spaces, such as skylights in the concourse by providing a band of clerestory glazing. Privacy is indeed important, so a good strategy is to locate clerestory glazing up high. Equipment, such as aircraft de-icing rigs, that may pass nearby bathroom windows should be considered because equipment operators are often at a height where they can see in. Even if equipment operators cannot see in, the perception that someone can see inside the restroom may make occupants uncomfortable. Shadows at night can also be unintentionally revealing, so lighting should be placed to avoid casting shadows on the glass.

For full-height glazing, ensure the translucent white film reaches the height where modesty is no longer a concern. Thin, clear strips of the solid film can foster a connection to the outdoors without compromising privacy.

4.3.8 Technology

Some of the greatest customer service potential resides in the rapidly advancing realm of technology. Ironically, because the speed of change is so fast, incorporating cutting-edge products into restrooms may not be the most practical long-term strategy. Attempting to incorporate cutting-edge products into restrooms can create a constant state of obsolescence as newer versions of previous products and/or sensors rapidly become available. Standardization becomes nearly impossible. The key is to find a balance between innovation (especially in areas of sustainability) and long-term integration of updated products into the restroom spaces.

Fire Alarm

Fire alarm system design is based on national and local codes. However, the airport is encouraged to go above and beyond the code requirements for added protection and peace of mind. This typically means including smoke-detection coverage throughout the facility. One effective strategy is to locate manual pull stations adjacent to the restrooms as restrooms typically occur at standard intervals throughout terminals (see Section 2.5.2).



The alarm's visual strobe should be located so all occupants can see it, particularly those with hearing loss.

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Paging

Paging is used when informing travelers of flight status and making other important announcements. The volume needs to be loud enough to be intelligible, but not overpowering. In larger restrooms, ambient noise sensors might be utilized to increase/decrease the volume of pages. The paging speakers can reinforce announcements from some fire alarm systems.

Visual paging for persons with hearing loss is also a requirement of the ADA, and some airports have been sued for not providing this accommodation. With the clutter of accessories within the restroom, locating the visual paging device can be a challenge. Providing FIDS with a dedicated panel for visual paging outside the restroom (as recommended in Section 2.5.2) appears to be a reasonable accommodation; however, it has not been legally tested.

Ambient Music

Ambient music can be used to create a relaxed mood for travelers. In most facilities, the paging system can have an ambient music system input connected to it. Modest travelers appreciate the sound-masking properties of music as well.

Restroom Management System

Larger airports should consider incorporating an RMS. This system integrates several technologies, including some of the following:

- **Card reader.** Airports can leverage their existing card-reader and badge systems to implement an RMS (see Figure 4-3). When cleaning personnel present their airport security



Figure 4-3. RMS with information monitor and backlit room sign. The card reader to activate maintenance features is located behind back-painted glass below the sign.

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badge to a card reader, a screen is brought up that indicates that the restroom is being cleaned. The lighting and ventilation are then also increased through the airport's facility automation controls.

- **Occupant counter.** Airports can use restroom occupancy counters to schedule cleaning efforts based on the number of travelers that have visited a restroom. This helps the cleaning staff work more efficiently.
- **Video display.** A display unit can be used to indicate whether a restroom is open or closed for cleaning. This status can be generated on a screen that is integrated with the airport's card-reader system located within the entrance signage to a restroom. The screen should also indicate the location and direction of the nearest open restroom.
- **Sensor integration.** Product sensors for depleted paper, soap, and other products should, ideally, be connected to the RMS system to alert maintenance staff and track usage. This functionality is under development in the industry.

4.4 Components of Regulated/Essential Amenities

4.4.1 Companion Care Restroom

These spaces are typically fitted out like the main restrooms for continuity in design and to minimize the types of products, finishes, and so forth. In addition to the restroom recommendations presented in Section 4.3, consider the following amenities for companion care restrooms.

Door

Use a powered door operator with automatic locking and unlocking capabilities for touchless operation.

Mirrors

Include them at the sink and provide a full-height mirror. Position mirrors for stoma use.

Accessories

Provide a biohazard disposal receptacle and dispensers for feminine products and diapers. Consider a wall-mounted toddler seat or fold-down bench. If space permits, a child-sized toilet can speed up a visit. An emergency call button or fall-monitoring system is highly recommended. Stoma-friendly accommodations (e.g., a hook near the sink, shelf, and mirror) should also be provided.



Note that a child step is not recommended in this space as there is only one sink, and the fixed step makes the sink inaccessible for wheelchairs.

Curtain/Divider

A ceiling-hung curtain or folding screen on wheels can provide privacy if a traveler has a companion or children within the room. Hospital cubicle curtains are a good option for this use because they have the gap at the top to allow for proper fire protection sprinkler coverage. However, the material needs to be durable. If a wheeled divider is used, a little extra space will likely be needed to store it out of the way when it is not in use.

When planning to incorporate a curtain or divider in the room, consider locating the toilet close to the door for safety reasons. For instance, if the curtain is located between the toilet and door, a small child could potentially quickly open the door and head off on an expedition.

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4.4.2 Changing Table Restroom

The changing table restroom is essentially a companion care restroom that provides additional diaper changing space for any person too large to fit on a baby changing table (see Figure 4-4). For the changing table restroom, all the recommendations noted in Section 4.4.1 apply, as well as the following:

Changing Table

Often referred to as an adult changing table, this type of table also accommodates toddlers to teens. These tables are powered, and the height should be adjusted to a comfortable level for the person being changed (this is especially important if he or she self-transfers from a wheelchair without the use of a hoist) and to the right height for the person assisting with the change. There are often power buttons along the front edge of the table, and sometimes tables come with a push-button operator on an extended cord that allows the operator to move around to help the person being changed while adjusting the height. The cord is vulnerable to damage, so a wall hook should be provided to encourage users not to leave the control on the bed. While the table can be folded up against a wall when not in use, assume the down position for accessibility clearances when planning a space for the table. Tables typically will accommodate a person up to 400 lb (capacity varies by model and will likely increase as use becomes more common). California building codes require 36 inches of clearance at the head and foot ends for easy access around the table. Consider providing this clearance because it will likely be a common requirement in the future.



A feature that folds the table up automatically when a person climbs off will likely be available for consideration soon. With this type of table, a sensor should be installed to indicate through the RMS when the table is in the down position so maintenance can dispatch a cleaner.

It is recommended to have a spray hose available to rinse off the table after each use. Changing tables typically have a drain with a flexible tube connection down to a floor sink (a heavy-duty



Figure 4-4. Changing table restroom (LHR).

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floor drain). To create a clean system, do not use corrugated flexible drain tubes with internal ridges. Only smooth flexible tubing should be used for changing table drainage.

Hoist

A hoist mounted on a ceiling track (with X-Y maneuverability) is essential for assisting people with transferring from a wheelchair either to the toilet or to the changing table. Hoists typically feature a suspended "handlebar" with a hook on each end to attach a sling. The hoist is operated by a hand-held operator with buttons that enable movement in different directions as well as up and down. The operator is connected to the hoist body by a cord that can be pulled out as far as needed.

People who need to use a hoist typically bring their own sling. There are a variety of types of slings that accommodate different abilities and disabilities. Those with upper-body mobility can manage attaching the sling on their own. Those who need assistance typically travel with a companion. Airports are not expected to provide a stock of slings.

The hoist is self-powered via a low voltage strip along the moving guiderail that continually charges the hoist. If a person touches the strip, they will not feel the electrical current. Most hoists have a switch to return the hoist to an out-of-the-way dock in a corner. This switch may be a button located on the hoist or a remote button on a wall.



When planning the track and rail locations, take care to position ceiling-mounted fixtures and access panels in a location that does not conflict with the hoist movement. In addition, the hoist is a device that is prone to misuse; therefore, the restroom and amenities team should discuss liability and safety issues with the airport's legal team to plan for and prevent damage or injuries. Some airports in the United States and Canada have installed changing tables only, forgoing the installation of hoists (or hand-held showers). While this setup is not ideal, it is nonetheless a major improvement when compared to changing adult-sized family members on restroom floors.

Hand-Held Shower

A hand-held shower is another essential feature for a person being changed. If soiling is beyond the capabilities of wipes, the shower can be used to rinse the person off either on the table or if they can stand and walk, at the shower. Therefore, take care to place the shower in an open area with enough room for a person to stand without anything nearby getting damaged by water spray. The shower can also be used to rinse off the table by the maintenance staff. One good practice that some airports recommend to prevent mischief is to require a code received via phone call to operate the shower.

Diaper-Wipes Dispenser

Locate this dispenser near the changing table.

Adult-Diaper Dispenser

Locate this dispenser near the changing table.

Fold-Down Seat

A seat may help people dress themselves or be dressed by others.

Fall-Monitoring System

One high-tech option is to install an infrared sensor in the ceiling that projects an invisible grid on the floor (2-ft by 2-ft squares are recommended) to monitor whether someone has fallen

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using the changing table or hoist, has become incapacitated, has had a heart attack, and so forth. For the monitoring system, a typical standing person would show a two-square pattern on the floor. A person laying on the floor would show a larger, irregular-shaped pattern. If squares are motionless for a set period (1 minute is recommended), an alert is triggered through the sensor to notify facilities management via the building management system (BMS). A two-way intercom system will allow airport staff to ask if a person is okay. If there is no response, an emergency team can be immediately dispatched. Signage should be provided to explain to occupants how and why the space is monitored.

A simpler, but less comprehensively accommodating option than the infrared sensor is to provide an emergency call button, pull cord, or touch strip along the floor like those found in hospitals to call for help. These tend to invite mischief, so false alarms are likely to occur. Fall monitoring is highly recommended in other single-use toilet compartments including accessible stalls and companion care restrooms.

Other Signage

Be sure to provide instructions for using the change table and hoist. Post a notice that the room is monitored for fall prevention.

4.4.3 Service Animal Relief Area

While a SARA should have a sink node with a single position sink and the same components as a restroom, it is a unique space with specific finishes and additional components.

Artificial Turf

Since most service animals are dogs, SARAs are focused on dogs in their design. Service dogs are trained to relieve themselves on real grass and/or concrete/asphalt (in urban areas). As these surfaces are impractical in an interior airport space, artificial turf that is specifically manufactured for pet relief areas is recommended. The "grass" blades in the turf should be woven onto a drainable mat that can be stored in rolls like a carpet. Ample attic stock of turf rolls cut to size should be readily available to manage busy periods. The attic stock should be stored in a nearby closet. The turf typically sits on a grid of raised interlocking plastic tiles with legs to keep an open space below for flushing. The turf is secured around the perimeter with strong Velcro, so it is easily removed for cleaning. Pheromones are typically applied to the grass to inspire the dogs to relieve themselves; however, these need periodic reapplication to remain effective. Some dogs will be reluctant to go in SARAs because of the distracting scent of previous animals or because they are trained to go on concrete (city dogs) and not on carpet (dogs can tell that the turf is not real grass). In addition, some airports provide a litter box or two for cats (usually up on a shelf away from curious canine noses).

Turf Area Flushing

Turf should ideally be rinsed with a hose and flushed after each use. This puts an extra burden on the airport cleaning staff but is necessary to keep the turf smelling fresh and inconspicuous to people and animals. A hose reel stored in a locked recessed cabinet can be a convenient solution. Flushing systems made of a loop of perforated pipe that lines the perimeter below the turf are available as part of the turf system. Raised tiles supporting the turf mat are also available with arched gaps in the supports so the pipework can run through. The water should flush from the pipe perforations down a subfloor that slopes to one or more floor drains below. Flushing should be activated by a wall-mounted push button located on a wall that is remote from the turf.

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Epoxy Flooring

Because of the frequent cleaning of this space, a monolithic epoxy floor coating with integral wall base up 4 inches is highly recommended. The animals will have accidents, so a tile floor with grout is discouraged. Be mindful that sealed concrete can be slippery when wet.

Jointless Wall Surface

Like the floor, the walls should also be free of grout joints. Monolithic, impervious wall panels or coatings (e.g., quartz or epoxy paint) are recommended. Use the same surface finish throughout the space to minimize transition joints.

Mop Sink

In addition to being used for cleaning, a mop sink can be handy for rinsing an animal's paws. Mops and cleaning supplies should be stored in a small closet within the space. This can also be a good place to store extra turf rolls.

Accessories

In addition to the accessories at the sink node, a dispenser should be provided with free disposable, biodegradable bags to pick up an animal's leave-behinds. Some airports also provide a "pooper-scooper" to make cleanup easier. A vented waste can with a sealing lid will minimize odors. A refilling water bowl is a welcome feature for travelers with pets. A fold-down seat/bench is very useful, especially for people with reduced mobility, to sit and clean off a dog's paws or sit with their dog a few moments to give it a treat or attention. The seat/bench should be located by the mop sink. In addition, an electronic air purifier can help keep the air smelling fresh.

Exhaust

Doubling the typical level of exhaust (or more) is recommended to manage the odors from pets and humidity from frequent cleaning. More air should be exhausted from the room than supplied to it to create a slight negative pressure. Pulling air in from the public spaces prevents odors from escaping. A sensor can be used to trigger the exhaust when the door is opened. A louver in the door will help air intake but may inadvertently let odors leak out to public spaces.

4.4.4 Lactation

The American Institute of Architects has a detailed "Best Practices" document for lactation rooms that is recommended (see Figure 4-5 for a photo of an airport lactation room). While it is geared more toward corporate-type work settings, the information related to components is useful for airports as well. The recommendations below are largely based on this document.

Baby Diaper Changing Node

This room should feature a typical restroom baby diaper changing node with a changing surface. The faucet for this sink node should be a taller gooseneck type for easy cleaning of the lactation pump parts. The sink should be near the counter for quick access to soap, towels, etc.

Upright Chair

A simple-to-adjust task chair on wheels (like those found at office workstations) with durable, easy-to-clean fabric should be provided. The wheels are important so the chair can easily be

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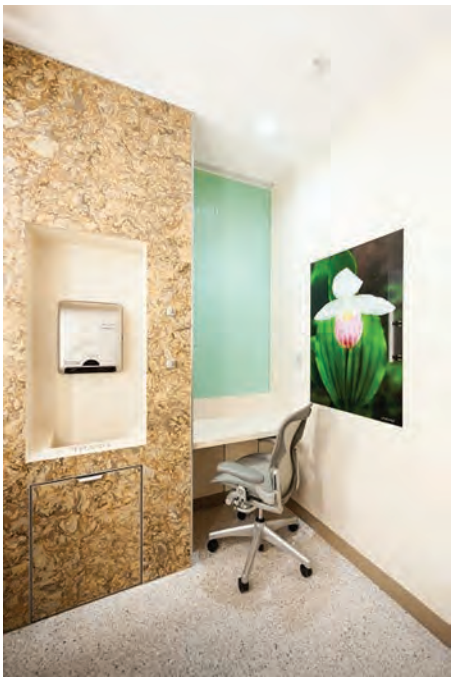


Figure 4-5. Lactation room (MSP).

moved out of the way for wheelchair users. If space permits, a second chair for an older child is a nice accommodation.

Full-Length Mirror

In addition to the mirror at the sink, a tall mirror is important for the mother to check her clothes before leaving.

Materials

Finishes should be calming and durable. Cork, linoleum, and low-pile, easy-to-clean carpet tile are good options for wheeled chairs. Large-format tile and terrazzo are more durable and easier to maintain but can make the room "loud" and a bit utilitarian feeling.

Lighting

In both lactation rooms and spaces for nursing mothers, lighting is very important. The light should not be too bright, and the ability to dim the light allows users to create their desired comfort level. Task lighting should be provided at the counter and sink. Ambient lighting should be dimmable.

Other Recommendations

Provide a lockable door with an occupancy indicator if possible. Other amenities to consider include coat/bag hooks, a diaper dispenser, a milk/bottle warmer, acoustic privacy to promote a calm environment, a comfortably warm temperature (or user-operable thermostat), and good exhaust.

4.4.5 Nursing Mothers

The space for nursing mothers (see Figure 4-6) will need similar nodes and components as the lactation room with the following differences.

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Figure 4-6. Nursing mothers' area (MSP).

Comfortable Chair(s)

Nursing mothers sit with their child cradled in their arms, so a softer chair with good arm support is preferred. A variety of chair styles for different body types can be considered; however, for ease in sitting down and standing up, the chair should not have wheels. While a rocking motion is a welcome feature, traditional rocking chairs should be avoided, as they are difficult for people with mobility impairments to get out of, especially while holding a baby. Chair materials should be easy to clean and should not have deep crevices around the seat cushion, which are prone to trapping crumbs and spills. Multiple chairs are recommended, depending on available space.

Side Tables

It is recommended to provide a side table between pairs of chairs or within a chair grouping on which a baby bottle, purse, and/or phone can be placed. An easy-to-reach electrical outlet on the side of the table is useful for phone charging (provide a USB port if possible).

Children's Play Activities

If a nursing mother has young children in tow in addition to her nursing baby, an activity area can be a good distraction for them. A few durable, easily cleanable toys and/or a small table and chairs with games can be provided, or if space is limited, a wall-mounted activity center works well.

Privacy Curtain

If there are multiple chairs for nursing, a retractable curtain (e.g., a hospital cubicle curtain with sprinkler clearance at the top) will provide mothers with a little privacy if desired.

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4.5 Considerations for Waiting- and Layover-Related Amenities

As described in Chapter 2, spaces for layovers and waiting provide activities and distractions for a traveler who has a significant layover (often including an overnight wait) or who is waiting a short time for a flight. Most layover-related amenities are provided by airport tenants; therefore, airport input into the layouts, nodes, and components of these spaces may be minimal. The following provides items to consider for the waiting-related amenity spaces.

4.5.1 Children's Play Area

There are many types of play activities that can be provided to expel the energy of bored or antsy children. Everything from kid-sized reading carrels to sprawling collections of slides and climbing structures can be installed in a children's play area (see Figure 4-7). Consider local artisans or craftspeople for these features. An essential component for the play area includes code-compliant flooring materials to soften falls (always verify local code requirements). Good lighting is essential to make all hazards visible and to make the play area inviting.

4.5.2 Sensory Room

This new type of amenity space for people with various sensory needs or conditions is often a joint planning project involving the airport and local relevant organizations. Each new iteration is a learning experience and testing laboratory for understanding which elements are effective and which are counterproductive. Variety is a key feature in terms of activities and visual and aural stimulation. Flexibility is another key element. Materials in these spaces need to be durable and easy to clean. The activities should be accessible. LGW, for example, has an area of floor cushions that are easy to move so that a wheelchair can easily be navigated around the room (see Figure 4-8). It is also good practice to have a controlled entrance to prevent abuse of the space and prevent overcrowding, which can overstimulate users. See Chapter 2 for more considerations.

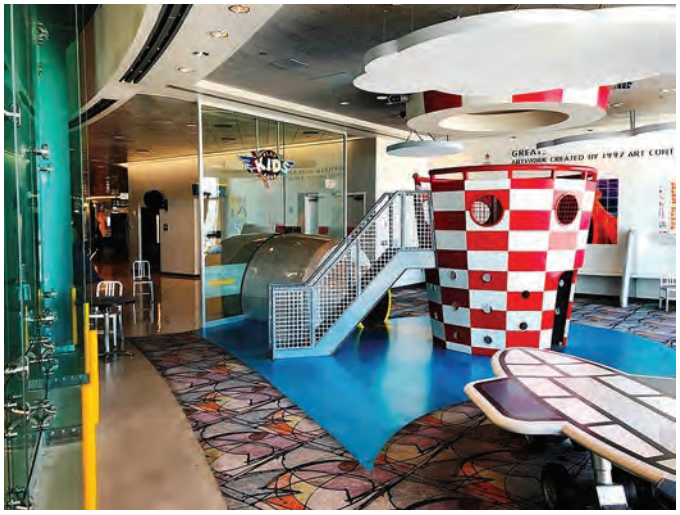


Figure 4-7. Children's play area—McCarran International Airport (LAS).

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Figure 4-8. Sensory room (LGW).

4.5.3 Meditation/Quiet Area

Calm is the desired ambiance for this space (see Figure 4-9). Soft, comfortable seating of various types (couch, lounge chair, bean bag, and so forth) for different body types and mobility is desirable along with carpeted flooring, a comfortable temperature, soft lighting, and whenever possible, privacy screens to break up the space into smaller, less-distracting areas. Quiet music may be provided in the background to mask sounds from nearby active spaces. Other provisions to consider include a vending machine with sleep masks and water, as is found at Hamad International Airport (DOH).

4.5.4 Yoga Area

Like meditation/quiet spaces, the yoga area is a calming space. Sometimes this space is included with meditation/quiet spaces. Desirable features include soft lighting, a comfortable temperature, no noisy distractions, and, possibly, privacy screens. A drinking fountain is an appreciated provision as well. A storage area for mats, balls, and bands (if provided) should be nearby. Since the yoga space tends to be out of view, good wayfinding signs are especially important.

4.5.5 Worship

The most basic component of this amenity space (see Figure 4-10) is seating—whether chairs or benches, fixed or movable. Screens or dividers can be used to configure the space for different services or events. From there, it depends on the mission and/or denomination(s) to make additional provisions. There may be a pulpit and one or more altars. A place for prayer rugs and racks could be considered as well as shelves for removed shoes, as there may be a space for ablution (see Section 4.5.6). Shelves may be needed for a library of books. In addition, office space with a desk, a few chairs (two to three), file storage, and storage space for an

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Figure 4-9. Meditation/quiet area—San Francisco International Airport (SFO).



Figure 4-10. Worship area—Toronto Pearson Airport (YYZ).

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Figure 4-11. Ablution area (LGW).

assortment of supplies may be needed. Specific requirements will be unique to each airport and its spiritual director. It is important to involve this person in the planning process.

4.5.6 Ablution

When provided, a space for ablution (see Figure 4-11) typically features one or more benches for removing shoes and socks upon entering the space, as well as lockers to store them. There are numerous types of footbaths available, from off-the-shelf fixtures to custom-built troughs with multiple positions. At the foot-washing stations, a seat, floor mat, soap dispenser, and hand-drying options should be provided. There should be at least one wheelchair-accessible position.

4.5.7 Smoking

Rarely provided in Western airports, except outdoors on the non-secure side, a smoking space (see Figure 4-12) can be as simple as an outdoor area (planned or impromptu) with (ideally) a container for spent smoking materials. When a smoking area is combined with other spaces, it may require separation from those spaces and may need to be equipped with adequate ventilation. Smoking rooms should be equipped with an air filtration system as well as air changes to help keep the air in the space fresh. The space should be tightly sealed from adjacent areas.

4.6 Cost Analysis

4.6.1 Overview

In general, airport restrooms are considerably more expensive than other types of public restrooms. The regulated/essential amenities can be costly as well. If restrooms are designed and constructed for longevity, as recommended in this guidebook, they will require more space per person and will be constructed with higher-quality materials. Managing the budget can be

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Figure 4-12. Smoking area (LAS).

a challenge. To help find the optimum balance among customer service needs, life-cycle costs, and constraints on construction costs, it is recommended that a professional estimator or contractor familiar with airports and the local building climate be included on the restroom and amenities team.

The matrix presented in Appendix A provides information on making initial selections of products with numerous variations based on costs, life spans, and features. Section 4.7 provides worksheets to help prioritize the components. Coupled with periodic cost estimates/updates, these tools can help keep the scope of the project within the allocated budget.

4.6.2 Estimates

Regardless of whether the restroom is part of a larger building project or is a standalone project, the first estimate should be prepared at the completion of the schematic design phase (when the general design concept is complete). The first estimate should be a high-level estimate to test the scope based on system or square-foot costs. If the total is too high, the scope and quality of materials can be re-evaluated to meet the budget, or the airport can elect to increase funding.

About midway through the architect's development of the construction documents, it is prudent to have a second estimate prepared. This estimate should be detailed, itemizing the labor and materials for each element in the project. This detailed estimate is the last chance to verify that the scope and quality are in conformance with the budget. It is much easier (and less costly) to make modifications at this point than after the bids are submitted.

4.7 Prioritization

With so many components to consider, it can be an overwhelming task to narrow a project's scope. Some components—floor materials, stalls, and so forth—are easy to include because they are must-haves. Others—grooming nodes, for example—are extras that might be nice to have, but only if there is enough space and a large enough budget.

Tables 4-1 through 4-3 show worksheets that list the components for restrooms described in this chapter (editable versions of these worksheets can be found on the TRB website by

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Table 4-1. Restroom Prioritization Worksheet 1 (refer to Section 4.1 for information on worksheet organization).

MUST HAVE	WISH LIST	NOT INTERESTED	RESTROOM COMPONENTS
			SIGNAGE
			Concourse Wayfinding
			Room Location Sign
			Room Name and Number
			Accessible and Ambulatory Stalls
			Door Numbers
			Employees Wash Hands
			Advertising
			Severe Weather Shelter
			SURFACES
			Floor
			• *Porcelain Tile*
			• *Porcelain Stoneware Slab*
			• *Epoxy Terrazzo*
			Wall and Base
			• *Porcelain Tile*
			• *Porcelain Stoneware Panel*
			• *Quartz*
			• *Epoxy Terrazzo Cove*
			Ceiling
			• *Gypsum Board*
			• *Metal Panel*
			• *Linear Metal*
			• *Acoustic Ceiling Tile*
			Stall Partition - Type
			• *Ceiling Hung*
			• *Floor Mounted / Overhead Braced*
			• *Floor Mounted*
			• *Floor and Ceiling Mounted*
			Stall Partitions / Urinal Screens
			• *Powder Coated Steel*
			• *Solid Plastic (HDPE)*
			• *Phenolic*
			• *Stainless Steel #4*
			• *Stainless Steel Textured*
			• *Recycled Paper - Resin Core*
			• *Solid Surface*
			• *Quartz*
			• *Stone*
			• *Glass*
			Counters / Backsplash
			• *Stone*
			• *Quartz*
			Solid Surface
			Room Door
			• *Stainless Steel*
			• *Painted Metal*

MUST HAVE	WISH LIST	NOT INTERESTED	RESTROOM COMPONENTS
			ACCESSORIES
			Mirrors
			• *Sink Node*
			• *Grooming Node*
			• *Full-height*
			Paper Towel Dispenser
			• *Folded*
			• *Center Pull*
			• *Roll Type*
			Waste Receptacle
			• *Free Standing*
			• *Surface Mounted*
			• *Recessed*
			• *Enclosed*
			Hand Dryer
			• *Top Access*
			• *Through Access*
			• *Recessed*
			• *Faucet Mount*
			Child-step at Sink
			Toilet Paper Dispenser
			• *Surface - Standard Roll*
			• *Surface - Jumbo Roll*
			• *Recessed - Standard Roll*
			• *Recessed - Jumbo Roll*
			Stall Waste Receptacle
			• *Surface*
			• *Recessed*
			• *Combination*
			Seat Paper Dispenser
			• *Individual*
			• *Combination*
			• *Automatic*
			Shelf at Toilet and Urinal
			• *Wall Mounted*
			• *Fold Down*
			• *Built In*
			Hooks
			Biohazard Disposal
			• *Exposed Container*
			• *Surface Mounted Concealed Container*
			• *Recessed Concealed Container*
			Sanitary Products Vendor
			• *Surface Mounted*
			• *Recessed*

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Table 4-2. Restroom Prioritization Worksheet 2 (refer to Section 4.1 for information on worksheet organization).

MUST HAVE	WISH LIST	NOT INTERESTED	RESTROOM COMPONENTS
			Baby Diaper Changing
			• *Fold Down*
			• *Counter - Surface*
			• *Counter - Integral*
			Diaper Kit Vendor
			• *Surface Mounted*
			• *Recessed*
			PLUMBING
			Sink
			• *Wall Mount*
			• *Under Counter*
			• *Self Rimming*
			• *Integral*
			• *Trough*
			Faucet
			• *Automatic*
			• *Turbine Powered*
			• *Solar Powered*
			• *Manual*
			Soap Dispenser
			• *Automatic*
			• *Manual*
			Soap Type
			• *Foam*
			• *Liquid*
			Soap Container
			• *Bulk*
			• *Bottle*
			Toilet
			• *Blow Out*
			• *Siphon Jet*
			Toilet Flush Valve
			• *Manual*
			• *Automatic*
			• *Turbine Powered*
			• *Solar Powered*
			Urinal
			• *Standard Flow*
			• *Waterless*
			• *Low Flow*
			• *Pint*

MUST HAVE	WISH LIST	NOT INTERESTED	RESTROOM COMPONENTS
			Urinal Flush Valve
			• *Manual*
			• *Automatic*
			• *Turbine Powered*
			• *Solar Powered*
			Drinking Fountain
			Bottle-filling Station
			• *Manual*
			• *Electronic*
			Mop Sink
			Mop Sink Faucet
			Hose-bibb
			HVAC
			Radiation
			Air Supply
			Exhaust
			Diffusers
			Thermostat
			ELECTRICAL
			Maintenance Receptacles
			Grooming/Shaving Receptacles
			LIGHTING
			Ambient
			Entry
			Sink Node
			Toilet Stalls
			Occupancy Sensor Switch
			Lamps
			• *LED*
			• *Fluorescent*
			Daylight
			TECHNOLOGY
			Fire Alarm
			Paging
			Ambient Music
			Restroom Management System
			• *Card reader*
			• *Occupancy Counter*
			• *Video*
			EMERGENCY NODE
			Emergency/Courtesy Phone
			Emergency Pull Station
			Fire Extinguisher
			AED
			Blue Light

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Table 4-3. Regulated/Essential Amenities Prioritization Worksheet 3 (refer to Section 4.1 for information on worksheet organization).

MUST HAVE	WISH LIST	NOT INTERESTED	REGULATED / ESSENTIAL AMENITIES COMPONENTS
			COMPANION CARE RESTROOM
			Restroom Sink Node
			Toilet
			Child-size Toilet
			Full-height Mirror
			Fall Monitoring System
			Accessories
			Biohazard Disposal
			Feminine Products Disposal
			Baby Diaper Dispenser
			Adult Diaper Dispenser
			Wall-mounted Toddler Seat
			Hook Near Toilet for Stoma Bag
			Curtain/Divider
			Hooks
			CHANGING TABLE RESTROOM
			Restroom Sink Node
			Toilet
			Changing Table
			Hoist
			Hand-held Shower
			Fall Monitoring System
			Monitored Use System
			Room Instructions Signage
			Full-height Mirror
			Accessories
			Diaper-Wipes Dispenser
			Adult Diaper Dispenser
			Fold-down Seat
			Hooks

MUST HAVE	WISH LIST	NOT INTERESTED	REGULATED / ESSENTIAL AMENITIES COMPONENTS
			SERVICE ANIMAL RELIEF AREA (SARA)
			Artificial Turf
			Turf-area Flushing
			Hard Flooring Surface
			Jointless Wall Surface
			Restroom Sink Node w/o Changing
			Mop Sink
			Air-purifying System
			Increased Exhaust
			Accessories
			Pet-waste Bag Dispenser
			Vented, Sealed Trash for Animal Waste
			Self-filling Water Bowl
			Fold-down Bench
			LACTATION
			Restroom Sink Node
			Tall Goose-neck Faucet
			Counter Area
			Upright Chair on Wheels
			Full-height Mirror
			Receptacle for Pump
			Phone-charging Outlet
			Task Lighting at Counter
			User-operated Thermostat
			Monitored Use System
			Accessories
			Baby Diaper Dispenser
			Hooks
			Bottle Warmer
			Door with Occupancy Indicator
			NURSING MOTHERS
			Restroom Sink Node
			Comfortable Chair(s)
			Full-height Mirror
			Phone-charging Outlet
			Children's Play Activities
			Accessories
			Baby Diaper Dispenser
			Hooks

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searching on "ACRP Research Report 226"). These worksheets provide a tool the restroom and amenities team can use to classify potential features as (1) those that are must-have, (2) those that belong on a wish list for if/when resources become available, and (3) those in which the team is not interested. A version of this table can be created for whichever amenity spaces are included in a project. The orange headings (see Table 4-1) are the primary component groupings used for restrooms throughout this chapter. The blue headings indicate that the component has options (the bulleted items) that are compared in the component comparison matrix provided in Appendix A.

Components should be considered that will have the greatest impact on the airport's customer service needs, especially those that elicit frequent complaints. Maintenance should also be considered. For example, do desired components place additional burdens on the facilities staff in the long term? The component comparison matrix (Appendix A) can be used to narrow choices for those components with numerous alternatives. A preliminary cost comparison of the must-have and wish-list selections will add that important variable to the decision-making process.

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CHAPTER 5

Implementation

5.1 Construction

Typically, a restroom or amenities project will be part of a larger airport project. It may be within a new terminal or expansion, or it may be part of a renovation of a specific area of the airport, perhaps baggage claim. In these types of projects, construction is straightforward. The work occurs within the construction limits and is simply another space that is built out. On the other hand, renovation of individual restrooms, usually as part of a comprehensive restroom upgrade program throughout the airport, is a whole different world of complexity.

5.1.1 Phasing

Unless the project is a facelift with new finishes and replacement of fixtures and accessories, it is unlikely the project will be contained within the restroom's existing walls. The customer service driver of providing larger stalls and more circulation space implies enlarging the restroom space. Alternatively, the project may involve carving out tenant space for a new SARA. This will affect the neighboring spaces and likely require relocating fixtures and their associated plumbing.

For obvious reasons, it is not possible to renovate all of an airport's restrooms at once, so the restroom and amenities team will need to develop a phasing plan. It works best to spread out the work, so each phase includes no more than one restroom from any concourse or terminal location at a time, even if this approach is not the most efficient. The loss of even one restroom can have a significant impact on a concourse, especially if the fixture count is inadequate to begin with.

Construction barriers will temporarily shrink circulation spaces, a significant impact in concourses. The airport code official should be consulted to discuss how this condition (narrowed corridors) may create egress and accessibility concerns that may require the creation of an alternative path.

If fixtures are relocated, power panels added, and so forth, then utility shutdowns for plumbing, HVAC, sprinklers, and power will need to be planned for and coordinated between the contractor and airport managers.

Similar impacts occur if the restroom is being moved to a better location based on the master plan developed in Chapter 2. This kind of move will typically require the relocation of another tenant or function area, which will need to be completed before demolition and construction work can occur for the new restroom. Timing can be affected by lease terms,

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so implementation decisions should be made with a long view in the context of the airport's overall master plan.

5.1.2 Delivery Methods

Today's construction industry offers a variety of project delivery methods:

- Design-bid-build
- Design-negotiate-build
- Design-build
- Construction management
- Owner-build
- Integrated project delivery

The choice of project delivery method, however, is affected by an airport's unique legal requirements, which are based on the type of governance. In an ideal world, the general contractor is brought in at the same time as the architect to be part of the restroom and amenities team. Their knowledge about construction methods and costs is invaluable and makes for a more efficient project planning and design process. However, because most airports are public entities, public bidding is required.

In a bidding situation, it is recommended instead to bring in a construction manager early in the process. With a similar knowledge of the local construction industry, a construction manager can equally provide insight into construction and costs. A construction manager can also oversee the construction period, which can be especially useful for airports that are smaller or have infrequent projects. Regardless of the delivery method, it is vital that the owner, architect, and contractor (and/or construction manager) have a frank discussion up front about expectations of the project and construction process and associated limitations so the partnership is on the same page from the start.

5.1.3 Maintaining Standards

To maintain the project's design standards, it is recommended to sole source some or all products. However, on a publicly bid project, it is standard practice to provide three manufacturers for a given product. Sometimes this is a legal requirement within a jurisdiction or it could be a requirement of the airport's governance. Depending on the requirement, the number of manufacturers required may vary. While the intention of creating non-exclusionary bidding is laudable, it can wreak havoc on a project's design and the airport's long-term maintenance, primarily by compromising product standards.

Allowing "similar" floor tiles over the course of several projects, for example, will complicate managing the airport's attic stock. Replacement tiles from a different manufacturer may not match the surrounding existing tiles in a future repair. Likewise, leaving the toilet manufacturer open to the capriciousness of the bidding climate can cause the airport to have to stock spare parts for multiple toilet models, not to mention having restroom and amenity spaces looking markedly different from each other.

A successful strategy used by some airports is to allow substitutions. In this scenario, the bid specifications will list the desired manufacturer and its product. Then, instead of listing comparable manufacturers, the specification indicates that substitution requests are permitted. Standard specification language puts the onus on the bidder to demonstrate that the proposed substitution is an equivalent match to the desired product. The argument can also be made

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legally that the cost of sole-sourced products is a small percentage of the overall construction cost as the labor cost is not included.

5.1.4 Prefabricated Restrooms

Prefabricated restrooms are a concept that evolved primarily from hospital projects where pre-assembled headwalls with all the equipment and finishes in place when shipped to the site were installed in a fraction of the normal construction time. While an entire prefabricated airport restroom might only be feasible in new construction, portions of the space can successfully be prefabricated. Restrooms have numerous, repetitive, modular elements that could be considered for prefabrication; these include pipe chase plumbing assemblies, sink walls, and, possibly, an entire family room. Here, too, finishes would already be complete. Shortening construction time significantly minimizes the disruption caused by one or more out-of-service restrooms.

5.2 Post-Occupancy Evaluation

Since airport restrooms and some amenity spaces, especially in large airports, are repeated throughout the facility, it is important to think of the spaces as living prototypes that are ever-evolving. As each instance is designed and completed, there should be a cycle of evaluation. In the construction industry, a valuable final phase of a project is a post-occupancy evaluation.

This evaluation is conducted near the 1-year anniversary of the project's completion, the milestone when the contractor's standard 1-year warranty period concludes. The evaluation is conducted by representatives from the airport, the architect, and the general contractor. The purpose is to verify that everything functions as planned. If not, the design might be tweaked, if not for the project being evaluated, then for the next project. If a product is failing, a replacement can still be made under warranty.

More important, product selection can be revisited to see if there is a better alternative available at the time of the evaluation. This is also the time to update the airport design and materials standard as well as the maintenance SOPs (see Appendix G for an example). Customer service comments from the past year should also be collected, analyzed, and discussed.

Ideally, the restroom and amenities team would be reconvened for this important conclusion to a project. With the wisdom of lessons learned, the team will be ready to begin the entire planning, design, and implementation process anew for the next restroom or amenities project.

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Abbreviations

ACAA	Air Carrier Access Act
ADA	Americans with Disabilities Act
ADG	airplane design group
AED	automated external defibrillator
ASQ	Airport Service Quality (ACI annual survey)
ATL	Hartsfield–Jackson Atlanta International Airport
BMS	building management system
CMU	concrete masonry unit
DFW	Dallas/Fort Worth International Airport
DLH	Duluth International Airport
DOH	Hamad International Airport
EQA	equivalent aircraft
FC	foot-candles
FEMA	Federal Emergency Management Agency
FIDS	flight information display system
GFCI	ground-fault circuit interrupter
GTC	ground transportation center
HVAC	heating, ventilating, and air conditioning
IATA	International Air Transport Association
ICC	International Code Council
ISO	International Organization for Standardization
JAC	Jackson Hole Airport
LAS	McCarran International Airport
LAX	Los Angeles International Airport
LED	light-emitting diode
LEX	Blue Grass Airport
LF	load factor
LGB	Long Beach Airport
LGW	Gatwick Airport
LHR	London Heathrow Airport
LoS	level of service
M&G	meeter and greeter
MSP	Minneapolis–Saint Paul International Airport
MUFIDS	multi-user FIDS
O&D	origin and destination
PHP	peak-hour passenger
PIT	Pittsburgh International Airport

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RMS	restroom management system
SARA	service animal relief area
SFO	San Francisco International Airport
SMF	Sacramento International Airport
SNA	John Wayne Airport
SOP	standard operating procedure
VAV	variable air volume
WW	well-wisher
YYZ	Toronto Pearson Airport

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Appendices A Through I

The following appendices are not provided herein but can be found on the TRB website by searching on "ACRP Research Report 226":

Appendix A: Component Comparison Matrix
Appendix B: Existing Restroom Evaluation Forms
Appendix C.1: Case Studies—Restrooms
Appendix C.2: Case Studies—Amenities
Appendix D.1: Stakeholder Outreach—Restrooms
Appendix D.2: Stakeholder Outreach—Amenities
Appendix E: Surveys
Appendix F: Restroom Design Guidelines/Standards Sample
Appendix G: Restroom Standard Operating Procedures Sample
Appendix H: Bibliography
Appendix I: Glossary

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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